ANNEX 1 (part A)

Innovation action

NUMBER — 776708 — HOUSEFUL

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1.1. The project summary

Project Number [1]	776708	Project Acronym [2]	HOUSEFUL

One form per project

General information

Project title [3]	Innovative circular solutions and services for new business opportunities in the EU housing sector
Starting date [4]	01/05/2018
Duration in months [5]	54
Call (part) identifier [6]	H2020-CIRC-2017TwoStage
Topic	CIRC-01-2016-2017 Systemic, eco-innovative approaches for the circular economy: large-scale demonstration projects
Fixed EC Keywords	Resources efficiency, Circular economy, Service innovation, Technological innovation, Energy Efficiency, Business model innovation
Free keywords	Social innovation, systemic service, housing sector, construction, water re-use, biowaste treatment, biogas production, wastewater, co-creation , circular economy business opportunities
Abstract [7]	

The housing sector is a major contributor to current global problems of resource depletion and climate change, representing one of the most important consuming sectors at EU level: 50% of all extracted materials, 40% of final energy consumption, 33% of water consumption and 33% of all produced waste. The lock-in to the linear business models of today is causing many environmental problems and is one of the major barriers in transition towards a circular economy. HOUSEFUL project proposes an innovative paradigm shift towards a circular economy for the housing sector by demonstrating the feasibility of an integrated systemic service composed of 11 circular solutions. HOUSEFUL will introduce solutions to become more resource efficient throughout the lifecycle of a building, taking into account an integrated circular approach where energy, materials, waste and water aspects are considered. This approach fosters new forms of co-creation, increasing the collaboration among stakeholders of the housing value chain to develop new circular solutions and services. HOUSEFUL concept will be large scale demonstrated at 4 demo-sites in Austria and Spain, adapting the concept to different scenarios, including in social housing buildings. HOUSEFUL solutions will be evaluated from an environmental (Life Cycle Assessment), economic (Life Cycle Cost) and social (Social Assessment) point of view. The results obtained will be used to define an integrated HOUSEFUL service which will be driven and promoted through a SaaS (Software as a Service). The SaaS will integrate a
Circularity Tool to quantify the circularity level of buildings and will include different circular solutions to be offered as services, encouraging the housing value chain to redesign traditional business models towards circular ones. 10 EU Follower buildings will be engaged with the support of a Collaborative Community of Housing Experts to replicate HOUSEFUL results and maximise the impact of the project.

1.2. List of Beneficiaries

Project Number [1]	776708	Project Acronym [2]	HOUSEFUL

List of Beneficiaries

No	Name	Short name	Country	Project entry month[8]	Project exit month
1	ACONDICIONAMIENTO TARRASENSE ASSOCIACION	LEITAT	Spain	1	54
2	INSTITUT DE TECNOLOGIA DE LA CONSTRUCCION DE CATALUNYA	ITeC	Spain	1	54
3	FUNDACION CARTIF	CARTIF	Spain	1	54
4	ALCHEMIA-NOVA GMBH	ALCN	Austria	1	54
5	AGENCIA DE L'HABITATGE DE CATALUNYA	AHC	Spain	1	54
6	VISUM LIMITED	VISUM	Ireland	1	54
7	SISTEMES AVANCATS DE ENERGIA SOLAR TERMICA SCCL - AIGUASOL	AIGUASOL	Spain	1	54
8	LGI CONSULTING SARL	LGI	France	1	54
9	ARCHITEKTURBURO REINBERG ZT GMBH	GWR	Austria	1	54
10	TURNTOO B.V.	Turntoo B.V.	Netherlands	1	54
11	GEMEINNUTZIGE BAU-, WOHN UND SIEDLUNGSGENOSSENSCHAFT NEUES LEBEN REGISTRIERTE GENOSSENCHAFT MIT BESCHRANKTER HAFTUNG	Neues Leben	Austria	1	54

12	COMITE EUROPEEN DE COORDINATION DE L'HABITAT SOCIAL AISBL	CECODHAS	Belgium	1	54
13	WATER, ENVIRONMENT AND BUSINESS FORDEVELOPMENT SL	WE&B	Spain	1	54
14	FONDAZIONE ICONS	ICONS	Italy	1	54
15	IDP INGENIERIA Y ARQUITECTURA IBERIA SL	IDP	Spain	1	54
16	HOMEBIOGAS LTD	HOMEBIOGAS LTD	Israel	1	54

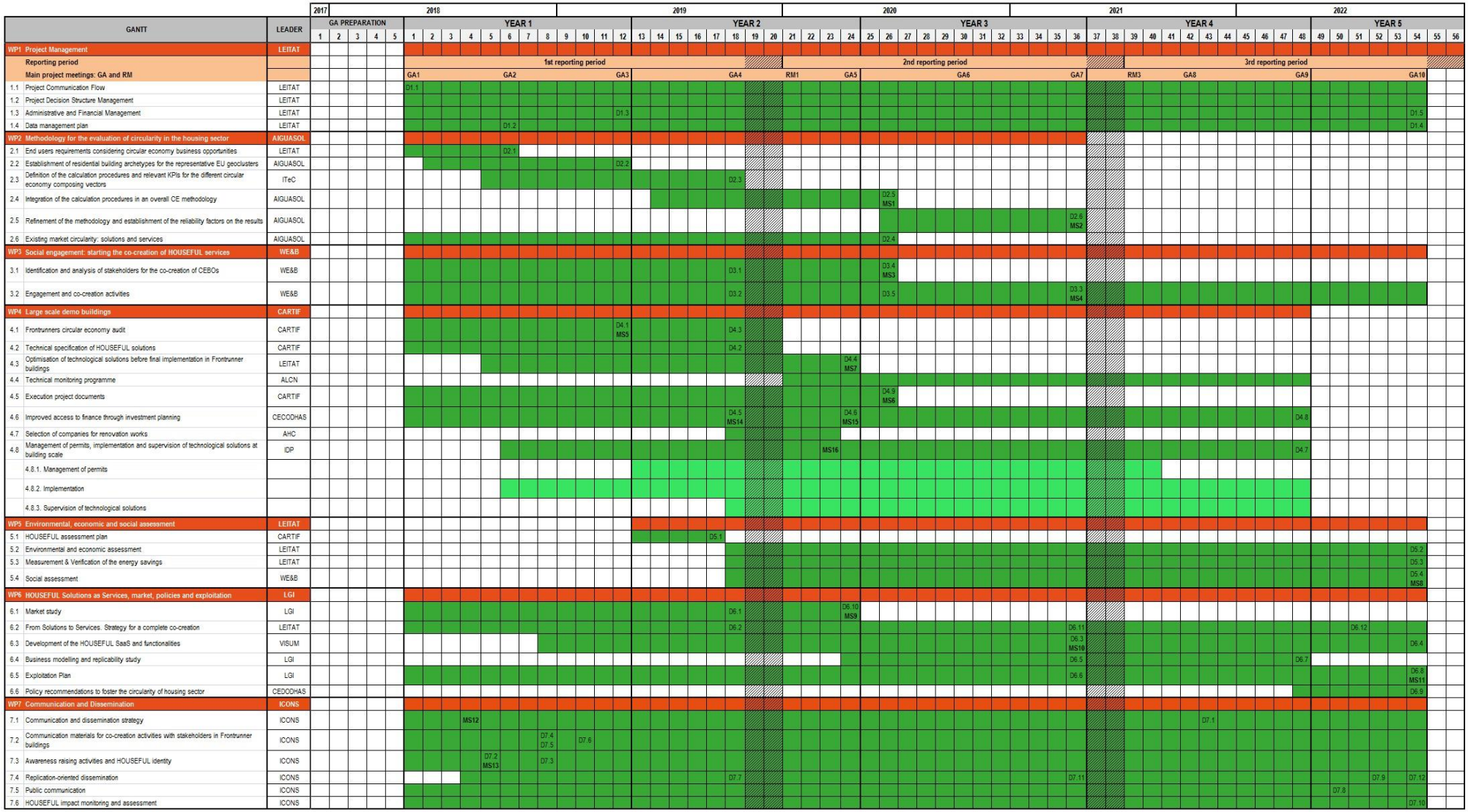
1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	**WP Title**	**Lead beneficiary**[10]	**Personmonths**[11]	**Start month**[12]	**End month**[13]
WP1	Project Management	1 - LEITAT		1	54
WP2	Methodology for the evaluation of circularity in the housing sector	7 - AIGUASOL		1	36
WP3	Social engagement: starting the cocreation of HOUSEFUL services	13 - WE&B		1	54
WP4	Large scale demo buildings	3 - CARTIF		3	48

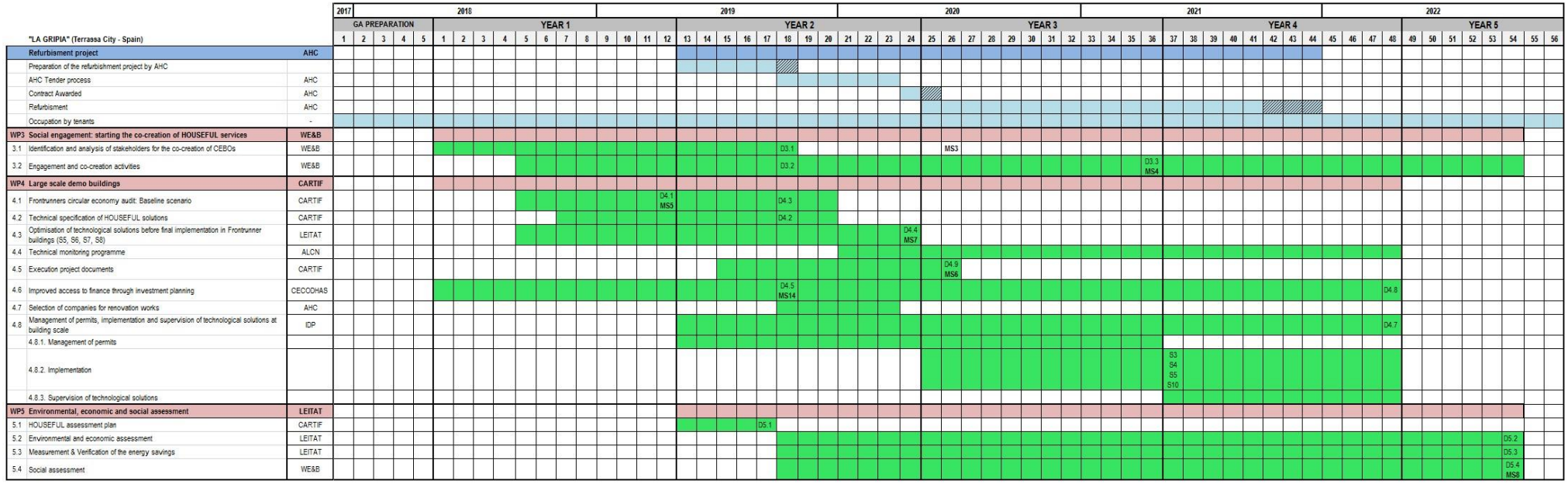
WP5	Environmental, economic and social assessment	1 - LEITAT		8	54
WP6	HOUSEFUL Solutions as Services, market, policies and exploitation	8 - LGI		1	54
WP7	Communication and Dissemination	14 - ICONS		1	54
WP8	Ethics requirements	1 - LEITAT	N/A	1	54
		Total			

.2 Gantt Chart



D: Deliverable. / M: Milestone / GA: General Assembly (main project meetings) / RW: Review Meeting

2017
2018
2019
2020
2021
2022
GA PREPARATION
YEAR 1
YEAR 2
YEAR 3
YEAR 4
YEAR 5
"ELS MESTRES" (Sabadell City - Spain)
Refurbisment project
Preparation of the refurbishment project by AHC
Identification of HOUSEFUL innovation works affecting the tender process
AHC Tender process
Contract Awarded
Refurbisment
Public tender for social housing occupation
Occupation by tenants
WP3 Social engagement: starting the co-creation of HOUSEFUL services WE&B
3.1 Identification and analysis of stakeholders for the co-creation of CEBOs WE&B
3.2 Engagement and co-creation activities WE&B
WP4 Large scale demo buildings CARTIF
4.1 Frontrunners circular economy audit: Baseline scenario CARTIF
4.2 Technical specification of HOUSEFUL solutions CARTIF
4.3 Optimisation of technological solutions before final implementation in Frontrunner buildings (S5, S6, S7, S8) LEITAT
4.4 Technical monitoring programme ALCN
4.5 Execution project documents CARTIF
4.6 Improved access to finance through investment planning CECODHAS
4.7 Selection of companies for renovation works AHC
4.8 Management of permits, implementation and supervision of technological solutions at building scale IDP
4.8.1. Management of permits
4.8.2. Implementation
4.8.3. Supervision of technological solutions
WP5 Environmental, economic and social assessment LEITAT
5.1 HOUSEFUL assessment plan CARTIF
5.2 Environmental and economic assessment LEITAT
5.3 Measurement & Verification of the energy savings LEITAT
5.4 Social assessment WE&B
D3.1
D3.4 MS3
D3.5
D3.3 MS4
D4.1 MS5
D4.3
D4.2
D4.4 MS7
D4.9 MS6
D4.8
MS16
D4.7
M4.5
S3 S4 S10
S5 S9 S11
S7 S8
D5.1
D5.2
D5.3
D5.4 MS8


2017
2018
2019
2020
2021
2022
GA PREPARATION
YEAR 1
YEAR 2
YEAR 3
YEAR 4
YEAR 5
"LA GRIPIA" (Terrassa City - Spain)
Refurbisment project AHC
Preparation of the refurbishment project by AHC
AHC Tender process AHC
Contract Awarded AHC
Refurbisment AHC
Occupation by tenants -
WP3 Social engagement: starting the co-creation of HOUSEFUL services WE&B
3.1 Identification and analysis of stakeholders for the co-creation of CEBOs WE&B
3.2 Engagement and co-creation activities WE&B
WP4 Large scale demo buildings CARTIF
4.1 Frontrunners circular economy audit: Baseline scenario CARTIF
4.2 Technical specification of HOUSEFUL solutions CARTIF
4.3 Optimisation of technological solutions before final implementation in Frontrunner buildings (S5, S6, S7, S8) LEITAT
4.4 Technical monitoring programme ALCN
4.5 Execution project documents CARTIF
4.6 Improved access to finance through investment planning CECODHAS
4.7 Selection of companies for renovation works AHC
4.8 Management of permits, implementation and supervision of technological solutions at building scale IDP
4.8.1. Management of permits
4.8.2. Implementation
4.8.3. Supervision of technological solutions
WP5 Environmental, economic and social assessment LEITAT
5.1 HOUSEFUL assessment plan CARTIF
5.2 Environmental and economic assessment LEITAT
5.3 Measurement & Verification of the energy savings LEITAT
5.4 Social assessment WE&B
D3.1
MS3
D3.2
D3.3 MS4
D4.1 MS5
D4.3
D4.2
D4.4 MS7
D4.9 MS6
D4.5 MS14
D4.8
D4.7
S3 S4 S5 S10
D5.1
D5.2
D5.3
D5.4 MS8

2017
2018
2019
2020
2021
2022
GA PREPARATION
YEAR 1
YEAR 2
YEAR 3
YEAR 4
YEAR 5
"Baumgarten" (Vienna - Austria)
Refurbisment project
Detail planning
Identification of HOUSEFUL innovation works affecting the tender process
Tender process
Contract Awarded
Refurbisment
Occupation by tenants
WP3 Social engagement: starting the co-creation of HOUSEFUL services WE&B
3.1 Identification and analysis of stakeholders for the co-creation of CEBOs WE&B
3.2 Engagement and co-creation activities WE&B
WP4 Large scale demo buildings CARTIF
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4.4 Technical monitoring programme ALCN
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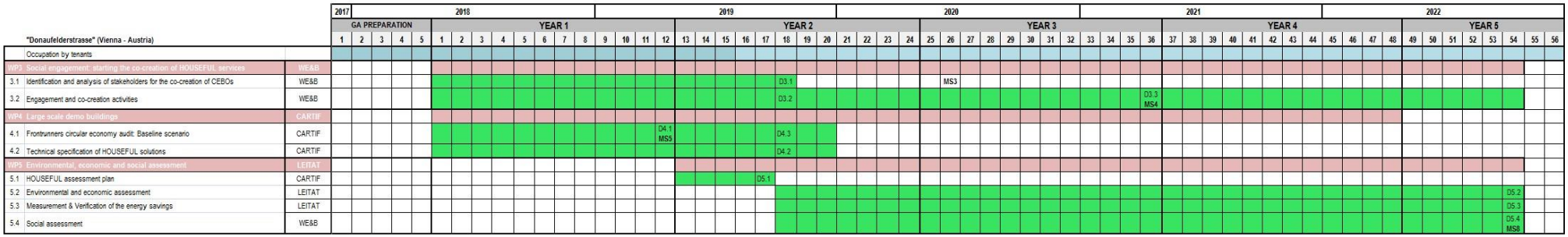
4.8.2. Implementation
4.8.3. Supervision of technological solutions
WP5 Environmental, economic and social assessment LEITAT
5.1 HOUSEFUL assessment plan CARTIF
5.2 Environmental and economic assessment LEITAT
5.3 Measurement & Verification of the energy savings LEITAT
5.4 Social assessment WE&B
D3.1
D3.2
MS3
D3.3
MS4
D4.1
MS5
D4.3
D4.2
D4.4
MS7
D4.9
MS6
D4.6
MS15
D4.8
D4.7
S3
S4
S10
S5
S6
S9
S11
S7
S8
D5.1
D5.2
D5.3
D5.4
MS8


2017
2018
2019
2020
2021
2022
GA PREPARATION
YEAR 1
YEAR 2
YEAR 3
YEAR 4
YEAR 5
"Donaufelderstrasse" (Vienna - Austria)
Occupation by tenants
WP3 Social engagement: starting the co-creation of HOUSEFUL services WE&B
3.1 Identification and analysis of stakeholders for the co-creation of CEBOs WE&B
3.2 Engagement and co-creation activities WE&B
WP4 Large scale demo buildings CARTIF
4.1 Frontrunners circular economy audit: Baseline scenario CARTIF
4.2 Technical specification of HOUSEFUL solutions CARTIF
WP5 Environmental, economic and social assessment LEITAT
5.1 HOUSEFUL assessment plan CARTIF
5.2 Environmental and economic assessment LEITAT
5.3 Measurement & Verification of the energy savings LEITAT
5.4 Social assessment WE&B
D3.1
D3.2
MS3
D3.3
MS4
D4.1
MS5
D4.3
D4.2
D5.1
D5.2
D5.3
D5.4
MS8

1.3.2. WT2 list of deliverables

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Project Manual	WP1	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	1
D1.2	Data Management Plan (DMP)	WP1	1 - LEITAT	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6
D1.3	First activity and management report	WP1	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.4	Final DMP	WP1	1 - LEITAT	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	54
D1.5	Minutes of management meetings	WP1	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	54
D2.1	Report on end-users requirements and CEBO on residential buildings	WP2	7 - AIGUASOL	Report	Public	6
D2.2	Circular economy repository for building archetypes in the geoclusters	WP2	7 - AIGUASOL	Report	Confidential, only for members of the consortium (including the Commission Services)	12

D2.3	Reference methodologies and KPIs in circular economy analysis	WP2	7 - AIGUASOL	Report	Public	18
D2.4	Existing market circularity solutions and product	WP2	7 - AIGUASOL	Report	Public	26

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D2.5	First approach of BCM methodology and implementation in theoretical scenarios	WP2	7 - AIGUASOL	Report	Public	26
D2.6	Description of the BCM methodology and implementation rules in real scenarios	WP2	7 - AIGUASOL	Report	Public	36
D3.1	Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version I)	WP3	13 - WE&B	Report	Public	18
D3.2	Social perceptions and beliefs in circular economy at building scale (version I)	WP3	13 - WE&B	Report	Public	18
D3.3	Social conditions for the co-creation of all HOUSEFUL services and co-created materials	WP3	13 - WE&B	Report	Public	36
D3.4	Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version II - final)	WP3	13 - WE&B	Report	Public	26
D3.5	Social perceptions and beliefs in circular economy at building scale (version II - final)	WP3	13 - WE&B	Report	Public	26
D4.1	Circular economy audit: Baseline calculation	WP4	3 - CARTIF	Report	Public	12

D4.2	Technical specifications and execution project of solutions in Spanish and Austrian demos (version I)	WP4	3 - CARTIF	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	18
D4.3	BIM models and material passport of the Spanish and Austrian demos	WP4	2 - ITeC	Report	Public	18
D4.4	Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safe-ty and monitoring plan for S7 and S8	WP4	1 - LEITAT	Report	Public	24
D4.5	Report on potential improvement in financial schemes and funding opportunities for Demo 2 Frontrunner buildings and replication activities	WP4	12 - CECODHAS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.6	Report on potential improvement in financial schemes and funding opportunities for Demo 3 Frontrunner buildings and replication activities	WP4	12 - CECODHAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D4.7	Report of demonstration and implementation of HOUSEFUL solutions at building scale. Includes the Technical monitoring programme	WP4	3 - CARTIF	Demonstrator	Public	48
D4.8	Report on financial schemes and funding opportunities for all Front-runner buildings and replication activities (final update)	WP4	12 - CECODHAS	Report	Confidential, only for members of the consortium (including the Commission Services)	48

D4.9	Technical specifications and execution project of solutions in Spanish and Austrian demos (version II - final)	WP4	3 - CARTIF	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	26
D5.1	HOUSEFUL assessment plan	WP5	3 - CARTIF	Report	Public	17
D5.2	Environmental and economic impact assess-ment	WP5	1 - LEITAT	Report	Public	54
D5.3	Measurement &Verification	WP5	1 - LEITAT	Report	Public	54
D5.4	Social assessment	WP5	13 - WE&B	Report	Public	54
D6.1	Market study (version I)	WP6	8 - LGI	Report	Public	18

Deliverable Number[14]	**Deliverable Title**	**WP number**[9]	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D6.2	Solution-to-Service monitoring plan (version I)	WP6	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.3	HOUSEFUL SaaS	WP6	6 - VISUM	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D6.4	HOUSEFUL SaaS – Final version	WP6	6 - VISUM	Other	Public	54
D6.5	Impact study of HOUSEFUL services on linear business models	WP6	10 - Turntoo B.V.	Report	Public	36
D6.6	Preliminary Exploitation Plan	WP6	8 - LGI	Report	Confidential, only for members of the consortium (including the Commission Services)	36

D6.7	Replicability analysis of new CEBOs	WP6	2 - ITeC	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D6.8	Final Exploitation Plan	WP6	8 - LGI	Report	Confidential, only for members of the consortium (including the Commission Services)	54
D6.9	Policy recommendations	WP6	12 - CECODHAS	Report	Public	54
D6.10	Market study (version II - final)	WP6	8 - LGI	Report	Public	24
D6.11	Solution-to-Service monitoring plan (version II)	WP6	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D6.12	Solution-to-Service monitoring plan (version III - final)	WP6	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	51
D7.1	Dissemination & Communication plan	WP7	14 - ICONS	Report	Confidential, only for members of the consortium (including the Commission Services)	43
D7.2	HOUSEFUL website	WP7	14 - ICONS	Websites, patents filling, etc.	Public	5
D7.3	Project presentation video	WP7	14 - ICONS	Websites, patents filling, etc.	Public	8
D7.4	Call-To-Action video	WP7	14 - ICONS	Websites, patents filling, etc.	Public	8

D7.5	Flyers (Communication materials for stakeholder engagement)	WP7	14 - ICONS	Report	Public	8
D7.6	Project leaflet	WP7	14 - ICONS	Report	Public	10
D7.7	Report on public communication, networking and clustering activities (version I)	WP7	14 - ICONS	Report	Public	18
D7.8	Multimedia tools: 1 PageFlow, 1 VNR and 1 web video produced and distributed	WP7	14 - ICONS	Websites, patents filling, etc.	Public	50
D7.9	Best Practices book	WP7	14 - ICONS	Websites, patents filling, etc.	Public	52
D7.10	Communication and dissemination activities impact assessment	WP7	14 - ICONS	Report	Confidential, only for members of the consortium (including the Commission Services)	54
D7.11	Report on public communication, networking and clustering activities (version II)	WP7	14 - ICONS	Report	Public	36

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D7.12	Report on public communication, networking and clustering activities (version III - final)	WP7	14 - ICONS	Report	Public	54
D8.1	H - Requirement No. 1	WP8	1 - LEITAT	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D8.2	POPD - H - Requirement No. 3	WP8	1 - LEITAT	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1

D8.3	GEN - Requirement No. 4	WP8	1 - LEITAT	Ethics	Confidential, only for members of the consortium (including the Commission Services)	3

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	1 - LEITAT
Work package title	Project Management		
Start month	1	**End month**	54

Objectives

1) To coordinate and manage the project and the efficient, smooth and timely execution of all planned pro-ject activities.
2) To revise and maintain project work plan
3) To ensure an accurate and on time communication flow among the project partners and with EC in order to trackthe project progress and meet its objectives
4) To complete the project within the agreed time schedule and budget.

Description of work and role of partners

WP1 - Project Management [Months: 1-54]
LEITAT, ITeC, CARTIF, ALCN, AHC, VISUM, AIGUASOL, LGI, GWR, Turntoo B.V., Neues Leben, CECODHAS, WE&B, ICONS, IDP, HOMEBIOGAS LTD
This work package deals with the coordination and management of all project activities and communication between partners and with the EC. The project will be coordinated by LEITAT (Project PC), with the help of a Management Committee, with a representative from each consortium partner, which will lead the management of the project, having the authority to make decisions on confidentiality and dissemination activities.

Task 1.1. Project Communication Flow (Leader: LEITAT; Participants: all) [M1-M54]
The PC will be in a continual contact with all the members of the Consortium and with the Work Packages leaders. The Management Committee meetings, defined in Management Section 3.2, will be also a communication context where administrative and financial assessment will be performed. In addition, the private area (intranet) of the project website described in the WP7 will be used for sharing management and financial related information, working documents, data, periodic reports and project deliverables which may be internally shared. Access to the project intranet will be limited to partners and the European Commission. The PC will also analyse potential and real conflicts between partners and propose agreements and make all necessary efforts to ensure a good and fluid communication process between all partners.

Task 1.2. Project Decision Structure Management (Leader: LEITAT; Participants: all) [M1-M54]
The decision structure is organised around the work package leadership. Each work package leader will be in charge of the specific tasks and deliverables for each work package. Each work package will be under the direct responsibility of the work package leader who will also draw guidelines for the work package-specific deliverables. This is expected to greatly assist the PC in the overall coordination of the project. The main decisions of the project will be taken by the Scientific Committee, which will be formed by representatives of each work package. These decisions will be implemented through regular meetings and correspondence among the consortium members, with 6 monthly updates sent to the consortium members through the PC. At each of the meetings, the consortium will discuss results of accomplished tasks, jointly develop a strategy for the next research step, and exchange general information regarding scientific work and its management. The day by day scientific progress of the project will be watched over by the Scientific & Technical Manager, who will as well review scientific reports and deliverables before being uploaded in the project's intranet. Any progress or alteration of the foreseen schedule will be reported to the Project PC, through a continuous and fluid communication, and to the Steering Committee, through half-yearly meetings to evaluate the Project's scientific development and the eventual correcting measures.

Task 1.3. Administrative and Financial Management (Leader: LEITAT; Participants: all) [M1-M54]

The PC will be responsible of the administrative coordination of the project. In the framework of this task, PC will monitor and steer the project implementation, paying special attention to the development of effective flows of communication within the Consortium and the efficient coordination of work between partners. The PC will collect all periodic activities and evaluation reports from all partners to be submitted and will inform all partners about suggestions and/or recommendations given by the Commission. The PC will also be responsible of the relations with the European Commission and of the correct distribution of the EC contribution between the partners. The PC will have to keep the Consortium informed of any information communicated by the European Commission. The PC will be responsible of the financial coordination of the project. This task will include the follow-up and delivery of all financial documents required periodically by the European Commission on time. The PC will also follow up the project expenses and track possible deviations. On the other hand, the PC will distribute the EC contribution amongst the partners according to the Grant Agreement and the Consortium Agreement.

Task 1.4. Data management plan (Leader: LEITAT; Participants: all) [M1-M54]
WP Leader will create a “Data Management Plan” (DMP) template using guidance provided by the EU. The aim of the DMP is to describe the data, generated and/or collected during the project life cycle and how it will be managed. Data refers to any kind of scientific result including statistics, results of experiments, measurements, observations resulting from fieldwork, survey results, interview recordings, images and bibliographic metadata used for the validation publications. According to EC recommendation, as part of making research data findable, accessible, interoperable and re-usable (FAIR), the DMP will include information on (1) the handling of research data during & after the end of the Project; (2) what data will be collected, processed and/or generated; (3) which methodology & standards will be applied; (4) whether data will be shared/made open Access; and (5) how data will be curated & preserved (including after the end of the project) in accordance with article 29.3 from the GA Option 1 for actions participating in the open Research Data Pilot, without changing the obligation to protect results, confidentiality, security and protection to personal data (Articles 27, 36, 37 and 39). HOUSEFUL project will always obtain informed consent from persons providing data which participation will be entirely voluntary. All activities procedures will be designed according to Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data. Within the context of the HOUSEFUL project, procedures for personal data must be processed in accordance with national and EU legislation, limiting the impact of the persons involved whilst ensuring data quality and confidentiality. Moreover, the DMP will include detailed information on the procedures for data collection, storage, protection, retention, and destruction.
WP Leader will draft a first version of the DMP, which will be delivered within the first 6 months and will be in charge to update it as data sets are produced during the project life. In order to improve and maximise open access and re-use of research data the PC, will update the DMP at a regular basis. It will take into ac-count the need to balance openness and protection of scientific information, commercialisation and Intellectual Property Rights (IPR), privacy concerns, security as well as data management. The implementation of the DMP will be coordinated and monitored by LEITAT, and the implementation will be done in coordination with LGI, since it is the partner in charge of the IPR management. The DMP will not have an impact on IPR, because the decision or whether to publish Open Access documents come after the more general decision on whether to first seek protection for intellectual property rights. Accordingly, research results can be published only after a patent application has been filled.

Participation per Partner	
Partner number and short name	**WP1 effort**
1 - LEITAT	
2 - ITeC	
3 - CARTIF	
4 - ALCN	
5 - AHC	
6 - VISUM	
7 - AIGUASOL	

8 - LGI	
9 - GWR	
10 - Turntoo B.V.	
11 - Neues Leben	
12 - CECODHAS	
13 - WE&B	
Partner number and short name	**WP1 effort**
14 - ICONS	
15 - IDP	
16 - HOMEBIOGAS LTD	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D1.1	Project Manual	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	1
D1.2	Data Management Plan (DMP)	1 - LEITAT	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	6
D1.3	First activity and management report	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	12
D1.4	Final DMP	1 - LEITAT	ORDP: Open Research Data Pilot	Confidential, only for members of the consortium (including the Commission Services)	54

D1.5	Minutes of management meetings	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	54

Description of deliverables

Deliverables: D1.1) Project Manual [M1, LEITAT] / D1.2) Data Management Plan (DMP) [M6, LEITAT] / D1.3) First activity and management report [M12, LEITAT]. / D1.4) Final DMP [M54, LEITAT] D1.5) Minutes of management meetings [M54, LEITAT]

D1.1 : Project Manual [1]

The Project Manual aims to aid project partners on their tasks and how to fulfil them. It is a dynamic reference document that will be updated as frequently as necessary. It provides details about: project organization, management structure, deliverables submission procedure, reports, intellectual property, publications and communication information. This deliverable is related to Task 1.1.

D1.2 : Data Management Plan (DMP) [6]

DMP aims at ensuring that the digital research data generated in the action will be made findable, accessible, interoperable and reusable in accordance with article 29.3 from the GA Option 1 for actions participating in the open Research Data Pilot, without changing the obligation to protect results, confidentiality, security and protection to personal data (Articles 27, 36, 37 and 39). IPR will be taken into account to ensure that making data accessible does not go against the interests of any beneficiary with regards of protection of intellectual property. HOUSEFUL project will always obtain informed consent from persons providing data which participation will be entirely voluntary. All activities procedures will be designed according to Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data. Within the context of the HOUSEFUL project, procedures for personal data must be processed in accordance with national and EU legislation, limiting the impact of the persons involved whilst ensuring data quality and confidentiality. This deliverable is related to Task 1.4.

D1.3 : First activity and management report [12]

A summary of the overall project status and specific state of progress of each WP will be prepared for each year of the project. The report will include possible deviations encountered and identified actions to achieve the project objectives. The Deliverable 1.3 will also provide an overview of the resources committed and explanations of major deviations in relation to the DoA. This deliverable is related to Task 1.3.

D1.4 : Final DMP [54]

Summary of the actions taken according to the first version of the DMP provided in D1.2 during the project execution period, evaluating its adequacy and the degree of fulfillment of foreseen objectives and procedures. Deliverable 1.4 will serve as an update of D1.2 and also will report on the degree of fulfillment of DMP within project execution. This deliverable is related to Task 1.4.

D1.5 : Minutes of management meetings [54]

Minutes of consortium meetings will be made accessible to all partners outlining the decisions taken at the meeting and the agreed plants for the next action of the project. Minutes will contain the signed list of attendees and will be submitted as D1.5 after each consortium meeting. This deliverable is related to Task 1.5.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

Work package number [9]	WP2	**Lead beneficiary [10]**	7 - AIGUASOL
Work package title	Methodology for the evaluation of circularity in the housing sector		
Start month	1	**End month**	36

Objectives

The main objective of WP2 is to develop a robust, reliable and day-to-day market usable methodology to quantify the degree of circularity of buildings, both pre- and post-refurbishment, which could be used by designers but also construction companies, promoters and financial institutions. The methodology will be developed considering the main circular economy principles related to recyclability, reusability and waste savings of the materials and active solutions defining the buildings, and also other key factors such as the feasibility of circular solutions (WP5) offered to stakeholders as services (WP6). The methodology will con-sider all the existing or brand new methodologies related to the key vectors of circularity, including LCC, LCA, S-LCA methodologies or the next future brand new EU Level(s) one. The methodology will result in an easy-to-use tool as HOUSEFUL Solution: S1, which will be the basis of SaaS developed in WP6.

Specific Objectives:

- To define the main weighting factors and KPIs representatives of the recyclability and reusability of materials, of theenergy and water life cycle, and of the social and economic performance;
- To define a set of rules and procedures from the existing bibliography and methodologies, and combine them in acomplete and comprehensive circularity calculation methodology;
- To establish some recommended and predefined values, both for representative parameters and also for required inputs,from residential building archetypes representing different European locations;
- To refine the methodology for the evaluation of circularity using the results obtained in the real demo scenarios andthe other WPs, establishing a high degree of reliability.

Description of work and role of partners

WP2 - Methodology for the evaluation of circularity in the housing sector [Months: 1-36]
AIGUASOL, LEITAT, ITeC, CARTIF, ALCN, AHC, VISUM, GWR, Turntoo B.V., HOMEBIOGAS LTD
Task 2.1. End users requirements considering circular economy business opportunities (Leader: LEITAT; Participants: AIGUASOL, ITeC, AHC, VISUM, ALCN, GWR) [M1-M6]
Task 2.1 aims to define the needs and requirements that the methodology for evaluating the circularity of buildings has to meet in order to provide the maximum value to the end-users. A valuable-by-design service will be therefore developed following a structured approached for defining customer needs and translating them into specific designs, according with the Quality Function Deployment (QFD) methodology. The “voice of the customer” will be collected via different channels such as: direct discussion or interviews, surveys, focus groups, customer specifications, observation, warranty data, field reports, etc. The output will be then elaborated in a development planning matrix called “house of quality”. This matrix will be used to con-vert high level needs into lower level features designed to satisfy these needs. VISUM will lead the implementation of the QFD methodology, providing templates, questionnaires and building the house of quality, while the other partners involved in the task will get in contact with the potential end-user and stakeholder in order to gather the information needed. AHC will provide input and feed-back from the end user perspective with the aim of providing a useful tool for management of their property and housing programmes. ALCN will provide detailed input from the end user perspective as the office of ALCN is situated in the demo building “Baumgarten”. ALCN will support the survey of end user requirements concerning the demo building “Donaufelderstrasse” by organising communication with property developers (NEUES LEBEN) and dwellers. GWR will give input on building requirements and user acceptance and demand from everyday business. AIGUASOL will provide from end user requirements from other re-lated projects (see Plug&Harvest or 4RinEU) and will adapt them to the HOUSEFUL frame, considering the expected new developments and SaaS services.

Task 2.2. Establishment of residential building archetypes for the representative European geoclusters (Leader: AIGUASOL; Participants: CARTIF) [M2-M12]

The aim of Task 2.2 is to establish the main circular economy conditions of building archetypes for the representative European geoclusters. The so-called circular economy main conditions will include the establishment of the main environmental aspects for each case (energy, water, materials and waste) that will be cross-referenced with the respective geoclusters, to obtain a set of representative circularity scenarios to be used in the following tasks of WP2, but also considered as a reference for other purposes related to the business model opportunities (WP6). Originating from at least 6 defined geoclusters (including climatic, economic and social parameters), which aim to be representative of different latitude realities for the housing sector, a selection of building archetypes will be established by AIGUASOL and CARTIF. These building archetypes will consider existing information of reference European projects for Single Family Houses and Multi Family Houses (i.e. Tabula, Entranze or 4RinEU). Initial requested information on archetypes will be completed with other relevant data from official European (Eurostat) and national information sources related to building national regulations, socioeconomic conditions and local user patterns or other relevant references (PlugN-Harvest). From the collected and derived information, a first draft version of a structured repository (geodatabase CARTIF) will be implemented and populated ready to be used in the following tasks.

Task 2.3. Definition of the focussed calculation procedures and relevant KPIs for the different circular economy composing vectors (Leader: ITeC; Participants: AIGUASOL, TURNTOO, CARTIF, ALCN, LEITAT) [M5-M18]
The aim of this task is to properly and reliably establish reference calculation methodologies for the different circular aspects, and define its main KPIs. To ensure replicability a set of common core indicators is needed for making environmental, social and economic improvements to be used in decision-making through the entire life-cycle of buildings. To support this development, the task will be aligned to:

- The results from the working groups involved in the common EU framework of indicators for the environmentalperformance of buildings. (i.e. Communication 'Resource Efficiency Opportunities in the Building Sector' COM (2014)
445)
- The LCA methodology as defined in ISO 14040/44 norms and existing software for LCA; The LCC methodology asdefined in ISO 15686-5 and other relevant documents The standards developed/under de-velopment by CEN/TC 350 - Sustainability of construction works (e.g. EN 15840; EN 15643; EN 15978; EN 16309; EN 16627) , based on LCA and LCC methodology for construction products and buildings.
- Product Environmental Footprint (PEF); Leesman and verification tools as KPMG true value; Cost op-timalmethodology as established in EPBD recast 2010;
- Level(s) new buildings sustainable methodology developed by the European Commission.

As a minimum, the following issues will be tackled: energy, water and material consumption; durability of construction products; design for deconstruction; CDW; recycled content in construction materials; recycla-bility and reusability of construction materials and products. ITeC and TURNTOO will focus on defining environmental and economic KPIs related to materials and CDW, whereas LEITAT and CARTIF will define the KPIs focused on water, waste and energy aspects such as life cycle Global Warming Potential (kg CO2 eq), Ozone Depletion Potential (kg CFC eq),water resource depletion (m3 water eq), Water Exploitation in-dex (WEI+), cumulative energy demand (MJ), cost by life cycle stage (€/ m2/yr), etc. ALCN will contribute to define KPIs by researching local framework conditions (local standards, rules and legislation), contacting local and global initiatives concerning circularity, and conducting interviews with experts from ALCN´s network. AIGUASOL will contribute to consider, adapt and define the specific energy calculation procedures and KPIs', and proposing the matching of these with other indicators and procedures related to the other above mentioned environmental aspects (water and material consumptions, waste, etc.). In task 3.1 the stakeholders will be engaged, it will be at this juncture at the project start that stakeholders will be requested to give inputs into other potential indicators that can be used to measure the social impact (task 5.4) and to give their feedback on the current indicators.

Task 2.4. Integration of the calculation procedures in an overall circular economy methodology (Leader: AIGUASOL; Participants: CARTIF, ITeC) [M14-M26]
From the results previously obtained in Task 2.2 (reference data for the building archetypes on the representative geoclusters), and Task 2.3 (main calculation procedures and KPIs for the different vectors composing the circularity of buildings), in Task 2.4 an overall methodology to determine the degree of circularity for residential buildings will be defined. This would be a multi-dimensional model of quantitative as well as qualitative parameters, ready to be used at the initial or decision stages of new or retrofitted building definition. The aim of the Building Circularity Methodology is to establish a clear approach to a calculation method that will offer valuable indicators of the degree of potential circularity business opportunities of a building. By that the BCM will establish a global calculation procedure able to determine the potential of reusability and recyclability of the materials, physical solutions, water and waste, and the energy life cycle, under a common approach and considering potential interrelations. As the first approach to the BCM, the methodology will be implemented by AIGUASOL and CARTIF in some theoretical scenarios defined in Task 2.2 to obtain a first database of results. This database will be used in a mathematical

sensitive analysis to be developed by AIGUASOL and CARTIF to establish the weighting factors of the main KPIs aiming to simplify the BCM as much as possible.

Task 2.5. Refinement of the BCM methodology and establishment of the reliability factors on the results [HOUSEFUL Solution: S1] (Leader: AIGUASOL; Participants: CARTIF, AHC, ALCN) [M26-M36] The aim of Task 2.5 is to refine the BCM preliminary version defined in Task 2.4, in a loop process with other direct tasks related to the real demo cases (WP4 by CARTIF), environmental and economic assessment (WP5) and business models (WP6), but also considering oriented feedbacks coming from the follower buildings and benchmarking with other existing or on-going EU projects on the same scope, as well as stakeholder inputs. AIGUASOL will be in charge to define an easy to use template to fill in all the required refinement data to be provided by other partners or third parties. AIGUASOL and CARTIF will also be in charge, jointly with the demo owners and the demo implementation WP leader, to collect these data and, by using the results obtained in Task 2.5, compare it with the first draft version. Homeowners/responsible partners of the frontrunner buildings (AHC, and ALCN) will participate in relation to their demo building, providing data and input from the building and feedback on the usefulness and usability of the methodology. From there, a final version of the reliable factors will be introduced and the overall methodology will be adapted by AI-GUASOL and CARTIF.

Task 2.6. Existing market circularity: solutions and services (Leader: AIGUASOL; Participants: LEITAT, CARTIF) [M1-M26] Task 2.6 includes a desk research on existing market circularity solutions in order to establish a comprehensive framework of the European current situation. This research will also facilitate the potential engagement of companies that supply these solutions and services, supporting Task 3.1 (identification and analysis of stakeholders). The successful circular solutions and services identified will be included in the SaaS developed in WP6 to promote replication in housing sector.

Participation per Partner

Partner number and short name	**WP2 effort**
1 - LEITAT	
2 - ITeC	
3 - CARTIF	
4 - ALCN	
5 - AHC	
6 - VISUM	
7 - AIGUASOL	
9 - GWR	
10 - Turntoo B.V.	
16 - HOMEBIOGAS LTD	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]

D2.1	Report on end-users requirements and CEBO on residential buildings	7 - AIGUASOL	Report	Public	6
D2.2	Circular economy repository for building archetypes in the geoclusters	7 - AIGUASOL	Report	Confidential, only for members of the consortium (including the Commission Services)	12

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D2.3	Reference methodologies and KPIs in circular economy analysis	7 - AIGUASOL	Report	Public	18
D2.4	Existing market circularity solutions and product	7 - AIGUASOL	Report	Public	26
D2.5	First approach of BCM methodology and implementation in theoretical scenarios	7 - AIGUASOL	Report	Public	26
D2.6	Description of the BCM methodology and implementation rules in real scenarios	7 - AIGUASOL	Report	Public	36

Description of deliverables

Deliverables: D2.1) Report on end-users requirements and CEBO on residential buildings [M6, AI-GUASOL] / D2.2) Circular economy repository for building archetypes in the geoclusters [M12, AI-GUASOL] / D2.3) Reference methodologies and KPIs in circular economy analysis [M18, AIGUASOL] / D2.4) Existing market circularity solutions and products [M26, AIGUASOL] / D2.5) First approach of BCM methodology and implementation in theoretical scenarios [M26, AIGUASOL] / D2.6) Description of the BCM methodology and implementation rules in real scenarios [M36, AIGUASOL]

D2.1 : Report on end-users requirements and CEBO on residential buildings [6]

Information gathering regarding end-users needs related to the CEBO on residential buildings will be done. This information will be used to define a valuable-by-design service through implementing a Quality Function Deployment methodology, which will be the basis for the development of the rest of the work in WP2. This deliverable is related to Task 2.1.

D2.2 : Circular economy repository for building archetypes in the geoclusters [12]

Definition of the main circular economy conditions of building archetypes for representative European geoclusters by cross-referencing main environmental aspects (energy, water, materials and waste) against at least six of the most representative geoclusters (including climatic, economic and social parameters). This deliverable is related to Task 2.2.

D2.3 : Reference methodologies and KPIs in circular economy analysis [18]

Establishment of the most relevant calculation methodologies and KPIs to be considered in the circular economy analysis to be deployed during the project. The technically pre-selected KPIs will be then considered when engaging stakeholders in task 3.1., as well as for the circular economy baseline evaluation (D4.3) and for the validation of HOUSEFUL solutions at building scale (D4.6). This deliverable is related to Task 2.3.

D2.4 : Existing market circularity solutions and product [26]

Market analysis and prospection looking for available circularity solutions and products, establishing a comprehensive European framework, providing key data and descriptions of each of them, as well as giving support to task 3.1 regarding stakeholders identification and analysis. A coordination between this deliverable and D6.1 economic information regarding market analysis will be promoted. This deliverable is related to Task 2.6.

D2.5 : First approach of BCM methodology and implementation in theoretical scenarios [26]

The BCM will establish a global calculation procedure able to determine the potential of reusability and recyclability of the materials, physical solutions, water and waste, and the energy life cycle, under a common approach and considering potential interrelations. As the first approach to the BCM, the methodology will be implemented in some theoretical scenarios defined in Task 2.2 to obtain a first database of results. This database will be used in a mathematical sensitive analysis to establish the weighting factors of the main KPIs aiming to simplify the BCM as much as possible. This deliverable is related to Task 2.4.

D2.6 : Description of the BCM methodology and implementation rules in real scenarios [36]

Detailed explanation on BCM methodology developed during the project for identifying the degree of potential circularity business opportunities of a building, also considering real scenarios validation, will be developed. This BCM methodology will be fed by all the previous tasks included in WP2. This deliverable is related to Task 2.4 and Task 2.5.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS1	Methodology for evaluating the circularity level of buildings	7 - AIGUASOL	26	D2.5) The methodology defined in WP2 will be validated using real scenarios (all Frontrunner buildings) and integrated in the SaaS developed in WP6.
MS2	Desk research of existing circular solutions & services	7 - AIGUASOL	36	D2.4) Existing market circular solutions

Work package number [9]	WP3	**Lead beneficiary** [10]	13 - WE&B
Work package title	Social engagement: starting the co-creation of HOUSEFUL services		
Start month	1	**End month**	54

Objectives

WP3 has two objectives: (1) To analyse and map out the stakeholders' belief structures underlying their behaviour, attitudes, and societal concerns in order to address knowledge requirements for the demonstration and replication of the 11 HOUSEFUL solutions exploited as integrated services; 2) To effectively engage local and regional stakeholders from Frontrunner buildings in the co-creation of the proposed 11 HOUSEFUL Services according to their needs and behavioural patterns, making feasible and realistic the co-creation of new circular economy business opportunities (CEBOs) from developed solutions. The methodology used and developed in this WP will be the basis for the demonstration of the HOUSEFUL Solution S2.

Specific Objectives:

- Identifying and mapping the key stakeholders and their relationships in terms of knowledge, influence and interestin the Frontrunner buildings;
- Designing and implementing an effective engagement strategy for the successful co-creation and final definition ofservices offered to stakeholders in each Frontrunner building in order to increase the social awareness and engagement of services co-created for covering their needs.
- Analysing the specific factors influencing behaviour and the choice making structures of people and the potential socialrisks and the barriers affecting the successful demonstration of the proposed HOUSEFUL solutions as new services (identification of the individual and collective needs, definition of common goals and best consensus scenario);
- Implementing an innovative engagement strategy for the successful co-creation and final definition of services offeredto stakeholders in each Frontrunner building in order to increase social awareness and engagement of services co-created for covering their needs.

Description of work and role of partners

WP3 - Social engagement: starting the co-creation of HOUSEFUL services [Months: 1-54]

WE&B, LEITAT, ITeC, CARTIF, ALCN, AHC, AIGUASOL, GWR, Turntoo B.V., Neues Leben, CECODHAS, ICONS

Task 3.1. Identification and analysis of stakeholders for the co-creation of CEBOs (Leader: WE&B; Participants: TURNTOO, LEITAT, ALCN, AHC, CARTIF, AIGUASOL, ICONS, GWR, NEUES LEBEN) [M1-M26]

The co-creation and demonstration of 11 HOUSEFUL services in selected 4 Frontrunner buildings requires the engagement of different actors forming part of the current linear value chains in the housing sector. The core HOUSEFUL project team (WE&B, TURNTOO, LEITAT, ALCN, AHC, NEUES LEBEN, GWR, CAR-TIF, AIGUASOL and ICONS) together with the Frontrunner buildings will undertake a stakeholder identification, mapping and analysis of the social values and beliefs of the circularity building models as well as the needs, concerns and expectations of HOUSEFUL services. Some stakeholders will influence the implementation of solutions and services in Frontrunner buildings, others have interest in the project concerning key players at each demo site and identified stakeholders to be engaged. A total of 32 national and international stakeholders have already been identified and have sent a LOI showing their interest in the project and or participating in the CCHE board – these initial stakeholders will aid in snowballing the stakeholders' identification in the context of the frontrunner building and in the international community levels. Local and regional key stakeholders will be identified and mapped through a social network analysis (SNA) method. In combination, a deeper social analysis will be approached by qualitative methods through a series of sectorial interviews and/or focus groups to be undertaken at each of the Frontrunner buildings. This will serve to start identifying insights of stakeholder's behavioural decisions for building circularity (including perceived risks of technologies, materials, etc.) and to answer the question of what drives stakeholders' decision to use CEBO services (attitudes, beliefs and concerns). This will aim to further identify the potential social barriers and enablers to implement the CEBO services. These activities will require an initiation of awareness activities regarding the project that will be coordinated with the communication activities in WP7. Therefore, AHC and NEUES LEBEN will be directly involved in this task to create communication materials for the qualitative analysis implying an initial engagement. With this approach, more qualitative information will be gathered on what the stakeholders could provide to HOUSEFUL, how to effectively engage with them and to better identify their communication channels, etc. The main outputs from this task will be recommendations and inputs for the final co-creation and replication that will result in an engagement strategy for each building site including the initial list of stakeholders that will be engaged (in task 3.2).

Task 3.2. Engagement and co-creation activities. [HOUSEFUL Solution: S2] (Leader: WE&B; Partici-pants: ICONS, LEITAT, CECODHAS, AHC, NEUES LEBEN, ALCN, AIGUASOL, ITEC) [M1-M54]

Following the stakeholder analysis defined in Task 3.1, the analysis of social and cultural barriers and enablers from the previous task will follow a process of engagement using social qualitative methods with the identified key stakeholders at each HOUSEFUL Frontrunner buildings. This task, lead by WE&B, will also deepen in the analysis of factors underlying the stakeholders' behaviour, values and belief, and attitudes and perceptions towards the use of these buildings. Outputs will be used to define the detailed technical specifications of circular solutions to be demonstrated in WP4.

Subtask 3.2.1 Co-creation workshops: developing a strategic plan for the HOUSEFUL solutions and services (Leader: WE&B) [M1-M54]

The engagement activities will follow a backcasting process with the selected stakeholders, due to its problem solving character. . This process will implement three types of co-creation workshops to start the engagement process in each one of the regions where Frontrunner buildings are located. WE&B will support the leaders of the frontrunners buildings who will be the partners responsible to organise these workshops. The partners involved in this task will support end-users to better understand the implications of incorporating circular solutions in their day-to-day life. The first workshop will be structured on the basis of the identified factors conditioning social behaviour and acceptance. This workshop will aim at identifying and validating barriers and enablers obtained in Task 3.1 but in a participatory approach, and co-creating the future scenarios of the social conditions of the CEBOs that stakeholders would desire. The second workshop will be designed based on the results from the first workshop to generate ideas and knowledge for co-creation of communication materials (messages, identification of resources and tools, contents) regarding the services provided for the front-runner buildings. Stakeholders, especially tenants, can ensure that their concerns are addressed and knowledge is generated by co-creating together the contents of future communication of the services.The output of this activity will be the basis for the co-creation of social validated HOUSEFUL services and also further consideration towards behavioural changes after effective engagement (indicators will be framed in the social assessment from WP5). Close to the end of the project, a third workshop will be undertaken with regards to the provision of the feedback and validation of the final services demonstrated in the frontrunner buildings as well as to present the SaaS created in WP6 and to co-create a roadmap for further uptake beyond the project. In parallel to this workshop, an Open House event will be organised as a more technical approach with a professional audience and not only with the concerned community of the frontrunners buildings. The Open House event will be organised in each Frontrunner building to enable a wider community of citizens and stakeholders (not directly involved in the demonstration) to see the buildings' implementations and to share the benefits of the different implemented services. Lead by WE&B, the partners AHC, ALCN and NEUES LEBEN will be involved in this task in relation to the identification and validation of the challenges and opportunities presented by the social housing aspect of the project and the socio-cultural-economic profile of the new tenants and dwellers of the buildings. Collaboration with the local social and/or voluntary social services will be promoted. The output of this activity will be the basis for the co-creation of all HOUSEFUL services (Month 30).

Participation per Partner	
Partner number and short name	**WP3 effort**
1 - LEITAT	
2 - ITeC	
3 - CARTIF	
4 - ALCN	
5 - AHC	
7 - AIGUASOL	
9 - GWR	
10 - Turntoo B.V.	
11 - Neues Leben	

12 - CECODHAS	
Partner number and short name	**WP3 effort**
13 - WE&B	
14 - ICONS	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D3.1	Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version I)	13 - WE&B	Report	Public	18
D3.2	Social perceptions and beliefs in circular economy at building scale (version I)	13 - WE&B	Report	Public	18
D3.3	Social conditions for the co-creation of all HOUSEFUL services and co-created materials	13 - WE&B	Report	Public	36
D3.4	Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version II - final)	13 - WE&B	Report	Public	26
D3.5	Social perceptions and beliefs in circular economy at building scale (version II - final)	13 - WE&B	Report	Public	26

Description of deliverables

Deliverables: D3.1) Social engagement strategy for the co-creation of HOUSEFUL solutions as new services [M18, WE&B including results from Demo 2, Demo 3, Demo 4] (update in M26 including results from Demo 1 - D3.4) / D3.2) Initial social decisions and behaviour in circular economy at building scale [M18, WE&B, including results from Demo 2, Demo 3, Demo 4] (update in M26 including results from Demo 1 - D3.5) / D3.3) Social conditions for the co-creation of all HOUSEFUL services and co-created materials [M36, WE&B] / D3.4) Updated and final version of D3.1 [M26, WE&B] / D3.5) Updated and final version of D3.2 [M26, WE&B]

D3.1 : Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version I) [18]

A detailed review on the stakeholders' identification and analysis process for the different CEBOs and DEMO locations will be done. Definition of the social engagement strategy that will assure proper co-creation actions during the project, establishing a clear roadmap for Tasks 3.2, as well as defining the basis for the demonstration of S2 solution. This deliverable is related to Task 3.1. To be updated in M26 including results from Demo 1 (D3.4)

D3.2 : Social perceptions and beliefs in circular economy at building scale (version I) [18]

A report regarding initial social decisions derived from the needs, concerns and expectations of the previously identified stakeholders (Task 3.1) involved in the co-creation process will be presented. This report will be used as an initial comparison reference to be consulted in subsequent stages of HOUSEFUL project, leading to the final demonstration of S2 solution. This deliverable is related to Task 3.2. To be ppdated in M26 including results from Demo 1 (D3.5)

D3.3 : Social conditions for the co-creation of all HOUSEFUL services and co-created materials [36]

A detailed review of the social and cultural restrains, synergies and drawbacks found during the social engagement process for the co-creation of CEBOs, evaluating the results obtained through the social network analysis and the qualitative methods applied, defining an adequate strategy for demonstrating S2 solution. This deliverable is related to Task 3.2.

D3.4 : Social engagement strategy for the co-creation of HOUSEFUL solutions as new services (version II - final) [26]

Updated and final version of D3.1.

D3.5 : Social perceptions and beliefs in circular economy at building scale (version II - final) [26]
Updated and final version of D3.2.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS3	Social engagement plan	13 - WE&B	26	Stakeholder analysed process finished for all buildings, related to D3.1) Social engagement strategy.
MS4	Co-created services	13 - WE&B	36	Engagement workshop organised, related to D3.3.

Work package number [9]	WP4	**Lead beneficiary [10]**	3 - CARTIF
Work package title	Large scale demo buildings		
Start month	3	**End month**	48

Objectives

This WP aims at the successful demonstration of the circular economy integrated solutions in the four Front-runner buildings (two buildings located in Spain and two buildings in Austria). In detail, this WP aims to demonstrate the technical feasibility of the technical solutions behind the integrated HOUSEFUL service from material (S3, S4), water (S5, S6), waste (S7, S8, S9) and energy (S10, S11) vectors.

Specific objectives:

- To develop Material Passports of 4 Frontrunner buildings including a detailed inventory of the physical state of eachbuilding and the elements and materials it consists of by using a 3D model (BIM). Demonstration of S5.
- To ensure BIM interoperability along the WP providing all material, element, and system specifications based on thestandard IFC4 (ISO 16739:2013 - Industry Foundation Classes (IFC) for data sharing in the construction and facility management industries).
- To develop the technical specifications of the technological solutions: S6 and S7 for water vector; S8 and S9 for wastevector and S10 and S11 for energy vector.
- To search additional financial resources for proper dimensioning of TS and replication of proposed solutions asservices.
- Selection of companies for renovation works for Frontrunner buildings owned by public entities.
- To manage the required permit for the initiation of demonstration activities in Frontrunner buildings.
- To validate the feasibility of proposed solutions at lab scale before scaling up and demonstration at large scale.

WP4 - Large scale demo buildings [Months: 3-48]
CARTIF, LEITAT, ITeC, ALCN, AHC, VISUM, AIGUASOL, GWR, Turntoo B.V., Neues Leben, CECODHAS, IDP, HOMEBIOGAS LTD
The demonstration character of this WP4, indeed, will start in WP3, when engagement of stakeholders is initiated and social and cultural barriers are identified and tackled using activities planned in Task 3.1 and 3.2. The development of the technical project in this WP includes the demonstration of technological solutions in Frontrunner buildings following a simple process: detailed technical specification of all solutions with technological character behind the Integrated Systemic HOUSEFUL Service (Material (S3, S4), Water (S5, S6), Waste (S7, S8, S9) and Energy (S10, S11)). In parallel, permits for technological implementation of proposed solutions in Demo buildings will be sought for approval. From a technical point of view, a strong supervision of the works and commissioning of systems to be installed in Frontrunner Buildings will be carried out for technologies implemented from S5, S6, S7, S8, S10 and S11.

Task 4.1 Circular economy audit: Baseline calculation (Leader: CARTIF; Participants: ITEC, TURNTOO, NEUES LEBEN, ALCN, GWR, AHC, LEITAT) [M1-M20]
In this task is provided an overall and exhaustive circular economy audit of the initial status of the four Frontrunner buildings in all four vectors: water, waste, energy and materials and in four dimensions: techno-logical (LEITAT, CARTIF), social (ALCN, AHC), environmental (LEITAT), economic (ITeC, TURNTOO). A specific subtask will address materials deeply using Material Passports.
The aim is to facilitate, in the next steps, the technical definition, specifications (Task 4.2) and optimisation (Task 4.3) of the HOUSEFUL solutions and the monitoring programme (Task4.4) of each demo-case. The audit will take into account the initial set of KPI's and the calculation methods defined in WP2. A specific subtask will address materials deeply using Material Passports. The four Frontrunner buildings will be mod-elled with BIM and be analysed using Material Passport. Each constructive element of this model will incor-porate specific KPIs to analyse the circularity of buildings and facilitate proposals for improvements focused on the building circularity. Details are shown below.

Subtask 4.1.1. Material Passports integrating BIM for baseline definition and potential evaluation of deconstruction scenarios. [HOUSEFUL Solution: S3] (Leader: ITEC; Participants: TURNTOO, CARTIF, NEUES LEBEN, ALCN, GWR, AHC) [M1-M20]

The materials used in all four Frontrunner buildings will be recorded according to the planning schemes. The material passport will be created, as well as a full possible deconstruction plan for after life application. Before beginning the refurbishment of frontrunner buildings, hazardous building materials (e.g. asbestos, lead, mercury, polychlorinated biphenyls (PCB), chlorofluorocarbons, etc) will be identified with the aim to know which items need to be to safely removed, segregated and disposed. The materials located in this building will again be calculated through the BIM according to the plans of the building. The material pass-port will be compiled together with deconstruction scenarios and possible implementation of other innovative service based business models of HOUSEFUL. Each constructive element of this model will incorporate KPIs to analyse the circularity of buildings and facilitate proposals for improvements.

ITEC will carry out the BIM model of the 4 Frontrunner buildings, whereas TURNTOO will be in charge of producing a Materials Passports which is a digital document that records the identity of all construction materials used, documenting the products and raw materials used in a building. It also contains information about the quality of the materials and their location. This makes reusing and reclaiming materials during demolition or disassembly much simpler. This information will be specified under the standard IFC to assure the interoperability to include such information in BIM.

Task 4.2 Technical specification of HOUSEFUL solutions (Leader: CARTIF; Participants: LEITAT, ALCN, GWR, TURNTOO, AIGUASOL, ITEC, IDP, HOMEBIOGAS) [M1-M20].

Subtask 4.2.1. Technical specifications for solutions including technological implementation. [HOUSEFUL Solutions: S5, S6, S7, S9, S10 and S11] (Leader: CARTIF; Participants: LEITAT, ALCN, GWR, AIGUASOL, ITEC, IDP, HOMEBIOGAS) [M1-M20].

The main goal of this task is the definition of the technical specification (TS) of each HOUSEFUL solution including the technological implementation in each Frontrunner building considering the principles of the circular economy. KPIs defined in WP2 will be applied in the detailed specifications to cross-check the strategies for implementation. Each solution will be designed by specific partners with expertise on it, supervised by CARTIF in order to ensure that the circularity of the overall solution remains as a valuable Circular Economy Business Model. Each solution will be specified under the standard IFC to assure the interoperability to include such solutions in the BIM models.

Subtask 4.2.2. Technical specifications for solutions not including technological implementation. [HOUSEFUL Solutions: S4, S9] (Leader: CARTIF; Participants: ITEC, TURNTOO, AHC, GWR, NEUES LEBEN) [M1-M20]. This Subtask defines the technical specifications required for the proper demonstration of HOUSEFUL solu-tions without including a technological implementation (S4 and S9). In detail for S4, the type, number/quantity, origin an functionalities, among other main characteristics for those building materials available for renovation/refurbishment at local scale will be defined. Regarding S9, the results obtained in previous subtask 4.1.1 (material passport and BIM) will be used as basis information for the development of Design for Disassembly (DfD) evaluation of Frontrunner buildings. Technical specifications regarding constructive systems and materials to allow highly percentage of reusability of these components in other buildings or materials will be defined.

Task 4.3. Optimisation of technological solutions before final implementation in Frontrunner buildings (Leader: LEITAT, Participants, ALCN, IDP, HOMEBIOGAS, CARTIF, ITEC, AHC, NEUES LEBEN, AI-GUASOL) [M5-M24] Since specific solutions in Water (S5, S6), Waste (S7, S8) integrate innovative technologies for the treatment ofhousehold effluents and BWs, validation tests will be conducted firstly at lab scale (TRL 6) in order to adjust and optimise the proper configuration and operation of proposed systems, and then deployed at full scale in Frontrunner buildings (TRL7-8) in Task 4.8. In detail, the following Subtasks are planned to be executed:

Subtask 4.3.1. Optimisation of Water vector

S5 components (Leader: ALCN, Participants: IDP; LEITAT) The main goal of this subtask is the optimisation of NBS (as green facade or green roof) and O3/UV technologies for greywater and rainwater treatment for reuse in toilets in Frontrunner buildings 1 and 2. First approach includes the optimisation of NBS at laboratory (TRL5) before scale up (at TRL 7) and integration by ALCN (with the support of IDP in Task 4.8). First tests will include the selection of plants and analysis of their performance according to (Spanish and Austrian) climate conditions on the treatment of selected WW by analysing parameters such as COD, BOD, total Nitrogen, Suspended Solids, among other parameters. ALCN will provide calculations and experimental implementations concerning the size of treatment systems (i.e. m2 per litre influent), retention times, dilution factors and plant consortia. Once the NBS sys-tems are optimised, they will be scaled up according to water (greywater and rainwater) treatment needs in Frontrunner building 1 ("Mestres" - Sabadell) and 2 ("Gripia" - Terrassa). Potential water looses by evapotranspiration will be considered. The treated and collected water will be used for toilet flushing and irriga-tion. Then, first samples obtained by ALCN at laboratory scale will be treated by LEITAT at lab scale (TRL6) by using O3/UV technologies. Optimisation tests will include the analysis of O3 dosage, hygienisation time and power for hygienisation as well as microbiological tests such as E. coli or salmonella content. Once NBS are installed in Frontrunner 1 and 2 (Task 4.8), LEITAT will validate the optimal operating conditions of commercial O3/UV

technologies (TRL9) using the same equipments and methods used at laboratory scale. Results will be shared with IDP in order to select and adjust the proper operation of O3/UV commercial systems installed in Task 4.8.

LIST OF TECHNICAL SPECIFICATIONS

VECTOR: WATER

HOUSEFUL Solution: S5. Efficient treatment and reuse of rainwater and greywater streams using green and low-energy consuming solutions

TS 5.1.1: Greywater and blackwater separation (IDP) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.1.2: Greywater and blackwater separation (IDP) - DEMO 2: Spain "La Gripia" – 2 homes.

TS 5.2.1: Modifying greywater discharge system for treatment and reuse using innovative NBS (ALCN) and commercial O3/UV hygienisation (IDP) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.2.2: Modifying greywater discharge system for treatment and reuse using innovative NBS (ALCN) and commercial O3/UV hygienisation (IDP) - DEMO 2: Spain "La Gripia" – 2 homes.

TS 5.3.1: Integrating water meters at household and building level: controlling water reuse (IDP) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.3.2: Integrating water meters at household and building level: controlling water reuse (IDP) - DEMO 2: Spain "La Gripia" – 2 homes.

TS 5.4.1: Rainwater pre-treatment for reuse using NBS on facade or roof (ALCN) and integration in building (IDP, ALCN) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 5.4.2: Rainwater pre-treatment for reuse using NBS on facade or roof (ALCN) and integration in building (IDP, ALCN) - DEMO 2: Spain "La Gripia" – 2 homes.
TS 5.4.3: Rainwater pre-treatment for reuse using NBS on facade or roof (ALCN) and integration in building (IDP, ALCN) - DEMO 3: "1900 build" – 1 home.

HOUSEFUL Solution: S6. Efficient treatment and reuse of rainwater and greywater streams using green and low-energy consuming solutions

TS 6.1: Separation of liquid/solid fraction of wastewater and organic content. Treatment of liquid fraction with NBS indoor, at facade and outdoor - DEMO 3: "1900 build" – 1 home.

VECTOR: WASTE

HOUSEFUL Solution: S7. Efficient treatment and valorisation of BW and blackwater for biogas production

TS 7.1: Design and integration of AnMBR system (IDP, LEITAT) - DEMO 1: Spain "Els Mestres" – 8 homes.

TS7.2: Design and Integration of dAD system (HOMEBIOGAS, ALCN) - DEMO 3: "1900 build" – 1 home.

TS 7.3.1: Biogas cleaning for renewable heat production (CARTIF) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 7.3.2: Biogas cleaning for renewable heat production (CARTIF) - DEMO 3: "1900 build" – 1 home.

TS 7.4.1: Storing cleaned biogas in a safety and compact system. Final valorisation according to co-creation (CARTIF, IDP, ALCN, LEITAT) – DEMO 1: Spain "Els Mestres" – 8 homes.
TS7.4.2: Storing cleaned biogas in a safety and compact system. Final valorisation according to co-creation (CARTIF, IDP, ALCN, LEITAT) - DEMO 3: "1900 build" – 1 home.

HOUSEFUL Solution: S8. High quality fertilizer/compost of local origin

TS 8.1: Stabilisation of digestate for final use in local gardening (public parks, training activities in schools and/or neighbourhood). Specifications according to national regulation on compost use (LEITAT, ALCN) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS 8.2: Stabilisation of digestate for final use in local gardening (public parks, training activities in schools and/or neighbourhood). Specifications according to national regulation on compost use (LEITAT, ALCN) - DEMO 3: "1900 build" – 1 home.

VECTOR: ENERGY

HOUSEFUL Solution: S10. Energy efficient facades and roofs using modular and sustainable materials

TS10.1: Substituting asbestos roof (ITEC, AHC) - DEMO 2: Spain "La Gripia" – 41 homes.

TS10.2: Design of an energy efficient roof using sustainable materials with advanced functionality according to building/ community(dwellers) needs (ITEC, AIGUASOL, AHC) - DEMO 2: Spain "La Gripia" – 41 homes.

TS10.3.1: Design and improvement of energy-efficient facades using modular and/or more sustainable materials selected under circular criteria (ITEC,AIGUASOL, GWR, ALCN, AHC) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS10.3.2: Design and improvement of energy-efficient facades using modular and/or more sustainable materials selected under circular criteria (ITEC,AIGUASOL, GWR, ALCN, AHC) - DEMO 2: Spain "La Gripia" – 2 homes.
TS10.3.3: Design and improvement of energy-efficient facades using modular and/or more sustainable materials selected under circular criteria (ITEC,AIGUASOL, GWR, ALCN, AHC) - DEMO 3: "1900 build" – 1 home.

HOUSEFUL Solution: S11. Guarantee energy savings/ production in the building

TS11.1: Co-creation based on performance rather than ownerwhip for energy systems (i.e. ST, PV) finally installed in each building (Pay per performance business model). Monitoring of energy saving by active and passive measures from material/energy valorisation points of view (AIGUASOL) - DEMO 1: Spain "Els Mestres" – 8 homes.
TS11.2: Co-creation based on performance rather than ownerwhip for energy systems (i.e. ST, PV) finally installed in each building (Pay per performance business model). Monitoring of energy saving by active and passive measures from material/energy valorisation points of view (AIGUASOL) - - DEMO 3: "1900 build" – 1 home.

S6 components (Leader: ALCN)
The main goal is the optimisation of NBS indoor (TRL 6), at facade (TRL 5) and outdoor (TRL 6) of liquid fraction of WW with high organic content for water reuse. ALCN will provide calculations and experimental implementations concerning the size of treatment systems (m2 per litre influent), retention times, dilution factors and plant consortia adjusted to different climates. Results will allow the proper installation for bigger prototypes, finally integrated into real environment (TRL 7) and possibly serving the building with water cycles.

Subtask 4.3.2 Optimisation of Waste vector
S7 components (Leader: LEITAT, Participants: CARTIF, IDP, ALCN, HOMEBIOGAS)
The main goal is (1) to optimise the operation of the AnMBR system at laboratory scale for the treatment of blackwater and grinded BWs (potentially installed in Frontrunner building 2), and (2) the optimisation of dry anaerobic digestion (dAD) provided by HOMEBIOGAS for the optimal treatment of the solid fraction from non-segregated WW mixed with grinded BW (to installed in Frontrunner building 3). For both cases, biogas cleaning technology based on membrane contactors will be optimised for the efficient removal of CO2, H2S and impurities from biogas in order to adapt the final quality of gas to according to the predefined potential final use (CHP, grid injection, transport, etc.) in the different demos finally defined by co-creation activities.
First, LEITAT will design a compact and modular 5L AnMBR at lab scale. LEITAT will be in charge of its optimisation (biological and membrane systems). Biological reactor will be start-up using an inoculum from existing AnMBR. Operating parameters such as blackwater:BW ratio, particle size, organic load, hydraulic retention time and cellular residence time will be optimised for optimal COD, Nitrogen, TS and VS removal and biogas production. Commercial ultrafiltration polymeric membranes will be selected for optimal ultrafiltration of treated water from biological reactor. Optimisation of filtration performance (filtration and back-washing cycles) will be performed aiming to minimise membrane fouling and filtration cost (i.e. renovation, maintenance, energy cost). Biological and membrane systems will be connected for continuous operation. Potential reuse of treated water as irrigation water or use for toilets will require hygienisation. For that, LEITAT will performed again the S5 tests using the same lab scale O3/UV technologies and the methods. When laboratory trials are finalised, the system will be scaled up. LEITAT will design and construct a com-pact and transportable AnMBR unit able to treat the organic load rate of 8-10 inhabitant equivalents (black-water and grinded BW, in optimal ratio previously reported at lab scale). The design will be addressed for guaranteeing safe operating conditions and minor odour production. Hazard and operability analysis (HAZOP) and odour controls (by gas olfatometry gas and chemical analysis of samples by gas chromatography) will be carried out during the operation of AnMBR for optimisation of final design and odor control. The system will be started up and optimised in LEITAT facilities using real WW (from La Gripia building) and grinded kitchen wastes. Once safety guarantees of the system is ensured (by IDP, LEITAT, CARTIF and AHC) and O+M protocols are prepared, the system will be emptied and prepared for transportation to Demo 2 ("La Gripia" building) for final integration and operation at building scale (Task 4.8). Regarding dAD, after the separation of the dry and liquid fraction of domestic WW in Demo 3, the dry organic content will be transported via slides to the dry anaerobic digestor, placed close to the sewage pipe end and the burning system. The planned digestion system from HOMEBIOGAS will directly remove hydrogen sulfide and unpleasant odours from the gas outlet. Gas characterisation will be performed by ALCN. The system must further clean the gas for storing enriched methane in a gas storage balloon to save the biogas for the winter times, when it is mostly needed. Quality of biogas will be monitored by ALCN with the supervision of HOMEBIOGAS. Twice a year we will empty the sludge from the bottom of the tank with the fore-seen outlet and a pumping system directly sending it to the composting unit.

The liquid that can be removed permanently will also be processed in the composter (S8 component) and will further be collected as liquid fertiliser.

CARTIF will develop an innovative small scale biogas cleaning system based on (CO_2 and H_2S) absorption technologies (by gas washing plus activated carbon filters). Then gas chromatography tests will be per-formed aiming to optimise final gas quality according to operating parameters such as gas flow, temperature and pressure. The system will be scale up to ensure a biogas quality corresponding to natural gas grid quality specification (>95% CH_4, water vapour content <80mg/m3, etc.) for potential grid injection, use in CHP, combustion in a natural gas boiler or other valorisation routes proposed by stakeholders for co-creation at building scale. Installation of proposed equipments will be performed by CARTIF/IDP in Demo 1 and CARTIF/ALCN in Demo 3.

S8 components (Leader: LEITAT, Participants: ALCN, IDP)

Several technologies for the stabilisation of AnMBR digestate will be tested at laboratory for optimal selec-tion and scale up to large scale (considering as part of decision making the requirements and needs of stake-holders). In detail, final stabilisation techniques for AnMBR digestate such as drying (LEITAT), domestic co-composting by mixture with fresh BWs (LEITAT) and vermicomposting (LEITAT) will be previously studied at laboratory (TRL5) (treating 5 kg maximum per batch and technique) before optimisation (TRL6) and scale-up and demonstration at large scale by IDP (TRL7 in Demo 2). For mentioned technologies, physic-chemical analysis of compost characteristics such as pH, moisture, organic matter, soluble salts, C:N ratio, ammonia content, biodegradation, odour, maturity (Rottegrade index) and density will be performed and compared. Hygienisation degree will be also analysed (E. coli and Salmonella sp.) by LEITAT. On the other hand, a compost cultivator prototype will be developed by ALCN at TRL6 and optimised for composting fresh gardening wastes before mixing with dAD digestate at large scale in Demo 3. Optimisation of optimal ratio will be performed according to compost characteristics evaluated by ALCN.

Subtask 4.3.3 Optimisation of Energy vector

S10 components (Leader: ITEC, Participants: AHC, AIGUASOL, NEUES LEBEN, ALCN, CECODHAS)

The energy optimisation has to be tackled from two perspectives: active and passive/bioclimatic. The active system needs energy consumption but some active systems need less energy than others. Otherwise, passive systems don't need energy and became more efficient but sometimes is difficult to apply it in a residential refurbishment.

AIGUASOL and ITEC will work in the feasibility analysis of innovative active and passive systems potentially installed as improved elements in building structure in all Frontrunner buildings. Some examples of this active and passive systems improving the energy efficiency of buildings may include modular facades, roofs, cross ventilation, galleries, awnings, conditioning systems for radiant surfaces, evaporative cooling through water or vegetation films, Trombe walls, Canadian wells, and other solutions including energy generation from the building, among others. Co-creation between AIGUASOL and ITEC with stakeholders (including AHC, ALCN and NEUES LEBEN) will promote the design and potential installation of optimal sys-tems to be selected according to circular criteria as well as their cost (installation, maintenance and renova-tion cost). The final installation of selected objects in Demo 1, 2 and 3 will require the study of new financial resources to be searched in Task 4.6 (lead by CECODHAS). ITEC will be the partner in charge of the modular façade definition and AIGUASOL will help to define the energy properties of the developed modular façade. To do so, it will be used the EPBD established cost optimal procedure in order to check the best options to be developed and implemented, and considering different scopes (cost optimal buildings and nZEB solutions).

S11 components (Leader: AIGUASOL, Participants: AHC, ALCN)

Co-creation opportunities based on performance rather than ownership for energy systems (i.e. Solar Thermal (ST) and Photovoltaic systems (PV)) will be evaluated to be implemented in Frontrunner buildings. Pay per performance business models applied to renewable energy systems will be promoted. Under this model, an energy service company supplies, installs and maintains a ST or PV system, guaranteeing an annual solar production defined in a contract of 10 years. By using this model, it is guaranteed that ST or PV systems operate correctly at least during the contract signed between the energy service company and the own-er/neighbourhood community of the building. Renting or leasing options will also be considered when implementing renewable energies.

Task 4.4 Technical monitoring programme (Leader: ALCN; Participants: CARTIF, LEITAT, VISUM, IDP, AIGUASOL, HOMEBIOGAS) [M21-M48]

According to the technologies and solutions selected in the previous tasks, a monitoring program will be specified to ensure a correct impact assessment of the demonstration. Therefore, suitable sensors will be selected (photometric sensors, online microbial sensors, humidity sensors, temperature sensors, etc.).

Regarding solutions designed for WW treatment (S5 and S6), ALCN will include sensors for the evaluation of water quality after NBS treatment. Water meters will be installed by IDP for quantification of water reuse in toilets at home and building level after O3/UV treatment in Frontrunner building 1 and 2. Regarding Solution 7, VISUM will integrate sensors and monitoring required for the operation of AnMBR system foreseen in Frontrunner building 1.CARTIF will proceed with a simultaneous real-time monitoring of CH_4, H_2S and CO_2 after the cleaning stage in

order to confirm that the gas stream fulfils the quality requirements demand-ed. CH4, H2S will be measured by dual wavelength infrared cell with reference channel, while H2S will be measure by electrochemical cell. The data acquisition and the repository of data according to the KPIs will be done in cloud-based systems to oversee the data sets online at any time. The data monitoring can be linked to automation protocols to increase for example retention time of any treatment automatically if a threshold is exceeded. The cloud based system will provide the capability to link the static information from the BIM server (IFC compliant) to the dynamic data repository (BACN standards compliant) to show all detailed information by means of a dynamic BIM visualisation tool.

Task 4.5 Execution project documents (Leader: CARTIF; Participants: ALCN, LEITAT, IDP, AIGUASOL, ITEC, HOMEBIOGAS) [M1-M26]
Once the technical specification of the solutions in frontrunner buildings (Task 4.2) and the associated monitoring elements are defined (Task 4.3), the execution project documents will be produced. The documents will also be supported and enriched by the Material Passports and BIM models of each building (Task 4.1). This will serve to the following two tasks: generation of the financial scheme (Task 4.6) and after that the launching of the tender process (Task 4.7).

Task 4.6 Improved access to finance through investment planning (Leader: CECODHAS; Participants: ALCN, AHC, AIGUASOL, ITEC, LEITAT) [M1-M48]
Further than the EC contribution, other financial resources to create a feasible investment plan for proper dimensioning of technical specifications as well as for allowing the replication of 11 HOUSEFUL solutions marketable as services will be sought. There are several options that will be explored: Other national or re-gional funding schemes, RIS3, crowd funding through ethical banks, bank loans and/or guarantees through blending instruments from EIB (e.g. InnovFin, ESIF) and national banks, ERDF (if applicable). It is also foreseen to explore other alternative financial schemes, mainly those that will use as foundations an innovative public and private business models or even an only private model. The task includes the development of a financing plan, the progress monitoring of the implementation of the plan for each Frontrunner building, a risk/uncertainty assessment, mitigation plans and the evaluation of different potential scenarios and impacts for the project. New economic resources promoting the final definition and exploitation of HOUSEFUL solutions as services will be welcome and offered for successful replication activities in Follower buildings or new case studies after the end of project execution.

Task 4.7 Selection of companies for renovation works (Leader: AHC; Participants: IDP, ALCN, GWR, LEITAT, AIGUASOL, ITEC)[M18-M23]
Since the two Spanish Frontrunner buildings are owned by a public entity, a tender process will be necessary to the development of the works. For “Els Mestres” building, due to the immediate renovation works of the building, the tendering process and selection of companies will be initiated before HOUSEFUL execution. For “La Gripia building”, the tendering and selection of companies will be executed once the standard AHC refurbishment project is approved (M18). Therefore, in this last case, tendering documents will be prepared. AHC with the technical support of IDP, AIGUASOL, ITEC and LEITAT, will be in charge of this process, according to the Spanish rules and normative for public procurement processes. Within the other Austrian demo site the companies will be selected according to price and trust. ALCN with the support of GWR will be in charge of this selection process.

Task 4.8 Management of permits, implementation and supervision of technological solutions at building scale (Leader: IDP; Participants: ALCN, GWR, AHC, AIGUASOL, ITEC, LEITAT, CARTIF, HOMEBIOGAS) [M6-M48]
In this task, the company or companies selected will implement the different solutions and monitoring ele-ments in the building. Previously, all permits for technological implementation of proposed solutions in each Demo buildings will be sought for approval. Once approved, ALCN in Austria and IDP in Spain will be in charge the technical assistance and the supervision of the works related to the technological innovation shown in S5, S6, S7, S8 and S9. ITEC and AIGUASOL will be in charge of the technical assistance and supervision of works for solutions potentially installed in S10, evaluating the resulted energy performance in S11. Mentioned partners will periodically report to the consortium the progresses and potential delays or difficulties. These partners will follow the commissioning process of the system installed to ensure that the technical solutions are performing correctly. After this phase, the demonstration will be ready to start the environmental and socioeconomic impact assessment (WP5) of the solutions.

A detailed description of partner's role in WP4 has been included in each partner profile included in Part B, Section 4, in the description of the partner role in HOUSEFUL project.

Participation per Partner

Partner number and short name	WP4 effort
1 - LEITAT	
Partner number and short name	**WP4 effort**
2 - ITeC	
3 - CARTIF	
4 - ALCN	
5 - AHC	
6 - VISUM	
7 - AIGUASOL	
9 - GWR	
10 - Turntoo B.V.	
11 - Neues Leben	
12 - CECODHAS	
15 - IDP	
16 - HOMEBIOGAS LTD	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D4.1	Circular economy audit: Baseline calculation	3 - CARTIF	Report	Public	12
D4.2	Technical specifications and execution project of solutions in Spanish and Austrian demos (version I)	3 - CARTIF	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	18
D4.3	BIM models and material passport of the Spanish and Austrian demos	2 - ITeC	Report	Public	18
D4.4	Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safety and monitoring plan for S7 and S8	1 - LEITAT	Report	Public	24

D4.5	Report on potential improvement in financial schemes and funding opportunities for Demo 2 Front-runner buildings and replication activities	12 - CECODHAS	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D4.6	Report on potential improvement in financial schemes and funding opportunities for Demo 3 Front-runner buildings and replication activities	12 - CECODHAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D4.7	Report of demonstration and implementation of HOUSEFUL solutions at building scale. In-cludes the Technical monitoring programme	3 - CARTIF	Demonstrator	Public	48
D4.8	Report on financial schemes and funding opportunities for all Front-runner buildings and replication activities (final update)	12 - CECODHAS	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D4.9	Technical specifications and execution project of solutions in Spanish and Austrian demos (version II - final)	3 - CARTIF	Demonstrator	Confidential, only for members of the consortium (including the Commission Services)	26

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]

Description of deliverables

Deliverables: D4.1) Circular economy audit: Baseline calculation [M12, CARTIF] / D4.2) Technical specifications and execution project of solutions in Spanish and Austrian demos [M18, CARTIF] / D4.3) BIM models and material passport of the Spanish and Austrian demos [M18, ITEC] / D4.4) Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safety and monitoring plan for S7 and S8 [M24, LEITAT] / D4.5) Report on potential improvement in financial schemes and funding opportunities for Demo 2 Frontrunner buildings and replication activities. It includes a financing plan, progress monitoring of the implementation of the plan, risk/ uncertainty assessment, mitigation plans and different scenarios and impacts for the project. [M18, CECODHAS] / D4.6. Report on potential improvement in financial schemes and funding opportunities for Demo 3 Front-runner buildings and replication activities [M24] [CECODHAS] / D4.7) Report of demonstration and implementation of HOUSEFUL solutions at building scale. Includes the Technical monitoring programme [M48, CARTIF] / D4.8. Report on financial schemes and funding opportunities for all Front-runer buildings and replication activities (final update) [M48] [CECODHAS] / D4.9) Updated and final version of D4.2 [M26, CARTIF]

D4.1 : Circular economy audit: Baseline calculation [12]

This Deliverable reports the results of an exhaustive circular economy audit on the initial status of the four Frontrunner buildings in the four targeted vectors (i.e. water, waste, energy, materials) and in four dimensions (i.e. technological, social, environmental, economic), considering the identified KPIs and calculation methods set in WP2

(D2.3). The baseline data will be a key element for the technical definition specifications (task 4.2), optimization (task 4.3) and monitoring (task 4.4). This deliverable is related to Task 4.1.

D4.2 : Technical specifications and execution project of solutions in Spanish and Austrian demos (version I) [18]

This deliverable compiles the results from two interrelated HOUSEFUL activities: Task 4.2 and Task 4.5.

Consequently, the D4.2 will be divided in main two parts that will include: (1) Technical specifications of solutions including both technological implementation (S5, S6, S7, S9, S10 and S11) and those not including technological implementation (S4, S9) for the 4 Frontrunner buildings (Task 4.2). A first version of D4.2 including this part will be delivered in M18; (2) The execution projects will be related with the technological solutions (Task 4.5) and delivered as an updated version of D4.2 in M26 for Frontrunner building 1, 2 and 3 (D4.9). The execution project will be completed once results from optimization tests (at laboratory scale) are obtained and sensors/control system required by the monitoring programme are initially defined in Task 4.4. Both technical specification and the execution projects will be supported by the BIM models and material passport generated in Task 4.1 and the co-creation process promoted during WP3.

D4.3 : BIM models and material passport of the Spanish and Austrian demos [18]

The D4.3 will present a detailed review of the inventory of used materials of the four Frontrunner buildings considering both material passport strategies, the definition of deconstruction plans for afterlife applications and the BIM modeling for the demo sites. This task relates to the establishment of HOUSEFUL solution S3, but will be also used for the definition of technical specifications and execution projects in Task (D4.2). This deliverable is related to Task 4.1.

D4.4 : Validation tests of technologies in S5, S6, S7 and S8 solutions at lab scale. Includes the Safe-ty and monitoring plan for S7 and S8 [24]

The D4.4 summarizes the results from the laboratory tests deployed for the validation of S5 (rainwater and greywater treatment with NBS and O3/UV commercial system), S6 (unsegregated water treatment with NBS and O3/UV commercial system), S7 (blackwater and biowaste treatment for biogas production through AnMBR, dAD and absorption solution for biogas cleaning), and S8 (stabilized biowaste from AnMBR ad dAD used for producing high quality fertilizer/ compost). A safety and monitoring plan for S7 and S8 solutions will be included. This deliverable is related to Task 4.3 (Subtask 4.3.1 and Subtask 4.3.2).

D4.5 : Report on potential improvement in financial schemes and funding opportunities for Demo 2 Front-runner buildings and replication activities [18]

The main goal of this deliverable is to report the strategy for the identification of financial opportunities for the co-financing of refurbishment works in Frontrunner buildings, as well as for the promotion and increase in the implementation of HOUSEFUL solutions in more homes per building. The search of co-financing will include funding from private and public origins. Local and national alternatives in Austria and Spain, as well as general European options, will be considered. For that, a Financing plan will be delivered in M18. The report shall include a progress monitoring plan for each co-funding opportunity (identified), a detailed risk/uncertainty assessment, mitigation plans and different scenarios and impacts. This deliverable shall include a detailed financing plan for Demo 2. This deliverable is related to Task 4.6.

D4.6 : Report on potential improvement in financial schemes and funding opportunities for Demo 3 Front-runner buildings and replication activities [24]

The main goal of this deliverable is to report the strategy followed for the search of funding opportunities for both the co-financing of refurbishment works in Demo 3 Frontrunner building, as well as for the promotion and increase in the implementation of HOUSEFUL solutions in other buildings. The search of co-financing will include funding from private and public origin. The report shall include a progress monitoring plan for each co-funding opportunity (identified), a detailed risk/uncertainty assessment, mitigation plans and different scenarios and impacts. This deliverable is related to Task 4.6.

D4.7 : Report of demonstration and implementation of HOUSEFUL solutions at building scale. In-cludes the Technical monitoring programme [48]

A detailed report regarding results during demonstration stage of HOUSEFUL technological solutions (S5, S6, S7, S8, S9, S10 and S11) in the Frontrunner buildings will be presented. These results will be directly related to the selected KPIs of D2.3, providing data for the environmental, economic and social assessment of HOUSEFUL solutions (WP5) and the subsequent exploitation and business models analyses of WP6. The deliverable will be divided into 3 main parts which will include: 1) Management of permits (Subtask 4.8.1). All permits for the technological implementation of HOUSEFUL solutions will be identified, managed and sought for approval in each Frontrunner building. For that, technical specifications for each HOUSEFUL solution in each Frontrunner building will be the basis for the management of these permits, considering specific risks, ethics concerns and project milestones. It is worth to mentioning that “Els Mestres” building (Demo 1) and “Baumgarten” building (Demo 3) have initiated the procedure for their refurbishment before the HOUSEFUL kick off. The refurbishment includes some of the HOUSEFUL solutions (S3, S4 and S10). Then, management of some permits has been initiated for the successful demonstration of S3, S4 and S10 during the refurbishment. 2) Implementation (Subtask 4.8.2): the report will include a description of the procedure used for the installation, start-up and optimization of HOUSEFUL solutions in each Frontrunner building (S5, S6, S7, S8, S9, S10 and S11), as well as and the

quantification of KPIs (defined in D2.3) demonstrating the achievement of project results. 3) Supervision of technological solutions Subtask
4.8.3 and Task 4.4): Includes the Technical monitoring programme of the project. This deliverable is related to Task 4.8.

D4.8 : Report on financial schemes and funding opportunities for all Front-runner buildings and replication activities (final update) [48]

This deliverable will include an updated versions of D4.5 and D4.6 reports, which describe the financial schemes and funding opportunities searched for all Demos. This deliverable is related to Task 4.6.

D4.9 : Technical specifications and execution project of solutions in Spanish and Austrian demos (version II - final) [26]

Updated and final version of D4.2

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS5	Baseline of Frontrunner buildings completed	3 - CARTIF	12	D4.1) Baseline calculation for 4 Frontrunner buildings
MS6	Execution project documents ready	3 - CARTIF	26	D4.2) and D4.3) Technical specifications and execution projects of Frontrunner buildings
MS7	Technological solutions ready to be validated at demonstration scale	3 - CARTIF	24	D4.4) Optimisation of technological solutions (Task 4.3), Financial scheme (task 4.6), and technical monitoring programme (task 4.8).
MS14	Financial plan for Demo 2	3 - CARTIF	18	In task 4.6, delivery of D4.5 due on M18 and updated version on M48 (D4.8). It includes a Financing plan, a progress monitoring of the implementation of the plan, risk/uncertainty assessment and mitigation plans.
MS15	Financial Close for Demo 3	3 - CARTIF	24	Complementary financial access for demo 3 is completed. Associated deliverables - D4.6 (in M24) and updated version D4.8 (M48).
MS16	Tenants occupy the "Els Mestres" building	3 - CARTIF	23	They begin to participate in co-creation activities (D3.2) for final definition of technical specifications (D4.2).

Work package number [9]	WP5	**Lead beneficiary** [10]	1 - LEITAT
Work package title	Environmental, economic and social assessment		

Start month	8	End month	54

Objectives

- To define the HOUSEFUL assessment plan for the three pillars: environmental, economic and social.
- To assess the potential environmental and socioeconomic impacts of the circular strategies demonstrated under a LifeCycle Thinking approach.
- To determine the energy savings, water reuse, material recycling and waste valorisation potential achieved andassociated to the operating phase of buildings by adopting circular economy principles.
- To test the Level(s) tool developed by the European Commission and provide recommendations for improvement.

Description of work and role of partners

WP5 - Environmental, economic and social assessment [Months: 8-54]
LEITAT, ITeC, CARTIF, ALCN, AHC, VISUM, AIGUASOL, LGI, GWR, Turntoo B.V., Neues Leben, WE&B, ICONS, IDP, HOMEBIOGAS LTD
Task 5.1. HOUSEFUL assessment plan (Leader: CARTIF; Participants: LEITAT, ITeC, AIGUASOL, WE&B) [M13M17] This task aims to deploy the assessment plan for the three pillars of the project: envi-ronmental, economic and social. The assessment plan will be tailor-made to the characteristics of each front-runner building. The first step before the assessment of each pillar in Task 5.2, Task 5.3 and Task 5.4 is to define the assessment plan. This task should be strongly aligned with both Task 4.3 (technical monitoring program) because the monitoring program should cover the requirements of the plan and Task 2.3 (defini-tion of the focused calculation procedures and relevant KPIs) since the KPIs should allow to evaluate the pillars of the project and take them into account during the development of the assessment plan. The HOUSEFUL assessment plan covers three subtasks in order to define the plan for each pillar (environmental, economic and social). The environmental pillar will be assessed in two different ways: 1) by applying a life cycle thinking approach considering energy, materials, waste and water impacts (task 5.1.1) and 2) fo-cusing only on the energy impact associated to the operating phase (task 5.1.2), since this represents the highest environmental burden in the life cycle of a building as revealed in LCA studies.

Subtask 5.1.1 Environmental and economic assessment plan (Leader: LEITAT)
The aim of this subtask is to specify the methodology to evaluate the environmental and economic impact of the demonstration activities by applying a life cycle thinking approach. Thus, the boundaries and guidelines for the different stages (production, construction/refurbishment, operation, maintenance and demolition) will be determined and the environmental and economic indicators (aligned with task 2.3) will be selected.

Subtask 5.1.2 Energy assessment plan (Leader: CARTIF) The energy assessment of energy conservation measures will be done under International Measurement and Verification protocol (IPMVP). IPMVP defines the plan to evaluate the energy savings. Thus, in this subtask a Measurement and Verification (M&V) plan will be carried out, consisting in identifying the independent variables, selection of the IPMVP option, base-line energy consumption and definition of the mathematical model. The aim of this plan is specify the situa-tion of buildings before of interventions.

Subtask 5.1.3 Social assessment plan (Leader: WE&B). The aim is to define the social assessment plan based on behavioural and social criteria (attitudes, values, beliefs and norms). Social assessment procedures (indicators and data gathering processes) will be created on the basis of results from Task 3.1, taking into account the characteristics of each building and owner. The output of this subtask will be a plan for social assessment data gathering that will show the social impact of the project.

Task 5.2. Environmental & economic assessment (Leader: LEITAT; Participants: ITeC, GWR, LGI, AI-GUASOL) [M18-M54]

Subtask 5.2.1 BIM-LCA&LCC assessment (Leader: LEITAT). The environmental and economic impact of the circular solutions demonstrated in Frontrunner buildings will be assessed in this task based on the assessment plan defined in task 5.1 and the BIM results obtained in WP4. Since the BIM objects from the ITeC's database integrate LCA related information, the environmental impact of the solutions will be assessed from the data obtained from BIM. In line with the guidelines defined by the European Platform on LCA, different impact categories will be considered for determining the environmental impact. In particular, BIM results will generate a set of 22 impact categories (e.g. depletion potential of the stratospheric ozone layer, acidification potential of soil and water, human toxicity potential, eutrophication potential, % of recy-cled material, hazardous waste, energy of fossil origin, etc). The impact associated to hazardous substances will be assessed by applying a life cycle approach. Hazardous waste can be found in construction and demolition waste. For example, old styrofoam insulation from buildings is considered hazardous waste in Germany or asbestos in old roof tiles is also hazardous. According to the harmonised classification and la-belling (CLP00) approved by the European Union, asbestos may cause cancer and causes damage to organs through prolonged or repeated exposure. Potentially hazardous substances should also never be used in circular construction, as the reuse or recycling options are very limited due to health constraints. Some impact assessment categories included in LCA (e.g. hazardous substances, human toxicity) will be used to assess the potential impacts of hazardous substances. GWR and AIGUASOL will contribute to this task by providing relevant/new data for calculating BIM-LCA related information in modified circular buildings. The methodology developed in WP2 to quantify the circularity level of buildings will be tested with data obtained from frontrunner buildings. LEITAT will analyse the results obtained from the impact categories for each demo site in order to compare the environmental impact of different scenarios. Therefore, the environmental impact of the baseline scenario will be compared with the Houseful scenario. A Life Cycle Cost (LCC) of the lifetime cost associated with the buildings' life cycle will be conducted by LGI from the BIM results with the aim to evaluate the economic feasibility of the Houseful model proposed in each Demo Site. Capital costs (design and project management, material costs, construction/refurbishment costs), O&M costs and end of life costs will be taken into account in the LCC process.

Subtask 5.2.2 Level(s) assessment (Leader: LEITAT)
The new Level(s) tool developed by the European Commission will be used to assess the environmental performance of frontrunner buildings. The Level(s) tool will be ready for testing in autumn 2017 and there will be a test period that will last for 2 years. Therefore, the aim of this task is to test Level(s) with real cases (Frontrunner buildingsand provide feedback and recommendations to the European Commission for improving the tool.

Task 5.3. Measurement & Verification of the energy savings (Leader: LEITAT; involved partners: GWR, AIGUASOL, CARTIF, NEUES LEBEN) [M18-M54]
A Measurement & Verification (M&V) process based on the IPMVP and the energy assessment plan defined in task 5.1.2 will be carried out to determine the energy savings achieved in the frontrunner buildings after the renovation process. Significant variables that affect the energy consumption will be considered such as weather conditions. IPMVP provides 4 options for determining energy savings (A, B, C and D). The options selected for each Demo are outlined below:

Demo 1. Building of the 1960's in Sabadell (Spain)
Option D. Calibrated simulation. Savings will be determined through simulation of the energy performance of the whole building.
Metered energy data for the baseline does not exist. A package of energy conservation measures are expected to be implemented, including envelope interventions.

Demo 2. Building of the 1970's in Terrassa (Spain)
Option C. Energy savings will be determined by measuring energy consumption at the whole facility utility meter level Metered energy data for the baseline exists. Energy efficiency measures will be implemented, including envelope interventions

Demo 3. Centenary building in Vienna (Austria)
Option C. Energy savings will be determined by measuring energy consumption at the whole facility utility meter level Metered energy data for the baseline exists. Energy efficiency measures will be implemented, included envelope and roof interventions.

Demo 4. New building in Vienna (Austria)
Option D. Calibrated simulation. Savings will be determined through simulation of the energy performance of the whole building.
New constructed building with brief baseline data. Theoretical energy calculations for every part of the building exist

Certified Measurement & Verification Professionals (CMVP) from LEITAT and AIGUASOL will assure that the M&V process of the energy savings is conducted in compliance with the IPMVP. LEITAT will be in charge of

comparing the pre and post refurbishment scenarios of demos 2&3 following the option C of the IPMVP, whereas AIGUASOL will be in charge of carrying out the energy simulations (Option D of the IP-MVP) of demos 1&4. GWR and NEUES LEBEN will provide the technical data requested by LEITAT and AIGUASOL.

Task 5.4. Social assessment (Leader: WE&B; Participants: ICONS, AHC, ALCN, NEUES LEBEN) [M18-M54]
To measure the social impact of the HOUSEFUL services, it will be important to not only engage with the HOUSEFUL stakeholders from the very beginning of the project (task 3.1.) but to continually engage with them as the project continues. Feedback from the end users (especially tenants) will be vital to not only tailor the solutions to their needs but to know if these solutions have had an impact on their daily lives. The task will work closely with WP3 and more specifically task 3.1 on stakeholders' identification and analysis and task 3.2 on the engagement of stakeholders.

The social indicators defined within task 5.1 will be assessed with the aim to reveal the final impact of the social aspects on the local and regional population wellbeing and engagement. First, a baseline measurement of these indicators will be undertaken by WE&B, this will serve as a reference point to the future measure-ments and an indication of the potential positive social impact of the HOUSEFUL project.
The measurement of the social and behavioural indicators will be strategically planned in Task 51, although we can envisage the use of the stakeholder consultation through surveys and interviews at two different scales, the stakeholders at the frontrunner building (including tenants) and at the level of the circularity building international community.In this sense, WE&B will run a survey to a wider audience and stakeholders at the start of the project, its midpoint and again towards the end of the project.
The measurement of the social impact at the start and mid-point of the project will allow for changes and adjustments to be made in the project activities to ensure that the public and end users are satisfied with the use of the solutions and the changes in behaviour they would need to make to adjust to a circular economy.
The project partners will combine both qualitative and quantitative measures to not only get feedback on social insights and behavioural patterns but to also to have numbers and statistics to show and compare the social impact from the start of the project throughout it and again at the end. ALCN will co-organise the stakeholder communication and will react concerning the design of NBS in alignment with the results of stakeholder processes concerning expectations, no-go criteria, etc. NEUES LEBEN as housing cooperative will subcontract this task to their social building support organisations.

Participation per Partner	
Partner number and short name	**WP5 effort**
1 - LEITAT	
2 - ITeC	
3 - CARTIF	
4 - ALCN	
5 - AHC	
6 - VISUM	
7 - AIGUASOL	
8 - LGI	
9 - GWR	
10 - Turntoo B.V.	
11 - Neues Leben	

13 - WE&B	
14 - ICONS	
15 - IDP	
16 - HOMEBIOGAS LTD	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D5.1	HOUSEFUL assessment plan	3 - CARTIF	Report	Public	17
D5.2	Environmental and economic impact assessment	1 - LEITAT	Report	Public	54
D5.3	Measurement &Verification	1 - LEITAT	Report	Public	54
D5.4	Social assessment	13 - WE&B	Report	Public	54

Description of deliverables

Deliverables: D5.1) Houseful assessment plan [M17, CARTIF] / D5.2) Environmental and economic impact assessment [M54, LEITAT] / D5.3) Measurement &Verification [M54, LEITAT] / D5.4) Social assessment [M54, WE&B]

D5.1 : HOUSEFUL assessment plan [17]

An assessment plan will be developed with the aim to define the principles and the framework to be considered for assessing the impact of the circular solutions implemented in each frontrunner building. The assessment plan will present the methodology to assess the impact of the solutions at three different levels: 1) environmental and economics; 2) energy; and 3) social. This deliverable is related to Task 5.1.

D5.2 : Environmental and economic impact assess-ment [54]

This report will present the environmental and economic benefits achieved after implementing the circular solutions in frontrunner buildings. Quantitative results of the environmental impact assessment will be presented. The results of the Life Cycle Cost (LCC) analysis will be presented with the aim to evaluate the economic feasibility of the HOUSEFUL model proposed in each frontruner building. This deliverable is related to Task 5.2.

D5.3 : Measurement &Verification [54]

A Measurement and Verification (M&V) report will be produced presenting the energy savings achieved after refurbishment processes in frontrunner buildings. The report will provide details about the IPMVP option used as well as the independent variables considered for conducting the energy assessment in each demo building. This deliverable is related to Task 5.3.

D5.4 : Social assessment [54]

A social assessment report will be prepared to present the social impact results of the circular solutions implemented in demo buildings. The results of the social and behavioural indicators assessed will be presented with the aim to reveal the final impact of the social aspects on stakeholders, and especially from tenants. This deliverable is related to Task 5.4.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS8	Environmental, economic and social assessment of the circular solutions demonstrated	1 - LEITAT	54	D5.2) Environmental and economic impact assessment; D5.3) Measurement &Verification; D5.4) Social assessment

Work package number [9]	WP6	**Lead beneficiary [10]**	8 - LGI
Work package title	HOUSEFUL Solutions as Services, market, policies and exploitation		
Start month	1	**End month**	54

Objectives

The main objective of WP6 is to prepare the market entry strategy of the most promising technological and nontechnological HOUSEFUL solutions as systemic services (including the SaaS) to facilitate the transition from a demonstration phase to a commercialisation phase (T6.2). First, a market study (T6.1) will identify, characterise and quantify most promising markets for the HOUSEFUL services, SaaS included. The development of the HOUSEFUL SaaS (T6.3) will provide a user-friendly interface for the exploitation of the building circularity calculator as well as for the replication of HOUSEFUL solutions as services co-created in new buildings. The business modelling (T6.4.1) and replicability study (T6.4.2) will analyse the impact of the HOUSEFUL services on existing linear business models of the housing value chain stakeholders and identify drivers/barriers to the transition to circular business models. The exploitation strategy (T6.5) will draw on all the collected information to provide the future HOUSEFUL service providers with the most adapted approach to ensure market penetration. Finally, policy recommendations (T6.6) will be done to foster the circularity of the housing sector.

Specific objectives:

- To ensure the marketability of HOUSEFUL solutions as systemic services from the very beginning of the project• To develop a SaaS to encourage the housing value chain to facilitate the transition from linear to circular business models, through the creation of a Circularity Tool (CT) and an interactive sharing platform.
- To deliver a market study highlighting promising markets for the most mature HOUSEFUL services
- To analyse the impact of HOUSEFUL services on existing business models
- To study the replicability of new circular business models under different conditions.
- To propose an exploitation strategy to ease the market entry of most promising and mature HOUSEFUL services• To deliver policy recommendations to foster the circularity of the housing sector

Description of work and role of partners

WP6 - HOUSEFUL Solutions as Services, market, policies and exploitation [Months: 1-54]
LGI, LEITAT, ITeC, CARTIF, ALCN, AHC, VISUM, AIGUASOL, GWR, Turntoo B.V., Neues Leben, CECODHAS, WE&B, ICONS, IDP, HOMEBIOGAS LTD
Task 6.1: Market study (Leader: LGI; Participants: AHC, NEUES LEBEN, CECODHAS) [M1-M24]
The objective of the market study is to identify, characterise and quantify promising markets for the most mature and promising HOUSEFUL services and list local businesses and stakeholders along the housing value chain that may be interested in using them. The study will mainly focus on Spain and Austria (where the four demo-sites are located) with the support of the two building owners, respectively AHC and NEUES LEBEN. A preliminary market study and discussions with the project partners, notably CECODHAS, will be performed to define a strategic scope extension towards other EU countries (Ireland, Belgium, The Netherland, France, etc.). For each geographic market a volumetric estimation of future projects will be per-formed with a particular attention for the public social housing. Markets will be characterised thanks to common key indicators such as the type of building (private or social), size or the construction date (non-exhaustive list) to identify most promising market. Furthermore, an analysis of existing and foreseen policy and regulation will be performed to complete the study. Finally identified markets will be compared one to another to highlight the most promising markets. Focusing on this shortlist of markets the study will identify local businesses and stakeholders that may be interested in implementing the HOUSEFUL services. Local businesses working along the housing value chain, from equipment providers to building owners, will be identified and characterised. The established list could be used considered during further replication activities.

Task 6.2: From Solutions to Services. Strategy for a complete co-creation (Leader: LEITAT; Participants: all) [M1-M54] The main goal of this task is to guarantee the successful co-creation, implementation and demonstration of proposed HOUSEFUL solutions as new systemic services. From a detailed monitoring of all project activities (in WP2, WP3, WP4, WP5) affecting the concrete realisation of each solution, the identification market constrains and opportunities for each solution (Task 6.2) and the detailed definition of the exploitation plan for all solutions as potential services (Task 6.4), LEITAT will work to ensure that the effective co-creation of CEBO between stakeholders is feasible. First, for each HOUSEFUL solution, a Solution-to-Service Monitoring Plan (SSM Plan) will be developed. Each SSM Plan will include the following information: (1) Calendar including the main key steps to be achieve/achieved during project execution affecting the materialisation of each solution to offer as a service; (2) Key milestones including potential constrains during the demonstration phase; and (3) Detailed contingency plan. The SSM Plan will be created by LEITAT with the collaboration of all project partners at the beginning of project execution during first 3 months and will be updated every year, considering all outputs from all HOUSEFUL work packages and activities. When all key milestones in the SSM plan are successfully overcome and market study and exploitation plan for each solution are defined, detailed exploitation strategy and supply costs of each systemic services will be defined for their first potential replication in 10 additional Follower Buildings selected during project execution (minimum amount of buildings to be proposed by CCHE Board in WP7) or new ones from dissemination activities proposed in HOUSEFUL.

Task 6.3: Development of the HOUSEFUL SaaS and functionalities (Leader: VISUM; Participants: all) [M8-M54] A SaaS will be developed to support stakeholders (construction companies, technological and product suppliers, design teams, municipalities, regional/national housing agencies, owners, dwellers, tenants and policy makers) to bring residential buildings to the circular economy using solutions offered as services in HOUSEFUL. The SaaS will be embedded it into an ergonomic and tailored graphic user interface (GUI) in order to facilitate its use and exploitation. The GUI will be web-based and device responsive in order to allow the access on mobility via different devices. Since different types of users are expected to access to the HOUSEFUL service, different types of user accounts will be created to allow proper scalability of functionalities and personalisation according to the need of the user. The SaaS will be composed of two main modules: 1) a Circularity Tool (CT) to quantify the circularity level of residential buildings. The algorithm for the calculation of building circularity developed in WP2 is the cornerstone of the HOUSEFUL project as it allows a diagnostic of existing buildings before performing any action. Users will be able to work with data and calculation methods at two defined levels: a) a first level (free access-freemium business model) that will provide the simplest way to assess the circularity level of buildings based on few simple data and key KPIs and b) a second level with advanced functionalities (a pay per use) for professionals (i.e. design teams, construction companies, etc) that wish to assess the circularity level in a more accurate and reliable way, suggesting also different types of interventions that can be applied to improve it. It will be also possible to quantify the effect of new interventions or refurbishment on an existing building in order to choose the most suitable. In both levels, the GUI will allow the user to choose in which phase of the build-ing lifecycle apply the simulation and will guide him through a step process in which the input and calculation options have to be provided to obtain the expected output. The HOUSEFUL service will help the user throughout the process suggesting, whenever possible, precalculated values for the unknown parameters; 2) an interactive sharing platform to promote the co-creation of new CEBOs. The platform will contain successful circular solutions and services, the ones demonstrated in Frontrunner buildings (WP4) but also other successful experiences collected in WP2 (task 2.6), with the aim to promote the replication of these initiatives and facilitate the collaboration among stakeholders. This section of the GUI will be based on a dynamic mapping where dynamic filters can be applied to show only the results of interest. The SaaS will help the housing value chain work on their value proposition, encouraging the transition from linear to circular business models. The SaaS will be developed by VISUM with the technical contributions from all partners. Detailed functionalities of the SaaS will be defined by all the consortium during project execution.

Task 6.4: Business modelling and replicability study (Leader: LGI; Participants: TURNTOO, ITeC, WE&B, CARTIF) [M24-M48]
Globally the emergence of circular economy services leads to a fundamental transformation within impact-ed sectors as traditional linear business models of the stakeholders are progressively reviewed in favour of circular business models. Subtask 6.4.1: Business modelling (TURNTOO, ITeC, LGI, WE&B, CARTIF) [M24-M36]
This task aims at analysing the specific impacts of HOUSEFUL solutions as services on the business models of local stakeholders involved in the demonstration sites of the project (frontrunner buildings).
Taking advantage of the workshops organised with local communities in the WP3, business models will be described using the Business Model Canvas (Alexander Osterwalder, 2011). Major interactions and relations between different stakeholders as well as main business characteristics as value proposition, revenues sources or cost structure (nonexhaustive list) will be analysed in a transparent and visual way.
The multi-level perspective (Geels, 2002) will be used as a basis for the evaluation of the disruptive impact of the HOUSEFUL services on existing business models. Inputs from services design team (WP3 and WP4) and from the market study (T6.4) will be used to identify key drivers and barriers to the deployment of the HOUSEFUL services

within the existing, considering current business models as the regime and HOUSEFUL services as innovation niches. Selected tools recommended by the Climate KIC`s Climate Change Adaptation Innovation Approach will be considered (based on Climate KIC Pioneers toolkit, 2017). The involvement in the task of the WP3 (WE&B) and WP4 (CARTIF) leaders will facilitate interactions with local stakeholders necessary to successfully perform the analysis.

Subtask 6.4.2: Replicability study (ITeC, TURNTOO, LGI) [M36-M48]
The longevity and sustainability of new circular business models have to be assessed in a broader context and from different points of view. This task will use the market information of this WP, preliminary results of the co-creation (WP3) and demonstration activities (WP4) as well as the lifecycle analysis from WP5 to assessthe direct replicability of the HOUSEFUL services. An important contribution from the CCHE Board is expected for the description of the Follower buildings. Key success factors for replication as well as barriers will be identified in order to assess the replicability potential of the most promising and mature HOUSEFUL services. The outputs of this task will be used to write the exploitation plans (T6.5), the policy recommendations to foster the circularity of the housing sector (T6.6) and will be communicated during the replication oriented-dissemination in Follower buildings (T7.4.1). A roadmap including the necessary actions needed to ensure a successful replicability in Europe will also be included in this task.

Task 6.5: Exploitation Plan (Leader: LGI; Participants: all partners) [M1-M54]
This task aims at gathering all the information produced along the project to draft a market entry strategy. It will consider inputs from non-technical services (WP3), technical demonstration (WP4), environmental and socio-economic assessment (WP5), market analysis (WP6) to define the most adequate approach to a successful exploitation for the most promising HOUSEFUL services. Innovative methodologies will be used facilitate strategic discussions and overcome the barriers created by the multi-actor and multi-resource approach of the project. A LEGO© Serious Play© workshop with LGI certified facilitators will be organised within the first year of the project to better consider the different point of views, define a common vision and draft the first exploitation axes. Other methodologies such as Value Proposition Design (Alexander Osterwalder, 2011) may be used all along the project depending on encountered barriers.
The Exploitation task will consider different levels of activities, in particular:
- Evaluation of possible exploitation outcomes of demonstration results of the project
- Identification of IP right needs, monitoring and support on individual and joint actions
- Specific business model recommendations for each services
- Recommendation on effective exploitation strategies
- Economic and business assessment when necessary
- Additional general recommendations on technology management, financial engineering and market roadmap.The final plan shall cover all project related technologies & services that have been identified as mature enough and promising by the consortium.

Task 6.6 Policy recommendations to foster the circularity of housing sector (Leader: CECODHAS Participants: LEITAT,
AHC, NEUES LEBEN) [M48-M54]
The aim of this task consists in providing policy recommendations at local, regional, national and EU scale to increase the circularity level of the housing sector in our mainly linear economy. Pushing the supply and demand side by themselves is not sufficient to achieve a circular economy, and specific policies and regula-tions for the housing sector should be set to achieve a full circular transition.
Existing policies that promote renewable energies and energy efficiency in buildings need to be further strengthened and complemented with policies for resource efficiency, which look at a wider range of environmental impacts across the life-cycle of buildings. CECODHAS with the support of partners involved in this task will set EU recommendations on how to scale-up circular economy solutions and services in housing sector, including social housing. Legislation and regulatory barriers identified in WP3, during the market study (T6.2) and lessons learned during the project will be used to provide policy recommendations and strengthen the legal framework towards a circular economy. A better use of existing policy instruments will be proposed to promote circular procurement and the launch of circular tendering processes for public social housing which will scale-up new business models.

Task 6.7: Intellectual Property Rights management (Leader: LGI; Participants: All partners) [M1-M54]
This task aims to monitor and manage IPR related decisions derived from HOUSEFUL innovations. In close consultation with WP leaders, LGI as Innovation Manager will: i) Raise awareness of HOUSEFUL consortium members to the IPR rules set up in the Consortium Agreement that regulate the communication and exploitation of project results, in addition to the IPR provisions in the Grant Agreement and Rules for Participation of Horizon2020; ii) Ensure that the communication and exploitation activities comply with the IPR publication policies; iii) Assist the moderation of possible IPR related conflicts; iv) Monitor potential patent applications or other copyright related issues, being WP leaders responsible for the identification, monitoring and dissemination of the project outcomes (know-how, products, semi-products) and iv) Monitor and prepare for potential knowledge transfers between partners or external

providers during or after the project duration. All actions developed during this task and subtasks will be described and explained in Exploitation and Business Plan (D6.7).

Participation per Partner

Partner number and short name	WP6 effort
1 - LEITAT	
2 - ITeC	
3 - CARTIF	
4 - ALCN	
5 - AHC	
6 - VISUM	
7 - AIGUASOL	
8 - LGI	
9 - GWR	
10 - Turntoo B.V.	
11 - Neues Leben	
12 - CECODHAS	
13 - WE&B	
14 - ICONS	
15 - IDP	
16 - HOMEBIOGAS LTD	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D6.1	Market study (version I)	8 - LGI	Report	Public	18

D6.2	Solution-to-Service monitoring plan (version I)	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.3	HOUSEFUL SaaS	6 - VISUM	Other	Confidential, only for members of the consortium (including the Commission Services)	36
D6.4	HOUSEFUL SaaS – Final version	6 - VISUM	Other	Public	54
D6.5	Impact study of HOUSEFUL services on linear business models	10 - Turntoo B.V.	Report	Public	36
D6.6	Preliminary Exploitation Plan	8 - LGI	Report	Confidential, only for members of the consortium (including	36

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	
D6.7	Replicability analysis of new CEBOs	2 - ITeC	Report	Confidential, only for members of the consortium (including the Commission Services)	48
D6.8	Final Exploitation Plan	8 - LGI	Report	Confidential, only for members of the consortium (including the Commission Services)	54
D6.9	Policy recommendations	12 - CECODHAS	Report	Public	54
D6.10	Market study (version II - final)	8 - LGI	Report	Public	24
D6.11	Solution-to-Service monitoring plan (version II)	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	36

D6.12	Solution-to-Service monitoring plan (version III - final)	1 - LEITAT	Report	Confidential, only for members of the consortium (including the Commission Services)	51

Description of deliverables

Deliverables: D6.1) Market study [LGI, M18] [Updated version: M24 - D6.10] / D6.2) Solution-to-Service monitoring plan [LEITAT, M18] / D6.3) HOUSEFUL SaaS [VISUM, M36]. / D6.4) HOUSEFUL SaaS - Final version with completed functionality [VISUM, M54] / D6.5) Impact study of HOUSEFUL services on linear business models [Turntoo, M36] / D6.6) Preliminary Exploitation Plan [LGI, M36] / D6.7) Replicability analysis of new CEBOs [ITEC, M48] / D6.8) Final Exploitation Plan [LGI, M54] / D6.9) Policy recommendations [CECODHAS, M54] / D6.10) Updated version of D6.1 [LGI, M24] / D6.11) Updated version of D6.2 [LEITAT, M36] / D6.12) Updated and final version of D6.11 [LEITAT, M51]

D6.1 : Market study (version I) [18]

Detailed identification, characterization and quantification of most promising markets for the HOUSEFUL services, SaaS included (D6.3). Most important local business and stakeholders who could be interested in implementing HOUSEFUL solutions will be listed, coordinating with task 3.1. Furthermore, a strategic scope extension to other EU territories considering key indicators and future project volumetric estimations will be calculated. This deliverable is related to Task 6.1.

D6.2 : Solution-to-Service monitoring plan (version I) [18]

Definition of a Solution to Service Monitoring Plan (SSM Plan) for each of the HOUSEFUL services implemented. These SSM Plans will include a detailed calendar with the main key steps to be achieved, the key milestones and constrains during demonstration phase, and a detailed contingency plan. Through them, adequate co-creation strategies oriented to establish exploitable circular economy services will be guaranteed, supported by the Exploitation Plan definition of D6.4 and D6.7. This deliverable is related to Task 6.2.

D6.3 : HOUSEFUL SaaS [36]

A SaaS to promote the engagement of stakeholders to bring HOUSEFUL solutions to implementation will be developed. It will be based on two separated modules: a Circularity Tool to quantify the circularity level of residential buildings, based on the work deployed in WP2; and an interactive sharing platform to promote co-creation of new CEBO, where successful circular solutions and services will be included. This SaaS will consider different stakeholders typologies, allowing a broad degree of personalization. The first version of the HOUSEFUL SaaS will be delivered in Month 36. This deliverable is related to Task 6.3.

D6.4 : HOUSEFUL SaaS – Final version [54]

Delivery of the final version of the SaaS, with open access to all public at Month 54. This deliverable is related to Task 6.3.

D6.5 : Impact study of HOUSEFUL services on linear business models [36]

Establishment of the impact of circular solutions on local stakeholders linear business models by Business Model Canvas, multi#level perspective and s Climate Change Adaptation Innovation Approach. Synergies among D6.4, WP3 workshops and WP4 tasks will be capital to properly reach the local stakeholders. This deliverable is related to Task 6.4.

D6.6 : Preliminary Exploitation Plan [36]

Gathering of all the information obtained throughout the first 3 years of the project to set a preliminary draft of a market entry strategy. HOUSEFUL services data (WP3, WP4, WP5) and information related to market (D6.1) will be considered. First results regarding strategic discussion will be presented. This deliverable is related to Task 6.5.

D6.7 : Replicability analysis of new CEBOs [48]

Evaluation of the replicability of HOUSEFUL new CEBOs by analyzing the data generated in the project, mainly the market information of D6.1, preliminary results of the co-creation (WP3) and demonstration activities (WP4), and the lifecycle analysis from WP5; as well as the expert insight on the description of following buildings provided by the CCHE Board. Key success factors for replication and barriers will be identified and replicability potential for all HOUSEFUL services will be established. The outputs of this task will be used for exploitation plans (D6.5 and D6.7) and policy recommendations (D6.8), being also communicated during the replication oriented-dissemination in Follower buildings (T7.3.1). This deliverable is related to Task 6.4.

D6.8 : Final Exploitation Plan [54]

A definitive document establishing a final exploitation plan for each of the HOUSEFUL services, based on the previous work done for D6.5 together with more detailed discussion with each of the technological and service partners. Each HOUSEFUL service exploitation plan will include an evaluation of project outcomes, an identification of IP right needs, monitoring and support on individual and joint actions, specific business model recommendations, a proposal on effective exploitation strategies, and information regarding other economic and business assessment (if required). This deliverable is related to Task 6.5.

D6.9 : Policy recommendations [54]

Report on policy recommendations at local, regional, national and EU scale to promote the transition to circular models for the housing sector based on HOUSEFUL case study experience and data (WP3, WP4, WP5). Information regarding legislation and regulatory barriers identified in WP3 and D6.1 will also be integrated. This deliverable is related to Task 6.6.

D6.10 : Market study (version II - final) [24]

Updated and final version of D6.1.

D6.11 : Solution-to-Service monitoring plan (version II) [36]

Updated version of D6.2.

D6.12 : Solution-to-Service monitoring plan (version III - final) [51]

Updated and final version of D6.11 (being D6.11 an updated version of D6.2).

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS9	Launch of the market study	8 - LGI	24	D6.1) Market study

MS10	Launch of the SaaS	8 - LGI	36	D6.3) SaaS ready to be used by stakeholders
MS11	Complete list of potential HOUSEFUL services for stakeholders/end-users	8 - LGI	54	D6.7) Final Exploitation Plan

Work package number [9]	WP7	**Lead beneficiary** [10]	14 - ICONS
Work package title	Communication and Dissemination		
Start month	1	**End month**	54

Objectives

The core objective of WP7 is to manage and implement impact-based dissemination, communication and engagement activities with the aim to increase public and stakeholders' awareness, stimulate acceptance and foster replication of HOUSEFUL services from Frontrunner buildings towards Followers as well as supporting further uptake at European level and beyond. An integrated impact-driven approach will be adopted through a multi-stakeholder and multi-channel strategy, with the following specific objectives:

- To develop a Dissemination and Communication (D&C) Plan to design and manage an effective D&C strategyimplementation at EU and national level as well as guarantee public and professional coverage to enable widespread replication of the project's outcomes;
- To increase public awareness on the activities and the achievements of the project through the development of aneffective public communication strategy based on a multi-channel approach also exploiting HOUSEFUL partners' own communication channels;
- To build and engage with the HOUSEFUL community, implementing dedicated networking activities addressingEuropean citizens and stakeholders and replication-oriented actions addressing the Followers, through the support of the CCHE board;
- To measure the impacts generated by the implementation of D&C activities, through the intelligent use of outreachdata, quantitative performance indicators and key qualitative assessments;

Description of work and role of partners

WP7 - Communication and Dissemination [Months: 1-54]

ICONS, LEITAT, ITeC, CARTIF, ALCN, AHC, VISUM, AIGUASOL, LGI, GWR, Turntoo B.V., Neues Leben, CECODHAS, WE&B, IDP, HOMEBIOGAS LTD

Task 7.1. Communication and dissemination strategy. (Leader: ICONS; Participants: LEITAT, WE&B, CECODHAS, AHC) [M1-M54]

A Dissemination and Communication Plan will be delivered at the beginning of the project (M6). The main purpose of the plan will be to strengthen the vision and results of the project on a large scale based on a well-defined, impactbased communication strategy to support acceptance and replication at local level (involving both the players of the frontrunner buildings and the cities and EU level. The outreach of and engagement with HOUSEFUL contents will be continuously monitored to measure the effectiveness of the D&C strategy and to guarantee the achievement of the expected impacts. The Plan will outline targets, activities, channels, tools and timing. More specifically it will:

- Identify the target audiences, with the aim to build up a network of stakeholders representing the potential adoptersof the proposed HOUSEFUL solutions and methodology.
- Define key messages. At the beginning of the project a set of key messages will be developed to ensure the HOUSEFULconcepts are received by the target audiences.
- Define communication and dissemination formats, actions, channels and plan of activities.

Task 7.2. Communication materials for co-creation activities with stakeholders in Frontrunner buildings (Leader: ICONS; Participants: All) [M1-M48]. In detail, at local level, the following specific communication materials will be developed to guarantee the engagement of stakeholders for co-creation activities in WP3:

- A call-to-action (CTA) video, a specific video format based on animations and infographics aimed at involving thetarget audience to undertake an action, will be produced to support citizen engagement activities. The CTA will be produced with English voiceover and translated into local languages (Spanish and German) by the Local communication task groups.
- Flyers will be produced with simple and direct messages to inform and engage local citizens about the HOUSEFULsolutions demonstrated into the project, including the SaaS. The flyer will have a common visual identity, but contents may be customised according to the specificity of each Frontrunner building.
- A social media strategy definition, including identification of appropriate channels, messages and specific hashtagsto foster engagement of stakeholders.

Task 7.3. Awareness raising activities and HOUSEFUL identity (Leader: ICONS; Participants: all) [M1-M54]

Subtask 7.3.1 – Visual Identity (ICONS): The project visual identity will be implemented at the beginning of the project (M2), including the project logo, graphic elements and templates for presentations and reporting. A flyer will be produced in M10 to inform relevant target groups about the project, its objectives and expected impacts. A certain number of copies will be delivered to the partners for wide distribution at fairs, conferences and workshops.To ensure local outreach, the Local communication task groups will translate the flyer in local language.
A project web video will be produced at M8 to present the project, its objectives and the circular economy concepts in an easy-to-understand way based on animations and info-graphics, for online distribution via the website, social media, communication portals and platforms.
Subtask 7.3.2 – Website and social media (ICONS, all for content provision). The HOUSEFUL website (in English) will be designed and launched within the first five months from the project's start and it will be used as the primary online communication channel and interface towards different target audiences. Before the launch of the HOUSEFUL website, a simple project landing page will be online (M2) reflecting its visual identity, EU funding references and contact details. Direct link to the SaaS developed in WP6 (Task 6.3) for stakeholders and citizens will be established. In view of supporting local communication and engagement, dedicated sections will be developed in local languages, including updates and local news from the Local communication task groups towards local audiences, stimulating acceptance and engagement. Cross-linking and referencing with other external platforms and initiatives will be supported to increase clustering activities.The project website will be maintained for at least 24 months after the end of the project. Social media will be exploited as an additional communication, dissemination and engagement channel and as a participatory tool to foster dialogue, enhance public understanding, engagement and acceptance of stakeholders and citizens. A specific social media strategy will be developed to assess the most appropriate social media channels for HOUSEFUL.

Task 7.4 Replication-oriented dissemination (Leader: ICONS; Participants: all) [M3-M54]
A two level dissemination strategy will be implemented for replication purposes:
Subtask 7.4.1 Replication between Frontrunner and Follower buildings (Leader: LEITAT; ICONS; Local Task Groups). The Collaborative Community of Housing Experts (CCHE) will engage 10 additional demo-sites for replication of the proposed solutions in Europe. Throughout the execution of the project, dissemination activities and formats will be tailored in view of fostering networking and replication from the Front-runners to the Followers. Specific peer-to-peer learning and knowledge-sharing tools and activities will be set up: 1. dedicated replication-oriented contents, packaged in info-packs, periodic newsletters, detailed technical descriptions 2. Capacity-building activities will consist in the organisation of periodic webinars, where the Frontrunner buildings will update followers on the project progress and results, and two training sessions and study tours (one in Barcelona and another one in Vienna) organised by the Local Task Groups.
The 10 Follower buildings will be selected according to different criteria: 1) common in EU building archetypes; 2) located in different climatic conditions; 3) different socio-cultural-economic profile of tenants; and 4) interest and commitment of building owners to replicate the circular principles demonstrated in HOUSE-FUL frontrunner buildings. Follower buildings can undertake activities that complement and expand related project activities and these could be done in the same timeline and/or beyond HOUSEFUL. In any case the investments associated to the implementation of circular solutions in the follower buildings would be out of the scope of the project and therefore not funded within HOUSEFUL. However, HOUSEFUL will inform Follower buildings about funding and financing options for replicating the solutions demonstrated in the frontrunner buildings. HOUSEFUL will test and implement innovative circular solutions in frontrunner buildings, but with the scope of paving the way for the follower buildings to replicate successful solutions later on in their cities for their own benefit.

Subtask 7.4.2 Dissemination towards a broader stakeholder community (Leader: ICONS; Participants: all) to spread know-how and good practices in other EU cities and foster replication and uptake of the project results. All the potentially interested target groups identified as part of the D&C Plan (also taking into account the inputs from Task 3.1 on stakeholders' identification and the organisations who have already signed a LOI), and representing the potential adopters and replicators of the HOUSEFUL services will be approached through dedicated dissemination activities. The presence of EU and national housing associations in the partnership will maximise the HOUSEFUL results' dissemination potential through their existing communication channels and tools and addressing their communities and networks beyond the ones that will be approached via the project dissemination activities. In addition, the members of the CCHE board will also act as ambassadors of the project concepts and results towards their communities:
The Dissemination tools will include:
- e-Newsletter - a periodic e-Newsletter will be issued every six months and distributed to the stakeholders' database ofregistered users. It will be produced with innovative self-publishing tools enabling content publishing and aggregation from multiple external sources in the format of an interactive and easy-to-read e-zine.
- At least 6 info-packs (on both the specific areas: water, waste, energy and material resources and the implementedsolutions) will be produced and distributed to registered web users and key stakeholders and circulated through the partners' channels and promoted via social media;

- Dissemination webinars – four webinars will be organised on the four vectors and the related HOUSE-FUL solutions(water, waste, energy and material resources).
- HOUSEFUL Best practice book - in electronic and printable format, containing the main project achievements,is foreseen at the end of the project, providing information about the results, best practices, lessons learnt and recommendations derived by the implementation and demonstration of the HOUSEFUL services in the Frontrunner buildings as well as the challenges tackled. It will be widely promoted towards the HOUSEFUL community and CCHE board.
- Scientific and technical publications dedicated journals as well as conference papers will be performed by academicand technological partners. The objective will be to increase peer-to-peer dissemination and knowledge sharing among expert stakeholders in the circular economy area and to position the project al-so in professional and scientific journals and conferences.

Stakeholders relations, networking and clustering:

- Participation in fairs and conferences - HOUSEFUL partners will participate in conferences and fairs;
- Results from developed solutions will be disseminated in 4 technical workshops, covering advances in buildingcircularity on energy (M30) in Barcelona (Spain), waste (M33) in Terrassa (Spain), material (M36) in Netherlands and water (M50) in Vienna (Austria).
- Clustering activities with projects and initiatives on circular economy will be fostered through the organisation of jointdissemination activities, such as, organisation of common events, workshops, production of dedicated editorials;
- A final event (in the format of a workshop, conference, or roundtable) will be organised at EU level by LEITAT, inthe framework of another planned circular economy event, where the results and services developed and demonstrated during the project will be shared with other European stakeholders.

Task 7.5 Public communication (Leader: ICONS, LEITAT; Participants: all) [M1-M54]

This task aims at producing and distributing original HOUSEFUL contents through key and targeted messages to raise awareness, acceptance and impact towards wider audiences. Public contents will be widely distributed through EU and global information multipliers and online media (such as, Alpha Galileo, Physorg, Cordis Wire, youris.com etc.) and other information platforms and thematic portals, as well as through the partners' communication tools and will include:

- At least 8 independent articles and interviews around project-related topics, written by professional journalists will be produced and distributed to information multipliers, online newspapers, dedicated portals and promoted via social media networks.
- HOUSEFUL videos. The audiovisual strategy of the project will include: one Video News Release (VNRs) forTV distribution and four web videos. Focusing on innovative outcomes related to the Front-runners buildings and raising media and public interest the VNR will be produced and distributed to European and Word TV stations through youris.com audiovisual platform, managed by ICONS, and by exploiting the satellite gateways of the Eurovision department of the European Broadcasting Union. One short web video (30 seconds – 1 minute) including animations, infographics and real footage will be produced on the HOUSEFUL demonstration outcomes.
- Press and News Releases: at least 10 press and news releases focusing on specific project issues and milestones,promoting project events and progress, will be produced and distributed. They will be made available to local partners for translation into local languages and distribution through their existing channels.
- Pageflow – a digital storytelling tool, enabling to incorporate many editorial formats (background videos, images,text…) into a richer narrative illustrating and transforming the HOUSEFUL outcomes into market-able solutions.

Task 7.6. HOUSEFUL impact monitoring and assessment. (Leader: ICONS) [M1-M54]

Measuring outreach, community engagement and gathering citizens and stakeholders' feedback is essential to assess the effectiveness of the designed D&C strategy and HOUSEFUL impacts. A continuous monitoring activity of the effects of the widespread distribution of the HOUSEFUL contents on web and social media channels via analytics and dedicated state-of-the-art software tools or through direct data retrieval via the information multipliers' platforms will enable the project to measure its outreach. Outreach measurement is aimed at the definition of the absolute impact data, which provide an input to the Community Engagement Index analysis. The Community Engagement Index (CEI) measures the actual engagement of the project community with the project contents delivered on the internet and on social media. The CEI creates a univo-cal relation between any project content made available on the web and the actual interactions made by visitors coming across that content. The weighted aggregation of all CEI values calculated on single project con-tents will then generate the Project Engagement Index (PEI) representing the engagement capacity of the entire project communication and a fundamental key to the definition of the impacts of the project.

Participation per Partner

Partner number and short name	WP7 effort
1 - LEITAT	
2 - ITeC	
3 - CARTIF	
4 - ALCN	
5 - AHC	
6 - VISUM	
7 - AIGUASOL	
8 - LGI	
9 - GWR	
10 - Turntoo B.V.	
11 - Neues Leben	
12 - CECODHAS	
13 - WE&B	
14 - ICONS	
15 - IDP	
16 - HOMEBIOGAS LTD	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.1	Dissemination & Communication plan	14 - ICONS	Report	Confidential, only for members of the consortium (including the Commission Services)	43
D7.2	HOUSEFUL website	14 - ICONS	Websites, patents filling, etc.	Public	5

D7.3	Project presentation video	14 - ICONS	Websites, patents filling, etc.	Public	8
D7.4	Call-To-Action video	14 - ICONS	Websites, patents filling, etc.	Public	8
D7.5	Flyers (Communication materials for stake-holder engagement)	14 - ICONS	Report	Public	8
D7.6	Project leaflet	14 - ICONS	Report	Public	10

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D7.7	Report on public communication, networking and clustering activities (version I)	14 - ICONS	Report	Public	18
D7.8	Multimedia tools: 1 PageFlow, 1 VNR and 1 web video produced and distributed	14 - ICONS	Websites, patents filling, etc.	Public	50
D7.9	Best Practices book	14 - ICONS	Websites, patents filling, etc.	Public	52
D7.10	Communication and dissemination activities impact assessment	14 - ICONS	Report	Confidential, only for members of the consortium (including the Commission Services)	54
D7.11	Report on public communication, networking and clustering activities (version II)	14 - ICONS	Report	Public	36
D7.12	Report on public communication, networking and clustering activities (version III - final)	14 - ICONS	Report	Public	54

Description of deliverables

Deliverables: D7.1) Dissemination & Communication plan [M4 with internal updates at M18, M30, M43, ICONS] / D7.2) HOUSEFUL website [M5. ICONS] / D7.3) Project presentation video [M8, ICONS] / D7.4) Call-To-Action video (Communication materials for stakeholder engagement) [M8, ICONS] / D7.5) Flyers (Communica-tion materials for stakeholder engagement) [M8, ICONS] / D7.6) Project leaflet [M10, ICONS] / D7.7) Report on public

communication, networking and clustering activities [M18, ICONS] / D7.8) Multimedia tools: 1 PageFlow, 1VNR and 1 web video produced and distributed [M50. ICONS] / D7.9) Best Practice book [M52, ICONS] / D7.10) Communication and dissemination activities impact assessment [M54, ICONS] / D7.11) Updated version of D7.7 [M36, ICONS] / D7.12) Updated and final version of D7.11 [M54, ICONS]

D7.1 : Dissemination & Communication plan [43]

Definition and monitoring of a common and well organized communication strategy, identifying target audiences (closely related to stakeholders' identification and analysis in task 3.1), defining key messages to be conveyed per target group and establishing the most important communication formats, actions and channels, in accordance with article 28 of the GA ("Promoting the action-visibility of EU Funding"). The plan will be updated periodically throughout the project. This deliverable is related to Task 7.1.

D7.2 : HOUSEFUL website [5]

Design and launch of the HOUSEFUL website, where a description of the project (objectives and expected impacts), news, progress and milestones accomplished, direct connection with the SaaS solution developed in WP6 and other related external platforms and initiatives will be available for consultation. This deliverable is related to Task 7.3.

D7.3 : Project presentation video [8]

A short animation video addressing multiple stakeholders with key basic messages, facts and features of the HOUSEFUL project and its foreseen impacts in the circular economy of buildings will be produced to present the project in an easy-to-understand way. The video will be accessible on the website, promoted on social media and used in events to introduce the project. This deliverable is related to Task 7.3.

D7.4 : Call-To-Action video [8]

A specific video format based on animations and infographics aimed at engaging stakeholders in frontrunner buildings and inviting them to undertake actions will be produced to support citizen and stakeholder engagement activities. This deliverable is related to Task 7.2.

D7.5 : Flyers (Communication materials for stake-holder engagement) [8]

The flyer will inform local citizens and stakeholders about the HOUSEFUL solutions demonstrated in the project, including the SaaS, with simple and easy-to-understand messages. The flyer will have a common visual identity, but contents will be customized according to the specificity of each Frontrunner building. This deliverable is related to Task 7.2.

D7.6 : Project leaflet [10]

The HOUSEFUL leaflet, addressing European stakeholders at large, will provide information on the main characteristics of the project and its expected results and impacts. Key key information and messages will be adapted to properly engage the different stakeholders identified in Task 3.1. This deliverable is related to Task 7.2.

D7.7 : Report on public communication, networking and clustering activities (version I) [18]

Report regarding the activities related to communication strategies deployed during the project with respect to public communication, including articles and interviews, videos (D7.3, D7.4 and D7.9), press and news releases and pageflow (D7.8); networking and clustering activities, such as participation in fairs and conferences, activities with other circular economy projects and events attended and organized by the project. This deliverable is related to Task 7.4.

D7.8 : Multimedia tools: 1 PageFlow, 1 VNR and 1 web video produced and distributed [50]

Production of a PageFlow (digital storytelling tool, enabling to incorporate many editorial formats into a richer narrative to enhance HOUSEFUL marketable solutions), a Video News Release to disseminate the project's main outcomes addressing the broad audience via TV media and a web video summarizing the main project outcomes. This deliverable is related to Task 7.5.

D7.9 : Best Practices book [52]

The Best Practices book will provide an overview of the main project achievements, best practices, lessons learnt and recommendations. Contents will be retrieved from the main outcomes of the demonstration activities (described in the deliverables of the different WPs) and will be packaged in a direct and effective graphic layout. The Best Practices book will be available in both electronic and printable format. This deliverable is related to Task 7.4.

D7.10 : Communication and dissemination activities impact assessment [54]

Report on the impacts achieved through the implementation of the Dissemination and Communication Plan and Strategy established at the beginning of the project (D7.1). Continuous monitoring activity of outreach and community engagement (both online - through dedicated indicators - and off-line through qualitative analysis) will provide measures to assess the project impacts in terms of awareness, engagement and acceptance. This deliverable is related to Task 7.6.

D7.11 : Report on public communication, networking and clustering activities (version II) [36]

Updated version of D7.7

D7.12 : Report on public communication, networking and clustering activities (version III - final) [54]

Updated and final version of D7.11 (being D7.11 and update of D7.7)

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

MS12	Definition of the Dissemination and Communication Strategy	14 - ICONS	4	D7.1) The Dissemination and Communication Plan is finalised and approved at consortium level
MS13	Project website online	14 - ICONS	5	D7.2) The HOUSEFUL website is available in internet

Work package number [9]	WP8	**Lead beneficiary** [10]	1 - LEITAT
Work package title	Ethics requirements		
Start month	1	**End month**	54

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP8 - Ethics requirements [Months: 1-54]
LEITAT
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D8.1	H - Requirement No. 1	1 - LEITAT	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D8.2	POPD - H - Requirement No. 3	1 - LEITAT	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1
D8.3	GEN - Requirement No. 4	1 - LEITAT	Ethics	Confidential, only for members of the consortium (including the Commission Services)	3

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D8.1 : H - Requirement No. 1 [1]

The procedures and criteria that will be used to identify/recruit research participants must be submitted as a deliverable by the end of M1, before the start of activities involving humans. The applicant must clarify whether vulnerable individuals/groups will be involved, and the measures to protect them and minimise the risk of their stigmatisation must be submitted as a deliverable by the end of M1, before the start of activities involving humans.

D8.2 : POPD - H - Requirement No. 3 [1]

The informed consent procedures that will be implemented for the participation of humans and for the collection, storage, and protection of personal data, must be submitted as a deliverable by the end of M1, before the start of activities involving humans or collection of personal data. Templates of the informed consent/assent forms and information sheets (in language and terms intelligible to the participants) must be submitted as a deliverable by the end of M1, before the start of activities involving humans or the collection of personal data.

D8.3 : GEN - Requirement No. 4 [3]

An independent Ethics Advisor must be appointed to monitor the ethics issues involved in this project and how they are handled. The Advisor must be consulted at least on the following points: data handling, vulnerable people, consent and recruitment. A report by the Ethics Advisor must be submitted as preliminary scoping document at the end of month 6 to indicate the procedures and ethical requirements in the project. There must then be a final report to be submitted at M54. It is recommended that the ethics advisor is involved in the project throughout.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Methodology for evaluating the circularity level of buildings	WP2	7 - AIGUASOL	26	D2.5) The methodology defined in WP2 will be validated using real scenarios (all Frontrunner buildings) and integrated in the SaaS developed in WP6.
MS2	Desk research of existing circular solutions & services	WP2	7 - AIGUASOL	36	D2.4) Existing market circular solutions
MS3	Social engagement plan	WP3	13 - WE&B	26	Stakeholder analysed process finished for all buildings, related to D3.1) Social engagement strategy.
MS4	Co-created services	WP3	13 - WE&B	36	Engagement workshop organised, related to D3.3.
MS5	Baseline of Frontrunner buildings completed	WP4	3 - CARTIF	12	D4.1) Baseline calculation for 4 Frontrunner buildings
MS6	Execution project documents ready	WP4	3 - CARTIF	26	D4.2) and D4.3) Technical specifications and execution projects of Frontrunner buildings
MS7	Technological solutions ready to be validated at demonstration scale	WP4	3 - CARTIF	24	D4.4) Optimisation of technological solutions (Task 4.3), Financial scheme (task 4.6), and technical monitoring programme (task 4.8).
MS8	Environmental, economic and social assessment of the circular solutions demonstrated	WP5	1 - LEITAT	54	D5.2) Environmental and economic impact assessment; D5.3) Measurement &Verification; D5.4) Social assessment
MS9	Launch of the market study	WP6	8 - LGI	24	D6.1) Market study
MS10	Launch of the SaaS	WP6	8 - LGI	36	D6.3) SaaS ready to be used by stakeholders
MS11	Complete list of potential HOUSEFUL services for stakeholders/end-users	WP6	8 - LGI	54	D6.7) Final Exploitation Plan
MS12	Definition of the Dissemination and Communi-cation Strategy	WP7	14 - ICONS	4	D7.1) The Dissemination and Communication Plan is finalised and approved at consortium level

MS13	Project website online	WP7	14 - ICONS	5	D7.2) The HOUSEFUL website is available in internet

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS14	Financial plan for Demo 2	WP4	3 - CARTIF	18	In task 4.6, delivery of D4.5 due on M18 and updated version on M48 (D4.8). It includes a Financing plan, a progress monitoring of the implementation of the plan, risk/uncertainty assessment and mitigation plans.
MS15	Financial Close for Demo 3	WP4	3 - CARTIF	24	Complementary financial access for demo 3 is completed. Associated deliverables - D4.6 (in M24) and updated version D4.8 (M48).
MS16	Tenants occupy the "Els Mestres" building	WP4	3 - CARTIF	23	They begin to participate in co-creation activities (D3.2) for final definition of technical specifications (D4.2).

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Difficult in identifying the key parameters for quantifying the circularity level of residential buildings	WP2, WP6	A robust and reliable methodology to quantify the circularity level will be developed based on calculation methods and indicators which take into account a life-cycle approach. Partners' expertise will assure the successful definition and quantification of KPIs.
2	A robust and reliable methodology to quantify the circularity level will be developed based on calculation methods and indicators which take into account a life-cycle approach. Partners' expertise will assure the successful definition and quantification of KPIs.	WP3	A robust and reliable methodology to quantify the circularity level will be developed based on calculation methods and indicators which take into account a life-cycle approach. Partners' expertise will assure the successful definition and quantification of KPIs.
3	Low participation of stakeholders in co-creation workshops	WP3	A very thorough engagement strategy, framework and protocol will be developed at the start of the project to ensure a high level of engagement of the project stakeholders. If participation is nevertheless low, the engagement plan will be revised.
4	Delays in demonstration activities	WP4	All actions committed in HOUSEFUL are aligned to the action plans of the building owners/tenants. Long term plans supported by already committed funds and a solid ecosystem of private partners that are strongly committed to the project are the measures to mitigate the impact of this risk. When necessary, funding opportunities will be searched and applied (WP4-task 4.6). Regarding task 4.6, periodic review of the progress monitoring of the imple-mentation of the financing plan will be done, including a risk/uncertainty assessment and realisation of mitigation plans.
5	Planned investments not materialised in demo 1 and 2	WP4	Investment plans foreseen, in which innovative business models accompanying each specific action were included, demonstrate the strong commitment of both public bodies and private companies to implement the activities in three two Frontrunner buildings. Search of new funding sources, review of the financing plan, progress monitoring, risk/ uncertainty assessment and review of mitigation plans.
6	Implementation of NBS does not fit into vegetation period	WP4	Pre-grow plants in nurseries, adjustment of timetable

7	Low biogas production or quality	WP4	Review of lab tests regarding C/N/P ratios according to new blackwater:BW mixture. Testing of alternative inoculums. Review of biogas

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
			cleaning system for proper dimensioning of energy conversion system.
8	Digestate with low nutrient content	WP4	Use of digestate as raw material for cocomposting using gardening waste or kitchen wastes.
9	Low water reuse potential by high evapotranspiration	WP4	Review of NBS design. Selection of plants adapted to regional climate conditions where NBS are implemented. Better control of inlet water flow in NBS.
10	Lack of environmental data to assess the environmental impact of implemented solutions	WP5	In case of not obtaining enough environmental data from BIM, scientific literature and existing LCA databases such as Ecoinvent or GaBi Professional will be used.
11	Not sufficient inputs for the market assessment	WP6	Identify with all project partners a broad list of stakeholders to be consulted
12	Low participation of stakeholders in knowledge sharing activities (i.e. technical workshops, training sessions, study tours)	WP7	A very thorough communication strategy will be developed at the start of the project to ensure a high level of visibility of the project for the targeted audience and will seek for a strong involvement of the stakeholders. If participation is nevertheless low, the communication plan will be revised.
13	Technical failure in the demo-sites	WP4	Several demo-sites have been prepared in order to guarantee an efficient and smooth process of demonstration
14	Demonstrated technological solutions (water-S5,S6; waste- S7,S8,S9) includes high investment and/or O&M cost	WP4	Review of technical specifications, designs and dimensioning of each technology at pilot scale. Evaluation of alternative circular materials for construction. Reviewing O&M costs. Search of additional funding schemes for improved feasibility
15	AnMBR installation in buildings is not permitted due to biogas risk issues	WP4	A risk assessment will be conducted during unit design in order to define whether ATEX equipment is required in order to fulfil installation requirements. Additional CHP unit will be considered to avoid biogas storage, directly transforming it to electricity.

16	Low interest, lack of understanding from the stakeholders' perception (not acceptance) and misuse of new technologies and the SaaS.	WP3, WP6, WP7	The project will adopt a scalable approach to the dissemination and engagement strategy. Thanks to continuous monitoring process of dissemination, communication and social acceptance indicators it will be possible to implement corrective and customised measures (new dissemination tools, 1-to-1 communications, focussed meetings) to ease understanding and knowledge sharing and increase the level of interest among stakeholders
17	The project is ambitious and it could undermine the credibility	WP1, WP2, WP3, WP4, WP5, WP6, WP7	The extensive expertise of all partners of HOUSEFUL guarantees a high chance of success the achievement of project objectives. Moreover, a strong project management progress interaction

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
			(thorough three-monthly updates or throughout extra technical meetings or conference calls) will ensure the suitable progress of the project and thus the feasibility of HOUSEFUL. If any technical delay or problem is occurred during the project, scientific coordination along with advisory board will give the most proper solution for achieving the minor dispersion of the objectives.

18	Planned investment not materialised for Demo site 3	WP4	Several other co-financing schemes will be sought during the project execution including Vienna city funding, funding for being part of international fairs like the International Building Exhibition which will take place in 2022 in Vienna, other H2020 projects, further nationally funded projects with the Austrian Research Promotion Agency, bank loans with European guarantees, etc. If no additional funding would be available, most of the technology demonstration will be incorporated in other ways. For Demo site 3, only the innovative winter garden will be difficult to implement without deep refurbishment. All the other technologies in Demo 3, including biogas, green walls, green house, composting system, incorporation of nutrients, etc., will, in the worst case scenario, be realised in the same scale at the same site without full refurbishment of the building. We will still separate the wastewater and use all the organic solid residues of the whole building for biogas generation and further composting and all liquid residues in green infrastructures and nutrient use in urban farming. The incorporation of rainwater is planned mostly within the innovative winter garden, but if this cannot be installed the rainwater can only be used for conventional garden irrigation. In this case we would lose one innovation, which can also be realised in a smaller scale as double window embedment in about 0,5m2 surface area instead of originally planned 12m2. We can still use this window system as smart window and regulation of indoor air ventilation. It will not have the same cleaning effects as there will be space for three plants instead of 40 plants in the large setup. This innovation is currently being developed in the running project ReCO2ST (Residential Retrofit assessment platform and demonstrations for near zero energy and CO2 emissions with optimum cost, health, comfort and environmental quality. H2020-EeB-05-2017, grant agreement nr. 768576) and will be demonstrated in Denmark and the UK.

Associated with document Ref. Ares(2018)1767830 - 03/04/2018

4. MEMBERS OF THE CONSORTIUM

4.1 Applicants

Participant No.	**1**
Short Name	**LEITAT**
Country	Spain
Organization Type	**RTO**
Website	http://www.leitat.org/english/

Partner Profile

LEITAT is a Technological Center, member of TECNIO and recognized by the Ministry of Economy and Competitiveness, that aims to collaborate with companies and institutions by adding technological value both to products and processes, and focuses its activity on research, development and industrial innovation (R+D+2i).
As **Technological Partner**, the Center is clearly committed to adaptation to transform the technological challenges into economic and social value. Since its foundation in 1906, LEITAT has prioritized its vocation of proximity by strengthening the principles of professionalism and respect to people and environment at the same time. Leitat promotes the implementation of industrial innovation strengthening modernization of the productive structures and the development of new high technology value products in many different sectors.
Leitat, from dynamism, provides flexible management in an "**Open Innovation**" as collaboration and cooperation driving force for technology transfer. LEITAT takes part each year in many projects financed by the regional and national governments, participates in projects co-funded by the European Commission, and develops private R&D projects funded by industrial partners.
The Circular Economy Unit of LEITAT is formed by a multidisciplinary team capable of efficiently assess the demand of organizations on sustainability-related issues. It deals with Sustainable Consumption and Production, Life Cycle Assessment and Ecodesign, Environmental Communication and Training and environmental awareness. LEITAT applies the Life Cycle Assessment (LCA) as an environmental management tool to assess the environmental impacts of a product/process over its entire life cycle, "from cradle to grave", and promotes the incorporation of environmental criteria in design, in order to reduce the negative impact; therefore is fully capable of developing this task successfully.

Role in the project

In HOUSEFUL, Leitat's main role will be the technical and management coordination of the overall project, monitoring the accomplishment of objectives, control of deviations and application of contingency measures and establishment of corrective actions.

From the technical point of view, the main role will consist in testing different technologies at lab scale for the treatment of blackwater and grinded bio-wastes and for the optimisation of dry anaerobic digestion (dAD) for the optimal treatment of the solid fraction from non-segregated wastewater mixed with grinded biowaste (WP4). Several technologies for the stabilisation of AnMBR digestate will be tested at laboratory for optimal selection and scale up to large scale buildings.

LEITAT will be also the leader of WP5 "Environmental, economic and social assessment". LEITAT will assess the environmental performance of the solutions demonstrated in demo-sites (Life Cycle Assessment) and will test also the Level(s) tool created by the European Commission. LEITAT will carry out the Measurement & Verification of the energy savings achieved by applying the International Performance Measurement and Verfication Protocol (IPMVP-Option C) in two demo buildings.

The role of LEITAT will also include the collaboration in other tasks such as support in the definition of the building circularity methodology (WP2), participation in co-creation workshops (WP3), definition of new business models concerning circular economy services and solutions in the housing sector (WP6) and participation in communication and dissemination activities (WP7).

From the management point of view, the Administrative and Budgetary Manager of the project will be Lara Valentín, which will be responsible of the overall project administrative management and coordination as well as the overall budgetary management and coordination. Among other tasks, she will manage delivery and follow-up of administrative and financial documents to the project coordinator, who will transfer them to the EC, will ensure that the required resources are provided by each partner and that their work is consistent with the expenses and will organise project meetings.

Detailed description of LEITAT's role in WP4. *(Total Person-months: 40)* Associated with document Ref. Ares(2018)1767830 - 03/04/2018 **Task 4.1**: Characterisation of the technological dimension of the four vectors (water, was energy and materials) for all the Demo buildings of HOUSEFUL *(2 Person-months).*

Task 4.2: Definition of the technical specifications (TS) for the solutions including technological implementation: AnMBR system (S7.1), valorisation of biogas (S7.4) and composting (S8) *(1 Person-months).*

Task 4.3: Leader of this task, in charge of optimizing water, waste and energetic technological solutions at lab scale. LEITAT will specifically define the best operating conditions for commercial O_3/UV, AnMBR and composting strategies. LEITAT will also be in charge of designing and constructing a compact and transportable AnMBR unit able to treat the organic load rate of 8-10 inhabitant equivalents (blackwater and grinded BW, in optimal ratio previously reported at lab scale) *(10 Person-months).*

Task 4.4: Provide inputs to ALCN about water and waste solutions to adequately define and to update the technical monitoring programme to be deployed during Task 4.8 *(2 Person-months).*

Task 4.5: Preparing technical content of water and waste solutions for the definition of the technical execution projects in Demos 1, 2 and 3 *(2 Person-months).*

Task 4.6: Supporting the development of financing plans for demos *(3 Person-months).*

Task 4.7: Supporting AHC in the tender procedure of Demo 2, providing information about water and waste solutions *(2 Person-months).*

Task 4.8: Management of permits of the AnMBR pilot scale unit for its use in Demo 1 building. Supervision and support of the implementation and validation of water and waste solutions (validated at lab scale) in Demos 1, 2 and 3. LEITAT will follow the commissioning process of the systems installed to ensure that the technical solutions are performing correctly *(18 Person-months).*

Key personnel

Marta Escamilla (F) graduate in Environmental Science from the University of Girona, has a widespread experience in environmental management, especially in Environmental Management Systems for SMEs and large companies, audits, sustainable production, life cycle assessment and ecodesign, Ecolabel, environmental awareness and training workshops among others.

As the Sustainability Division Manager of LEITAT Technological Center, is in charge of coordinating projects and managing initiatives, at regional, national and European level. These projects promote a sustainable development, making a special effort on energy efficiency, rational natural resources consumption, waste management, to improve the environmental performance as well as contributing to substantial economic savings of the industrial sector.

Dr. Julia García-Montaño (F) holds a PhD in Chemistry (Universitat Autònoma de Barcelona) and is the Group Leader of the *Environmental & BIO Technologies Division in LEITAT*. Her professional career has been mainly centred in the study of alternative water and waste treatment & valorisation technologies: Advanced Oxidation Processes (AOPs), advanced electrochemical treatments, biotechnologies, nanotechnologies, membrane processes, high added value products recovery, water/waste-Energy nexus, among others. She has developed two research stays at the Plataforma Solar de Almería, has developed and coordinated several innovative national and international projects, has participated in various congresses and has numerous scientific publications in the water/wastes decontamination area. It is also noteworthy her educational experience as she combined her scientific career with her career as an assistant teacher in several Universities.

Dr. Lara Valentin (F) works as an International Project Manager at Leitat and has experience in managing international projects in the area of Environmental technologies. She holds a Bachelor's Degree in Environmental Sciences by the Autonomous University of Madrid and a PhD in Biotechnology by the University of Helsinki. She did her postdoctoral work in the Department of Chemical Engineering at Autonomous University of Barcelona where she was first introduced to project management.

Dr. Sergio Martínez (M) holds a PhD in Environmental Science (University of Girona) and his career has been mainly focused on the evaluation of biomass as a renewable resource, covering aspects like biomass quantification, studying biomass conversion technologies, the economic feasibility of power generation using biomass, Life Cycle Analysis, physico-chemical and thermal analysis of biomass and the evaluation of atmospheric emissions. He has directly and indirectly participated in several national and international R+D projects. He has also participated in numerous conferences and he published articles in national and international scientific journals demonstrating his experience from projects which has been working in. His scientific experience during his PhD has been combined with teaching experience in the University of Girona, where he punctually participated in some lectures in Environmental Science Degree. After finishing his studies, he worked as plant manager in a

biomass composting plant (treating organic fraction of urban solid wastes and sewage sludge). Nowadays Sergio

Associa
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develops his professional work in the R&D Projects Department in LEITAT Technological Center. He works as a researcher in R+D+I projects for the development of new technologies for solid waste and wastewater treatment and valorisation using advanced technologies such as anaerobic digestion, pyrolysis and pre-treatment technologies. He enjoyed a post-doctoral fellowship (Torres Quevedo, 2011-2013), and today, he works as a Senior Researcher in LEITAT in the preparation, negotiation, execution, dissemination and management of European, national and industrial projects. He is responsible of the Energy Valorisation and Waste recycling Unit in LEITAT.

Marc Torrentellé (M) is graduated in Environmental Science by the Universitat Autònoma of Barcelona and he holds a Master's Degree in Energy Efficiency and the Certified Measurement and Verification Professional (CMVP) qualification. Since 2009, he is a member of R&D Sustainability Division of LEITAT Technological Center. During this time, he has participated in several national and EU projects related to environmental issues. Highlight his involvement in energy efficiency and renewable energy projects, performing energy audits in industries and buildings, defining Action Plans for the energy renovation of buildings and monitoring the energy

savings achieved after implementing energy efficiency measures. He has also conducted studies for the certification of determined products and services to obtain eco-labels, such as EU Ecolabel.

Experience

Related publications:

- Hidalgo C. **Environmental analysis of innovative sustainable composites with potential use in aviation sector – A Life Cycle Assessment Review** SCIENCE CHINA Technological Sciences DOI: 10.1007/s11431-016-9094-y 2017 (Article).
- Escamilla M., Hidalgo C., Riera MR. **Using LCM to create shared value through healthcare and pharmaceutical supply chains** LCM Congres 2017 (Poster).
- Escamilla M., Puigmal L. **Sustainability assessment of innovative photovoltaic panel treatment to convert waste into resources promoting circular economy**. SETAC LCA Case Study Symposium. Montpellier (France) November 2016 (Poster).
- Escamilla M., Villaba R., Puigmal L. **More sustainable textiles and improved laundry processes for mitigating the microplastics impact in the marine environment. MERMAIDS project**. SETAC LCA Case Study Symposium. Montpellier (France) November 2016 (Poster).
- Claret A., Escamilla M., Hidalgo C. **Life Cycle Assessment as a tool to assess the sustainability of innovative materials: the case of second generation superconductors tapes.** SETAC LCA Case Study Symposium. Montpellier (France) November 2016 (Poster).
- Claret A., Escamilla M., Hidalgo C. **Sustainability Assessment of integrated innovative wastewater technologies in Mediterranean tourist facilities (demEAUmed project).** SETAC Europe. 22nd LCA Case Study Symposium. 2015 (Poster).
- Gattringer,H., Claret, A., Radtke, M., Kisser, J., Zraunig, A., Rodriguez-Roda, I., Buttiglieri, G. **Novel Vertical Ecosystem for Sustainable Water Treatment and Reuse in Tourist Resorts.** In: International Journal of Sustainable Development and Planing. Volume 11, No. 3 (2016) 263-274. ISSN: 1742-7601 (paper format), ISSN: 1743-761X (online).
- Claret, A., Antón, A. and Montero, J.I. **Eco-agro-industrial Park in Gran Canaria: conjunction between agricultural and industrial sector**. Ecotech & Tools: Environmental & Integrated Assessment of Complex Systems Conference. 2011 (Poster). **Related projects:**

RUN4LIFE [H2020-IND-CE-2016-17-730285] Recovery and utilization of nutrients 4 low impact fertilizer.

Run4Life demonstrates at large scale an alternative strategy for improving nutrient recovery rates, based on a decentralised treatment of segregated black water, kitchen waste and grey water combining existing wastewater treatments with innovative technologies.

demEAUmed [FP7-ENV-2013-WATER-INNO-DEMO-619116] Demonstrating integrated innovative technologies for an optimal and safe closed water cycle in Mediterranean tourist facilities

Project is aimed to demonstrate and promote innovative technologies for an optimal and safe closed water cycle in Mediterranean tourist facilities, leading to their eventual market uptake. LEITAT is also in charge of testing innovative technological solutions for water treatment.

RESSEEPE [FP7-EeB.NMP.2013-3 EeB.NMP.2013-3] REtrofitting Solutions and Services for the enhancement of Energy Efficiency in public buildings

RESSEEPE aimed to demonstrate innovative technologies to improve the energy performance of buildings.

Infrastructures and facilities Associated with document Ref. Ares(2018)1767830 - 03/04/2

LEITAT has all the necessary infrastructures, laboratories and arrangements to undertake the tasks envisaged in the project and fulfil its specific role. The following are main specific equipments that will be directly linked with activities of LEITAT within the project:

- Physical-Chemical Analysis and microbiological: ICP-MS, Ec-Meter Basic 30+ (ionic conductivity), Karl fisher (H2O content, Coulometer KF 831), TG Q500, DSC Q20 (thermal), Climatic chamber (- 30 < T < 100 ºC, only small device, security), Ionic Chromatography (ion identification), various membrane pilot plants (MF/UF/NF/RO), Total Organic Carbon (TOC) and Total Nitrogen (TN) analyzer (Simazdu).
- Synthesis, Manufacturing & Assembly: Glove box, Inkjet Ardeje, Vacuum Oven Chamber, and ovens (1000ºC, gas), Rotavapor, Plasma Air or atmospheric, Sonicator VCX750, planetary ball milling Pulverisette 7, Flat sheet membrane fabrication equipment, Microscopies, centrifugues, Microbiological cabines.
- Electrochemistry: Potentiostat/galvanostat (Autolab, VMP3 and SP portatil from Biologic, 7 channels 400 mA, 3 EIS, up to 20V, 1 channel 2A, 1 channel 10 A, and connected to glove box).
- Bioelectrochemical reactors for different applications, including, MFC, MECs, MDCS, among others.
- Anaerobic reactors, Methane potential analysers, Portable Biogas analyser (CH_4, CO_2, H_2, O_2), Micro GC for gas analysis.

LEITAT holds license of the main LCA Databases such as **Ecoinvent** and Gabi databases, and the rest of commercial Databases included in the softwares. LEITAT works with different impact methods according to European guidelines: ILCD method, CML, ReCiPe, among others.

Participant No.	**2**
Short Name	**ITeC**
Country	Spain
Organization Type	Private non-profit fundation
Website	www.itec.cat

ITeC Institut de Tecnologia de la Construcció de Catalunya

Partner Profile

ITeC is a non-profit private foundation working in the field of construction. Our **goals** as an organization to support innovation are:

- to be a benchmark institution in terms of information, knowledge, innovation and sustainability applied to the construction sector,
- the provision of technological services to improve the competitiveness of the construction sector agents: organizations, companies and professionals.

The ITeC Board put together the **main representative stakeholders from the construction sector in Catalonia**: regional public bodies, universities, professionals and builders' associations.

MAIN ACTIVITIES:

1. **European Technical Assessment Body** specialised in the assessment and certification of nonstandardised innovative products*. Active Member of the EOTA - European Organisation for Technical Assessment of non-standardised innovative products:
   - European Assessment Documents - EAD. -
   European Technical Assessments - ETA.

Industrialized systems for **façades and roofs, **Fire protective products**, **Internal Partition kits**, **Industrialized Constructions**, Technologies for **Waterproofing**, **thermal and acoustic insulating** materials, **timber** products, **anchors and fixing systems.***

1. Methodologies for a proper **calculation and management of the environment impact of buildings** through their entire life cycle.
2. **Market research (member of EUROCONSTRUCT)** combining the economic approach with our knowledge of the construction market from the technical, legal, environmental and commercial points of view.

Role in the project

Design and demonstration of circular strategies in maintenance operations and analysis of opportunities for the optimal use of materials during the life cycle of demo buildings (WP4). To carry out the BIM models of demobuildings.
Monitoring the environmental impact of the new circular business models using environmental indicators such as energy, water and materials (LCC and LCA).

WP's participation: WP1; WP2 and lead the task 2.3; WP3; WP4 and lead the Action 4.1.1; WP5; WP6; WP7

Detailed description of ITEC's role in WP4. ***(Total Person-months: 34)***
Task 4.1: Providing data (economic) related to the baseline calculation of the circular economy audit. Leader of the sub-task 4.1.1. Material Passports integrating BIM for baseline definition and potential evaluation of deconstruction scenarios. ITeC will be in charge of the BIM models of the 4 Frontrunner buildings and will give technical support to TURNTOO in the Material Passport definition and in the evaluation of the deconstruction scenarios *(10 Person-months)*.
Task 4.2: Supporting technical specifications of solutions S4, S9 and S10. Defining specifications to assure interoperability of Houseful solutions in the BIM models. Design for Disassembly (DfD) evaluation *(9 Personmonths)*.
Task 4.3: Leader of the subtask 4.4.3 'Optimisation of Energy vector' related to the S10 components: Feasibility analysis of innovative active and passive systems potentially installed as improved elements in building; Promoting the design and potential installation of optimal systems to be selected according to circular criteria as well as

their cost (installation, maintenance and renovation cost); Modular façade definition Associated with document Ref. Ares(2018)1767830 - 03/04/2018*(6 Person-months)*.
Task 4.5: Supporting the project documents production (e.g. Bill of quantities -BoQ- of innovative vs conventional solutions related to S4, S9 and S10 in the Spanish demos) *(2.5 Person-months)*.
Task 4.6: List of potential investors interested with the investment opportunity, generating the documentation to explain the investment opportunity at potential investors *(2 Person-months)*. **Task 4.7:** Technical supporting AHC in the tender procedure *(2.5 Person-months)*.
Task 4.8: Technical assistance and supervision of works for solutions potentially installed in S10 *(2 Personmonths)*.

Key personnel

Licinio Alfaro (M). Architect and technical architect, specialized in sustainability, master in architecture, energy and climate. Specialist in the calculation of environmental impact for the life cycle of buildings: process of Construction, use, maintenance and deconstruction.
Project manager of environmental software to calculate waste, CO2 and Kwh in construction and civil works. Experience in management and execution of sustainable building projects. Analysis of environmental aspects in the field of architecture construction, environmental ISO application in construction promoters, specialist in waste management in construction.
Gloria Díez (F). Gloria Díez. Degree in Architecture, School of Architecture of Barcelona, Polytechnic University of Catalonia (ETSAB - UPC). Specialist in construction and building sustainability with 15 years of experience. Expertise area in construction and demolition waste management as well as LCA of buildings and building products. Involved in 15 European and National co-funded projects as project coordinator/researcher, most of them focused on sustainable product development and environmental impact assessment. Member of the Spanish mirror committee CEN/TC 350 'Sustainability of construction works' (since 2004). Co-author of several publications related to sustainability of buildings.
José Lucas (M). Degree in Building Engineering, Barcelona School of Building Construction, Polytechnic University of Catalonia (EPSEB - UPC) and postgraduate in Project BIM Manager, Association of Surveyors, Architects and Construction Engineers of Barcelona (CAATEB). Specialist in modeling, analysis and BIM model management as well as LCA of buildings and building products. He has participated in the elaboration of the environmental information for the database BEDEC and in the development of the environmental management tool TCQiGMA of The Catalonia Institute of Construction Technology (ITeC). He has made more than 10 energy audits of public schools.
Núria Vila (F). Degree in Architecture, School of Architecture of Vallés, Polytechnic University of Catalonia (ETSAV - UPC) and Master in Facility Management, Polytechnic University of Catalonia (UPC). Experience of 25 years defining the state of conservation and maintenance of the buildings. Specialization in the field of maintenance planning and management of buildings with advanced technology as the software DicPla, PGMe and TCQi MNT, doing technical certifications, studying the problems of the concrete with aluminosis. Defining the informatic requirements for the software TCQi MNT. Coauthor in publications regarding Inspection, diagnosis and intervention in façades.
David Sabatés (M). Degree in Architecture, School of Architecture of Barcelona, Polytechnic University of Catalonia (ETSAB - UPC) and Master of Buisiness Administration for IESE Business School. Ten years of expertise advising innovation projects and companies in the field of construction. Specialist in European programs -H2020, LIFE, INTERREG- and mergers and acquisitions -M&A-.

Experience

- BEDEC database. The ITeC database supplies technical, environmental and economic information regarding all kind of elements used in every situation in the construction market (building and civil works, urbanization, rehabilitation, maintenance, security and health, quality control). It is updated every year and since 2004 it has environmental information related to CO2, energy consumption and waste.
- TCQGMA software. The mix of database BEDEC and the environmental software calls TCQGMA developed by ITeC, allows a to have the environmental impact of a construction process. The calculations are generated automatically. It is not necessary to have specific knowledge about the topic. **Related publications:**
- COAC - COAATB - ITeC (G.Díez). Sustainability Practices in Construction. ISBN 84-7853-495-4 (Spanish edition); 45 Pages; 2005
- F. Mañà; A. Cuchí; D. Castelló; G. Díez; A. Sagrera. Sustainability parameters. ISBN 84-7853-4611 (Spanish edition) / ISBN 84-7853-455-5 (Catalan Edition); 96 Pages; 2003
- F. Mañà Reixach; A. Cuchí Burgos; G. Díez Bernabé; C. Orgaz Tejedor. La Cubierta Captadora en los Edificios de Viviendas (Sustainable housing roofs); ISBN 84-7853-436-9 (Spanish edition) / ISBN 84-7853-435-0 (Catalan Edition); 128 Pages; 2002 **Related projects:**

- Project name: Development of the energy efficiency indicator TCU (Total Cost for the User) Associated with document Ref. Ares(2018)1767830 - 03/04/2018

  Description: The TCU takes into account the energy cost of the construction product during the useful life cycle. It is a data which add to the cost of acquisition. It allows to have a complete vision and choose the product with more criteria.
  Year: 2017
  Customer: ICAEN

- Project name: Environmental impact in public tenders.
  Description: Development of tools which modeling the building and allow to know the environmental impact and provide improvements. It has in consideration the CO2 emisions, energy consumption and water for each construction elements.
  Year: 2015
  Customer: private

- Project N°: LIFE98 ENV/E/000351
  Project Title: Programme of technical actions to encourage the evaluation, minimization and selection of waste originated at the work sites during construction and demolition.
  Year: 1998
  Amount of the co-financing: 196.815,58 €
  Duration: 1998-2000
  Coordinating beneficiary: ITeC (ES)
  Other partners involved: GREMIO DE ENTIDADES DEL RECICLAJE DE DERRIBOS, GERD (ES); DRAGADOS Y CONSTRUCCIONES SA (ES); CONTROL DEMETER SL (ES).

- Project N°: LIFE03 ENV/E/000150
  Project Title: RECONS - Reducing Construction Environmental Impact Year:
  2003
  Amount of the co-financing: 483,125.00 €
  Duration: 2003-2006 Coordinating beneficiary: GREMI DE CONSTRUCTORS D'OBRES DE BARCELONA I COMARQUES (ES)
  Other partners involved: ITeC (ES); DEPARTAMENT DE MEDI AMBIENT I HABITATGE DE LA GENERALITAT DE CATALUNYA (ES); UNIVERSITAT POLITÈCNICA DE CATALUNYA, UPC (ES); DEP. DE POLÍTICA TERRITORIAL I OBRES PÚBLIQUES DE LA GENERALITAT DE CATALUNYA (ES)

- H2020 Project Grant Agreement number: 636709
  Project acronym: HOMESKIN

  Year: 01/02/2015

  Amount of the co-financing: 4.517.567€

  Duration: 2015-2017

  Coordinating beneficiary: ENERSENS, Dr. Brice FIORENTINO

  Other partners involved: ITeC (ES); USTUTT (GE); ARMINES (FR); Trocellen (IT); Parex Group (FR); FLAG (IT); CEA (FR); SORANE (SW); Wavestone (LU).

Infrastructures and facilities

Office building of 3 floors where 75 workers are developing their tasks. Advanced software to modelling and managing the whole life cycle of a construction: Data Base of materials with embedded information economical, environmental, brands, etc.-; Software to elaborate documents and managing construction processes; Software to managing the phase of use of the constructions.

Participant No.	**3**	Associated with document Ref. Ares(2018)1767830 - 03/04/2018
Short Name	**CARTIF**	

Country	Spain
Organization Type	Research and Technology Org.
Website	http://www.cartif.com/en/

Partner Profile

Fundación CARTIF is a leading Spanish Applied Research Centre in terms of R&D and technology transfer activities created in 1994 formed up by some 150 researchers.
CARTIF participation in international activities is promoted through the International Projects Department and has been materialised in the current participation in 34 ova n-going international projects, including 16 projects from H2020 (4 of them coordinated) and 7 FP7 projects.

CARTIF has a wide experience with large scale demonstrations in buildings, since it coordinates the following H2020 Smart Cities projects**,** with several big demonstration sites in each of them: **mySMARTLife** "Smart Transition of EU cities towards a new concept of smart Life and Economy", **REMOURBAN** "REgeneration
MOdel for accelerating the smart URBAN transformation" (www.remourban.eu) and **URBAN GreenUP** "New Strategy for Re-Naturing Cities through Nature-Based Solutions". Also CARTIF takes part as a partner in the H2020 Project **SmartEnCity** "Towards Smart Zero CO2 Cities across Europe" and in the project **MAtchUP** "MAximizing the UPscaling and replication potential of high level urban transformation strategies".
Other coordinated Smart Cities FP7 projects are: **CITyFiED** "RepliCable and InnovaTive Future Efficient Districts and cities" and **R2CITIES** "Renovation of Residential urban spaces: towards nearly zero energy CITIES". CARTIF is one of the founding members and main promoters of the Smart City VyP initiative carried out in the Spanish cities of Valladolid and Palencia (http://www.smartcity-vyp.com).

In the area of biogas, CARTIF has developed several projects related with the **cleaning and upgrading of biogas**, such as **BIOFUEL_FOR_COOKING** (Spain+India) and **LIFE REVAWASTE** "Demonstration of an integral and sustainable system for multi-waste recycling and valorization" (EU). In order to remove CO2, adsorption and membrane processes has been used, including membrane selection, optimization of the operating conditions, and pilot plan design and construction.

Role in the project

- WP2 Circular profile & baseline scenario of Demo Sites ○ Calculation of Circularity profiles and baselines of Driver buildings.
- WP4 Large Scale Demo Buildings ○ Overall coordination and specially implementation of demos ○ Monitoring programmes
    - Biogas cleaning: Biogas generated in the anaerobic treatment will be treated in order to remove undesirable elements (as hydrogen sulphide, water traces, solid particles, siloxanes and VOCs) that could entail problems during biogas combustion.
    - The cleaning process will include the following steps: cooling to reduce humidity as well as partial removal of VOCs and siloxanes, gas washing to reduce acid gas (H2S) and ammonia (NH3) and complete removal of siloxanes and VOCs by activated carbon.
- WP6 Monitoring Sustainability and Socio-economic assessment ○ Measurement & Verification of the energy savings (task 6.4) ○ LCA
- WP7 Communication and Dissemination for social and market acceptance ○ Participation in scientific and technical congresses and dissemination events

Detailed description of CARTIF's role in WP4. ***(WORK PACKAGE LIDER) (Total Person-months: 34)*** **Task 4.1:** Task Leader. Audit performance of the initial status of the four demo buildings in all four vectors: water, waste, energy and materials from the technological point of view.
In order to align technical specification of all solutions to the further task of standardization under IFC4 (subtask 4.1.1), CARTIF will participate in producing the detailed technical specification of all solutions with technological character behind the Integrated Systemic HOUSEFUL Service (Material (S3, S4), Water (S5, S6), Waste (S7, S8, S9) and Energy (S10, S11)).
Collaboration in subtask 4.1.1, in two main aspects: a) specification (aspects related to standardization under IFC) of Material Passports and b) exporting to IFC the four Frontrunner buildings modeled with BIM. As far as

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each constructive element of this model will incorporate specific KPIs to analyse the circularity of buildings and facilitate proposals for improvements focused on the building circularity, all this information should be assured in IFC models.
CARTIF will be in charge of standardizing under IFC4 the materials information used in all four Frontrunner buildings as well as the BIM produced *(7 Person-months)*.
Task 4.2: Task Leader. Supervision of the different solutions designed. Definition of the technical specifications required for the proper demonstration of HOUSEFUL solutions.
4.2.2: CARTIF will be in charge of each solution that will be specified under the standard IFC to assure the interoperability to include such solutions in the BIM models *(5 Person-months)*.
Task 4.3: CARTIF is in charge of optimizing biogas cleaning and upgrading technological solutions at lab scale and implementation of these solutions in Demo 2 and Demo 3 *(8 Person-months)*.
Task 4.4: CARTIF will proceed with a simultaneous real-time monitoring of CH_4, H_2S and CO_2 after the cleaning stage in order to confirm that the gas stream fulfils the quality requirements demanded. CH_4, H2S will be measured by dual wavelength infrared cell with reference channel, while H2S will be measure by electrochemical cell.
In charge of linking the BIM server (IFC compliant) to the dynamic data repository (BACN standards compliant) to show all detailed information by means of a dynamic BIM visualisation tool *(8 Person-months)*.
Task 4.5: Task Leader. Production of the execution project documents. Collaborating on the Material Passports and BIM models enrichment *(3 Person-months)*.
Task 4.8: Supporting the implementation and validation of waste solutions (validated at lab scale) in Demo 2 and Demo 3 *(3 Person-months)*.

Key personnel

M Dolores Hidalgo Barrio, pHD, **(F)** is currently R&D Projects Scientific Manager at the Environment and Biofuels Area in CARTIF Technology Centre and associated researcher of the ITAP Institute (University of Valladolid). She is also Editor in Chief of two international journals. Dr. Hidalgo studied Chemical Engineering at Valladolid University where she received her Doctor degree in 1999 (theme: anaerobic digestion). She has a degree in Environmental Technology received in 1993 and a degree in Finances received in 2004. Coordinator of numerous research projects, she is the author of over 50 articles, more than 70 papers at international conferences, organizer of a number of workshops and inventor of 3 patents. Dr. Hidalgo has been working 17 years in the management, treatment and valorisation of organic waste.

Miguel Ángel García-Fuentes (M): Architect and MSc Arch, both from the University of Valladolid (Spain). He participated as Construction Manager in the self-sufficient Urcomante Solar House project presented to Solar Decathlon Europe 2010. Currently he is working as Project Manager and Researcher at the Energy Division of CARTIF Technology Centre since 2012, in European RTD Projects in the framework of Smart Cities and Nearly Zero Energy Cities, mainly related to energy performance improvement in new buildings and building/district renovation, Building Energy Performance Simulation (BEPS) tools and Building Information Modelling (BIM), focusing the gap of data interoperability. He has participated in the coordination teams of the EU-FP7 funded BaaS (Building as a Service Ecosystem) and R2CITIES (Renovation of Residential Urban Spaces: towards nearly zero energy CITIES) projects, and currently is the Technical Project Manager of REMOURBAN (Regeneration Model for accelerating the smart urban transformation) H2020 Lighthouse project.

Francisco Corona (M), Master of Science in Chemical Engineering by the University of Salamanca. Master of Engineering Research Processes and Systems. Master of Environment and Renewable Energy. He belongs to CARTIF Foundation since 2008 and he is a researcher of the Chemical Processes and Biofuels Area, participating in research tasks of processes and chemical products optimisation, waste valorisation, biogas process development, nutrient recovery, biomass pyrolysis and gasification, second generation biofuel, biorefineries and bioethanol and biodiesel obtaining processes. Besides, he has participated in regional, national and international proposals and projects. In addition, he has participated in the organisation of courses, workshops and seminars of waste valorisation as energy, bioproducts, etc. and he has participated as reviewer of JCR manuscripts.

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Dr. Ali Vasallo (M) graduated from the University of Valladolid in 2004 with a First Class Honours degree in Industrial Engineering with the specialization in Theory of Structures and Industrial Buildings. He completed his PhD Thesis in 2009 in the field of vortex-induced vibrations of slender structures. He works at the Energy Division in CARTIF Technology Center. He is a project manager and researcher currently involved in several initiatives dealing with Energy Efficient Buildings, Smart Cities and Dynamic behaviour of slender structures. He is regularly attending events in connection with several Technology Platforms from the fields of building efficiency, district heating systems and Smart Cities. He also participates as Project Coordinator of the EU-FP7 funded CITyFiED (Replicable and Innovative Future Efficient Districts and cities) project. Together with his activities in CARTIF Technology Center, he has been working in the School of Industrial Engineering of the University of Valladolid as Associate Professor of Industrial Constructions, Metallic Structures and Strength of Materials. He was appointed a Visiting Fellow at the School of Engineering at the University of Warwick (2011).

Laura Pablos (F). Chemical Engineer since 2008 and Master of Engineering since 2011, both from the University of Valladolid (Spain). Executive Master in Innovation from Spain's School for Industrial Organisation (Madrid) since 2011. Advanced knowledge and more than six years of a great experience in carrying out Life Cycle Assessment (LCA) studies, Carbon and Water footprint, Eco-labels, Environmental Management Systems, Environmental criteria for Green Procurement and Ecodesign (ISO 14006). Expert in assessing the environmental sustainability of different solutions via LCA studies, offering an added value in more than fifteen Regional, National and European projects, addressing the sustainability of the products and production processes involved. Three of them are: i) Eco-design for Eco-innovation. The Green Car case. Funded by STC SUDOE (Green-Car Eco-Design, Ref.:SOE2/P1/E326) from 2011 to 2013; ii) Sustainable design applied to industrial and business competitiveness. Funded by: Ministry of Industry, Tourism and Commerce of Spain (DGPYME – INNOEMPRESA Programme, SUPRA-REGIONAL SUP-030300-2010-12) during 2010; iii) Introduction of environmental criteria in the purchase of several equipment in airport terminals. Private client: TECNOMA, in collaboration with AENA, during 2010. Diverse publications in different workshops and relevant platforms.

Experience

Related publications:

- Hidalgo, D., Martín-Marroquín, J., Corona, F. (2017). "Biogas today, biomethane tomorrow". DYNA (Q4). DOI: dx.doi.org/10.6036/8482
- Hidalgo, D., Martín-Marroquín J., Corona, F. (2017). "The effect of feed composition on anaerobic codigestion of animal-processing by-products". Journal of Environmental Management (Q1). DOI: doi.org/10.1016/j.jenvman.2017.06.033
- Hidalgo, D., Martín-Marroquín, J., Corona, F. (2016). Biogas upgrading technologies: a mini-review. Advances in Chemical Engineering and Process Technology, Vol. 1 (1), pp. 1-10
- Hidalgo, D., Martín-Marroquín, J. (2015). Biochemical methane potential of livestock and agro-food waste streams in the Castilla y León Region (Spain). Food Research International (Q1). DOI: 10.1016/j.foodres.2014.12.044
- García-Fuentes et al. (2014). Residential áreas retrofitting towards nearly Zero Energy Districts (nZED). A case study: Valladolid-Cuatro de Marzo district. Poster. Proceedings of World Sustainable Building 2014. Barcelona, Spain.

Related projects:

- "Production of renewable natural gas from wet biomass and all types of organic waste for cooking and micro CHP" (BIOFUEL_FOR_COOKING). Spanish+Indian funds, Jan-17/Dec-18 - anaerobic co-digestion of organic waste streams + biogas cleaning and upgrading. Coordinated by CARTIF.
- "Valorization of pig carcasses through their transformation into biofuels and organic fertilizers" (LIFE VALPORC project). EU funds. Oct-14/Sep-17 – anaerobic co-digestion of organic waste streams.
- "Demonstration of an integral and sustainable system for multi-waste recycling and valorization" (LIFE REVAWASTE project). EU funds. Oct-13/Dec-16 – anaerobic co-digestion of organic waste streams – biogas cleaning. Coordinated by CARTIF.
- "Smart Transition of EU cities towards a new concept of smart Life and Economy" (mySMARTLife) (H2020SCC-1-2016-2017 - Smart Cities and Communities **lighthouse project**). EU funds. Nov-16/Dec-21 Urban Transformation Strategy to support cities in the definition of transition models, as a suitable path to reach high level of excellence in its development process, addressing the main city challenges and progressing to the smart people and smart economy concepts. www.mysmartlife.eu. Coordinated by CARTIF.

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- "RepliCable and InnovaTive Future Efficient Districts and Cities" (CITyFiED) (FP7 Smart Cities). Coordinated by CARTIF. CITyFiED project will concentrate its activities to develop a systemic vision and strategy to adapt European cities and its urban ecosystem into smart, sustainable and inclusive cities of the future. www.cityfied.eu

Infrastructures and facilities

- 1.2 m^3 two-phase anaerobic digestion pilot plant for off-site experiments.
- Lab scale membrane contactor for biogas upgrading assays.
- Thermostatic chamber for biodegradability test.
- Fully equipped chemical labs.

Participant No.	4	
Short Name	**ALCN**	
Country	Austria	
Organization Type	**SME**	
Website	http://www.alchemia-nova.net/en/	

Partner Profile
Scientific expertise on natural resources and agricultural by-products: Development of cascading utilisation strategi renewables, evaluation of renewable and seondary resources; Phytotechnologies: biomining, constructed wetlands, water and air treatment; Circular economy: Cradle to Cradle® projects including certification as official partner, resource recovery; Sustainable buildings: Including ecosystem services into the building, sutainable materials; Sustainable product design: Design criteria, made for recycling; Research project conception, coordination and realisation: H2020, LIFE, ERA-NET, national or regional funds, etc.
Role in the project

ALCN will coordinate the Austrian demonstration sites; organise all stakeholder processes concerning the Austrian demonstration sites; closing the nutrient loops in the demonstration buildings in Austria and Spain; be responsible for the implementation of nature-based solutions in Austrian and Spanish demo buildings; define new business models concerning circular economy services and solutions in the housing sector; form new value chains; evaluate circular economy financing opportunities. Involvement mostly in WP4, but also other WPs regarding circularity, dissemination, exploitation.

Detailed description of ALCN's role in WP4. ***(Total Person-months: 42)***

Task 4.1: Characterise social dimension on the 4 vectors (waste, water, energy, materials) in the demo buildings with focus on demo building 3. Provide necessary information and access for BIM Modeling and Material Passport creation for Demo building 3 *(4 Person-months)*.

Task 4.2: Definition of the technical specification (TS) for the solutions including technological implementation: Modifying greywater discharge system for treatment and reuse using innovative NBS (TS 5.2.1), Rainwater pretreatment for reuse using NBS on facade or roof (TS 5.4.1, 5.4.2) and integration in building (TS 5.4.3), separation of liquid/solid fraction with NBS indoor and outdoor (TS 6.1), design and integration of dAD system (TS 7.2) *(5 Person-months)*.

Task 4.3: Leader of the subtask "Optimisation of water vector": Optimisation of NBS (as green facade or green roof) for greywater and rainwater treatment for reuse in toilets in Demo buildings 1 and 2. Optimisation of NBS at laboratory (TRL5) before scale up (at TRL 7) and integration by ALCN (with the support of IDP in Task 4.8). Selection of plants, analysis of their performance according to (Spanish and Austrian) climate conditions on the treatment of selected WW. ALCN will provide calculations and experimental implementations concerning the size of treatment systems (i.e. m^2 per litre influent), retention times, dilution factors and plant consortia.

Subtask "Optimisation of waste vector": Gas characterisation (outlet from biogas plant) in Demo 3 will be performed by ALCN. Quality of biogas will be monitored by ALCN with the supervision of HOMEBIOGAS. Twice a year tank will be emptied and the sludge with the foreseen outlet and a pumping system directly sent to the composting unit. Installation of proposed equipments will be performed by CARTIF/ALCN in Demo 3. A compost cultivator prototype will be developed by ALCN at TRL6 and optimised for composting fresh gardening wastes before mixing with dAD digestate at large scale in Demo 3. Optimisation of optimal ratio will be performed according to compost characteristics evaluated by ALCN.

Subtask "Optimisation of energy vector": Co-creation between AIGUASOL and ITEC with stakeholders (including AHC, ALCN and NEUES LEBEN) will promote the design and potential installation of optimal systems to be selected according to circular criteria as well as their cost (installation, maintenance and renovation cost) *(14 Personmonths)*.

Task 4.4: Task Leader. Specifying a monitoring program to ensure a correct impact assessment of the demonstration. Selection of suitable sensors (photometric sensors, online microbial sensors, humidity sensors, temperature sensors, etc.). Regarding solutions designed for WW treatment (S5 and S6), ALCN will include sensors for the evaluation of water quality after NBS treatment.

Data acquisition and repository of data according to the KPIs will be done in cloud-based systems to oversee the data sets online at any time *(5 Person-months)*.

Task 4.5: Preparing technical content of water and waste solutions for the definition of the technical execution projects in Demos 1, 2 and 3 *(3 Person-months)*. Associated with document Ref. Ares(2018)1767830 - 03/04/2018 **Task 4** Development of a financing plan, progress monitoring of the implementation of the plan for demo building 3, risk/uncertainty assessment, mitigation plans and the evaluation of different potential scenarios and impacts for t project *(4 Person-months)*.

Task 4.7: ALCN with the support of GWR will be in charge of the selection process of companies for renovation works according to prize and trust *(2 Person-months)*.

Task 4.8: All permits for technological implementation of proposed solutions in Demo 3 will be sought for approval. Once approved, ALCN in Austria will be in charge of the technical assistance and the supervision of the works related to the technological innovation shown in S5, S6, S7, S8 and S9. ALCN will periodically report to the consortium the progresses and potential delays or difficulties. ALCN will follow the commissioning process of the system installed to ensure that the technical solutions are performing correctly *(5 Person-months)*.

Key personnel

Johannes Kisser MSc. (M), Degree in Chemical Engineering. Dissertation about recovery of metals from plants. Coordinator of Austrian research project on metal recovery from waste with plants. Environmental analysis at the Environmental Protection Agency in Austria. 3 Lectureships in Austrian Marketing University of Applied Science about sustainable product design. Working at alchemia-nova since 2002, since 2010 managing director of the re-founded alchemia-nova OG and from 2014 onwards alchemia-nova GmbH.
Theresa Heitzlhofer MSc. **(F),** Human ecologist. Degree in Biology/Ecology. Focus topics bio-geo-chemical nutrient cycles, microbial ecology, human ecology. Experience in Education for Sustainable Development, Participatory Approaches. Since 2017 staff member at alchemia-nova.
Gianluca Vassallo BSc. **(M),** Degree in Geography. Focus topics resources criticality analysis, GIS location analysis for renewable energies and hydrology. Master student in Natural Resource Management and Ecological Engineering at the University of Life Science Vienna (BOKU) and the Czech University of Life Science (CULS). Since 2017 staff member at alchemia-nova.
Andrea Zraunig MSc. **(F)**. Biologist (phytochemist). Many years of experience in the isolation and characterization of complex natural compounds from plants. Natural ecosystems in Thailand, Costa Rica, Ethiopia, Tanzania and Sumatra. Since early 2014 permanent full time staff member at alchemia-nova, responsible for water treatment with plants, botany related issues, natural compounds, and management of laboratory tests. **Heinz Gattringer MSc. (M),** Studies and degree in agriculture/horticulture from Geisenheim University in Germany. From 1996 to 2007 more than 10 years practical experience in commercial plantations and agricultural enterprises in South America (Ecuador), as well as international development cooperation. Since 2007 researcher at alchemia-nova in Vienna. Responsible for renewable resources, sustainable material sourcing and production. Commercialisation of the vertical ecosystem®.

Experience

Related publications:

Andrea Zraunig, Heinz Gattringer, Johannes Kisser, Manfred Radtke; Building-Integrated Indoor Vertical Ecosystem for Treatment and Recycling of Greywater; Sustainable Sanitation Practice, Issue 23/2015;
H. Gattringer et al., NOVEL VERTICAL ECOSYSTEM FOR SUSTAINABLE WATER TREATMENT AND REUSE IN TOURIST RESORTS; Int. J. Sus. Dev. Plann. (2016) 1–12; in evaluation
Patent AT516363: Gradual vertical constructed wetland for purifying waste water, industrial waste water and the like, H. Gattringer, J. Kisser, M. Radtke, A. Zraunig, 2015
H. Gattringer et al., "gugler! build & print triple zero - Integration of the Cradle to Cradle production into the flagship project Gugler", (2010), nachhaltigwirtschaften.at

Related projects:

- **Green INSTRUCT.** Green intergated sructural elements for retrofitting and new construction of buildings. H2020-EEB-04-2016, project no. 723825, 2016-2020 Greywater and storm water management, reduction of air pollution, mitigation of heat islands effects, additional insulation of the building, noise reduction, with facades. Pre-fabricated structural elements for water recovery using almost only recycled materials.
- **ReCO2ST.** Residential Retrofit assessment platform and demonstrations for near zero energy and CO2 emissions with optimum nature-based solutions. H2020-EEB-2016-2017/H2020-EEB-2017 Indoor air purification and climate control with cost, health, comfort, and environmental quality.
- **Gugler! Build & print triple** Federal Ministry for Including in-house water treatment of industrial

Associated w
document Ref. Ares(2018)1767830 - 03/04/2018

zero.	Transport, Innova-effluents with plants. Cradle to Cradle design of tion and Technol- building. Using only sustainable resources with ogy, Programme full dismantlability.	
	"Houses of tomorrow plus", 2010ongoing	
• **Wohnwagon**	Feasibility Study, FFG Austria, 2015	Closing resources-, nutrient- and energy cycles for mobile homes
• **demEAUmed.**	Demonstrating FP7 Water integrated innovative technologies Inno&Demo project no. 619116, 2014-2017	Development and validation of the vertical ecofor an optimal and safe closed system "vertECO", set-up in Hotel Samba, Lloret water cycle in Mediterranean tour- de Mar, Spain ist facilities
	Project co-financed by Linz (city), Up-	
• **Loop Linz.**	Model project with per Austria (state), retail stores, indus- trial companies, IBAW e.V.	Innovative packaging for supermarket products. Investigating strengths and weaknesses of the bioplastics. innovation.

Infrastructures and facilities

alchemia-nova GmbH runs a fully equipped laboratory for extraction and testing of all kinds of natural compounds as well as a vertical ecosystem laboratory test unit with electronic on-line sensors. A portable photometer and several offline sensors are also available for water quality measurements. A green façade element that cleans water is currently being developed. Modern computer and communications equipment is available to allow for international cooperation, teleconferences, cloud based management tools, and data sharing systems.

Participant No.	**5**
Short Name	Agència de l'Habitatge de Catalunya
Country	Spain
Organization Type	**Public**
Website	www.agenciahabitatge.cat/

Partner Profile

The Agència de l'Habitatge de Catalunya (Housing Agency of Catalonia- AHC) created by the Law 13/2009, of 22th July has the purpose to endow the Catalan Government with an agile administrative tool to facilitate the management of all public actions in terms of housing.
AHC has the aim to develop, implement and manage housing polices in Catalonia. The activity of the Agency is divided in three areas: Promotion of housing, Social Programs of Housing, Rehabilitation and Quality of Housing. It manages a social housing stock of 14.704 dwellings. It has established experience in retrofitting of the social and residential buildings. It has developed, during the past years, more than 400 refurbishment projects; 20% of them related to improvements of building envelopes. Owing to its past and ongoing experience in building retrofitting, AHC has played an important role in compiling the main refurbishment and buildings energy management best practices and barriers in Spanish context (legal, technical, social and financial barriers and best practices in refurbishment activities). In this sense, AHC staff has a large experience in social work as they are permanently in contact with public housing users.
AHC as a regional housing legislator, public housing manager and renovation promoter is a key stakeholder in Spain. It is a member of energy efficiency platform of Social Hosing Promoters of Spain (called AVS), Environmental Product Declaration (EPD) Spain Committee and GBC in Spain. It is a member of the Circular Economy think-tank group of the Catalan Government.
AHC is able to catalyse change in existing buildings and also influence behaviour amongst residents and housing organisations. It has direct influence through their realm of operation (housing stock, client portfolio etc.) and the capacity to influence other through involvement in relevant networks and associations. Also, as members of CECODHAS, AHC is able to access data from other organisations in order to extrapolate data and results for policy promotion purposes and also to influence other similar, private sector, through discussion, debate, and dissemination.

Role in the project

Tasks Assigned and Role in the Project: Provision of 2 demo sites (social housing buildings) and management of demonstration activities at these sites; communication with key stakeholders; dissemination to policy makers and politicians.

Detailed description of AHC's role in WP4. ***(Total Person-months: 20)***
Task 4.1: To introduce the required data in the Madaster platform to generate the Material Passport for Demo 1 and Demo 2: complete description of all product, components and raw materials. Adaptation of the BIM model for Demo 1 and Demo 2. Contribution in deliverable D4.3 *(5 Person-months)*.
Task 4.2: Facilitate the inspection and tests in the two demonstration buildings and the technical specifications for the recycling / reuse of materials at the end of their useful life *(2 Person-months)*.
Task 4.3: Provide knowledge for optimisation of Energy vector in Demo 1 and Demo 2 with a validated multicriteria model for energy renovation taking into account the different steps in energy refurbishment projects (audit, monitoring, measure selection, project development and implementation and energy performance validation) *(2 Person-months)*.
Task 4.6: Study the complementary financial mechanisms for Demo 2. For the intervention calendar, Demo 1 will serve as background for the definition of mechanisms in Demo 2 *(2 Person-months)*.
Task 4.7: Task Leader, in charge of adapting the tenders to the innovations of the "HOUSEFUL solutions S5, S7, S8 and S11" that will be implemented in the Demo 1 and Demo 2. Collaborate in the implementation of the solutions / services proposed during the useful life of the buildings and study the adaptation of the usual management of public administration *(2 Person-months)*.
Task 4.8: Management of the rehabilitation works, signature of the works certifications and responsible for ensuring the correct implementation and control of the HOUSEFUL solutions by the contractors *(7 Personmonths)*.

Key personnel
Ms. ANNA MESTRE (F). Technical Architect, Master in Rehabilitation Techniques of the Architectonic Patrimony. Postgraduate in Environmental Management. Responsible of External Action Service of the Housing Agency of Catalonia. Project coordinator of RELS –ENPI project, and Technical person in charge of an agreement of collaboration with Tunis. From February 1990, Technical Architect in Architectural Projects Department from Adigsa developing refurbishment projects from different social housing quarters in Catalonian region. **Ms. ANNA PUJOL (F).** Building engineer. Project developer of refurbishment project in AHC with more than 16 years of experience. She has been involved of the implementation technical team of the RESHAPE- IEE and RELS-ENPI MED projects. **Mr. JOSEP LINARES (M).** Technical Director of the Housing Agency of Catalonia, Head of Operating Unit of Housing Refurbishment and Improvement. Teacher in maintenance and rehabilitation issues in Institut Gaudí of Catalonia; co- director and teacher of Ecohabitat postgraduate in Elisava University (Barcelona); teacher of edification engineering in Elisava University (Barcelona); teacher in rehabilitation work and accessibility in SERT School of Professional Association of Technical Architects of Catalonia (COAC). Member of the Judicial Experts Association in Catalonia. **Ms. CRISTINA CARDENETE (F).** Degree in Environmental Sciences, Master's degree in Energy efficiency and renewable energies, Master's degree in Local management and civil service. Project manager of European and research projects, driving strategy implementation, by piloting and coordinating new international initiatives. Horizon scanning for new international opportunities (or threats), coupled with research and analysis on select topics to enable better delivery on the company strategy.
Experience

Associate
with document Ref. Ares(2018)1767830 - 03/04/2018

Related publications:

Associated with document Ref. Ares(2018)1767830 - 03/04/20

- "Why is a strategy for building energy renovation necessary in the Mediterranean?", The Parliament magazine, Issue 28, September 2013.
- The Catalan Strategy for Energy Renovation of Buildings has been drawn up as part of the MARIE European Project. This is due to the commitment of the Regions forming part of this project to achieve by December 2014 a long-term strategy to mobilise the renovation of public and private residential and tertiary. It was approved by Catalan Government on December 2013 and an Executive Committee was set up to implement it as a collective body in the department responsible for Housing. The Strategy Road Map aims to achieve UE's settled basic targets in 2020.
- The RELS methodology is a validated multicriteria model for energy renovation of Mediterranean buildings taking into account the different steps in energy refurbishment projects (audit, monitoring, measure selection, project development and implementation and energy performance validation), which has been implemented in 2 buildings of the AHC. Apart from the 20-20-20 EU objectives, it is focused on social dwellings and it incorporates the human behaviour as an added value. In the regions where the methodology has been implemented (Tunisia, Italy, Spain) it has figured out a singular context: most of the families cannot afford an acceptable comfort in their homes due to their low-income conditions. Indeed, the influence of users is important in the final energy performance of their homes.

Related projects:

- **4RinEU (H2020)** [Started: 01-10-2016 (48 months)] is about Robust and Reliable technology concepts and business models for triggering deep Renovation of Residential buildings in EU. AHC will contribute by following the implementation of 4RinEU deep renovation packages and providing support and feedbacks for the definition of business models, investigation of financial mechanisms and analysis of replication potential starting from our building stock.
- **RELS (ENPI CBC MED)** [Finished: 31-12-2015 (50 months] main objective was to develop a methodology for energy renovation in social housing and to implement and test it in pilot projects around the Mediterranean. AHC coordinated this project, which studied relevant factors to reach the integral approach for renovation was: the empowerment of the demand side (housing associations and tenants), including new business models / value cases, and a life-cycle view of the renovation process.
- **MARIE (MED)** [Finished: 31-12-2014] main objective was to develop a Mediterranean Strategy for Energy Renovation of buildings, focusing on proposing tools and financial mechanisms for energy renovation of Buildings, providing marketing and training programmes, an innovation laboratory and cooperation hub. AHC was on charge of the characterisation of the building stock for the Catalan region and it participated on the identification of the barriers and solutions for the promotion of energy renovation of housing.
- **GrowSmarter (H2020)** [Started: 15-01-2016 (60 months)] one of the Smart city projects which aims to transforming cities for a smart, sustainable Europe. AHC signed a collaboration agreement with Gas Natural, in order to propose buildings that fulfil the technical requirements developed in the project so they could be retrofitted according to the proposed GrowSmarter solutions.
- **REHABITAT (POCTEFA)** [Finished: 30-06-2012 (36 months)] Renovate specific pilots to achieve a better coexistence and maintenance of common areas throughout the combination of training and participation of citizens.

Infrastructures and facilities

The AHC has established experience in retrofitting of social and residential buildings. It has developed, during the past years, more than 400 refurbishment projects; 20% of them related to improvements of building envelopes. Owing to its past and on-going experience in building retrofitting, AHC has played an important role in compiling the main refurbishment and buildings energy management best practices and barriers in the Spanish context (legal, technical, social and financial barriers and best practices in refurbishment activities).

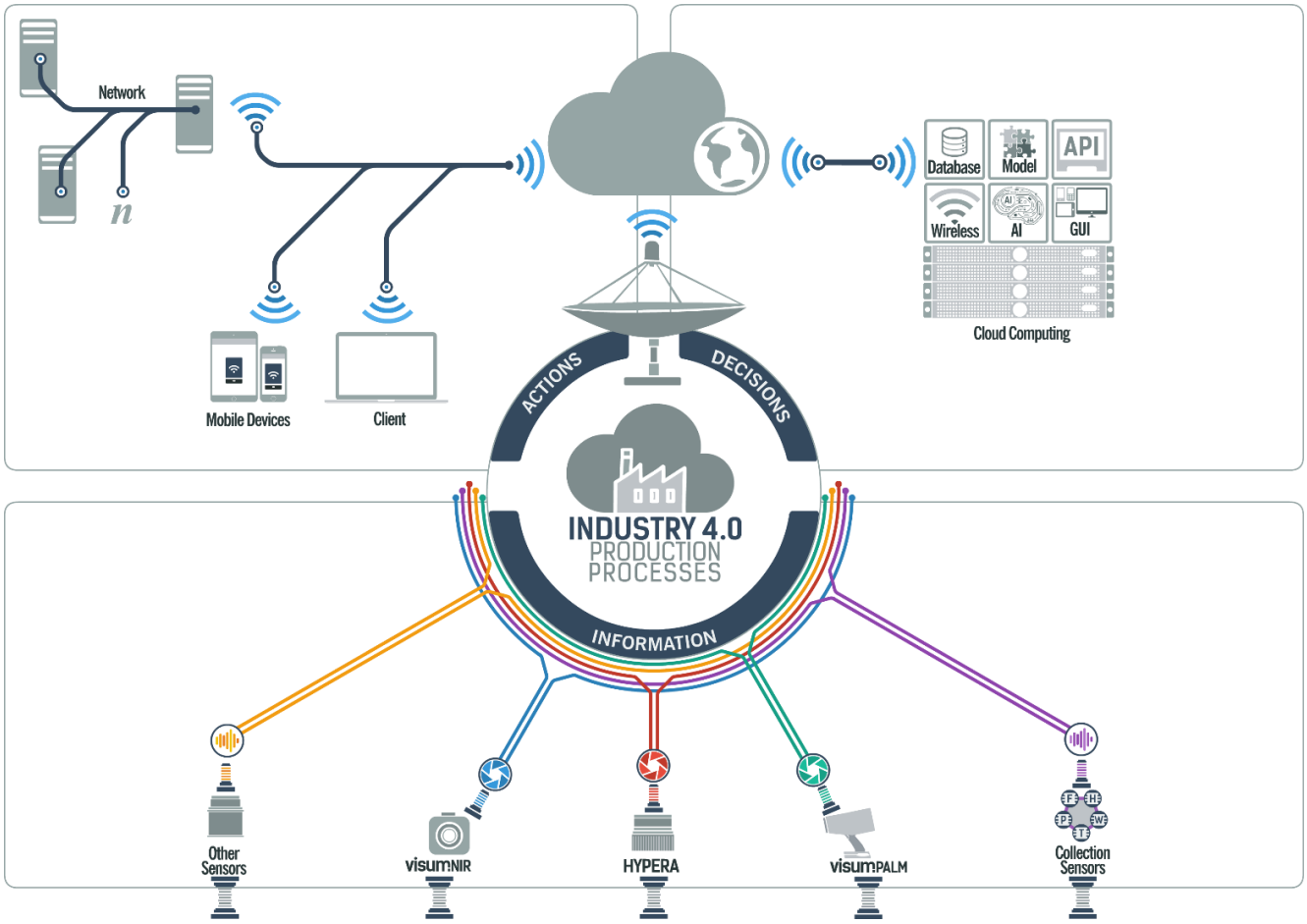
In addition to the staff expected to work directly on the project, AHC will call on technical staff and their extensive experience in the retrofitting of residential buildings that has knowledge about all the phases of the life cycle of a building, but specifically the use and maintenance.

Also, the External Action Unit, has expertise and experience in the coordination and the management of European and international projects.

Participant No.	**6**
Short Name	**VISUM**
Country	Ireland
Organization Type	SME
Website	www.seeingnewdata.com

Partner Profile

VISUM (www.seeingnewdata.com) is an optical engineering company from Dublin that specializes in the development and commercialization of Near Infrared Spectroscopy (NIRS)-based portable and inline analyzers and measurement systems for process monitoring and quality control applications in the food, pharmaceutical and chemical applications. The company was established in 2014 and currently employs 9 staff from its facilities in Nexus UCD (technology park at the University College Dublin campus). In 2016, the company launched the VISUMPalm (a handheld NIR measurement instrument) and has secured first sales. An integration platform (VisumNIR) for the inline version is currently under development in cooperation with early adopter clients from the food processing industry (dairy, meat and olive oil) and pharmaceutical industries. VISUM Ltd. is an SME Champion-awarded a Phase 1 and a subsequent Phase 2 Grant under H2020's SME Instrument.



Role in the project

VISUM will be responsible for the development of the SaaS that will be used to implement the HOUSEFUL service. For this reason VISUM will designing the architecture, coding the tools, establishing the API with the cloud provider and building the databases (data provided from other entities). A testing and troubleshooting system will be implemented to validate the platform. VISUM will also contribute their skills in knowledge/IPR management, as well as in commercial branding, reaching industry, the development of SaaS products and business models. They will also contribute to knowledge transfer and training activities and gearing up for the postproject phase. Finally VISUM has extensive expertise in collaborative R&D project (FP7, H2020, BBI) and industry-driven business-to-business actions. Their project management procedures are ISO and UNE certified, and they have an internal PMO that has the necessary resources and skills to ensure smooth administrative and financial reporting to the project coordinator.

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Detailed description of VISUM's role in WP4. ***(Total Person-months: 4)***

Task 4.4: VISUM will integrate sensors and monitoring required for the operation of AnMBR system foreseen in Frontrunner building 1. According to the design and operational need provided by LEITAT and IDP, VISUM will propose a monitoring system to ensure the proper functioning of the AnMBR and evaluate its performance over the time *(4 Person-months)*.

Key personnel

DR. TIMOTHY KEHOE (M) PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acousto-optical metrology techniques for nanotechnology applications, in particular for characterisation of nanoscale structures fabricated by nanoimprint lithography and directed self-assembly. Other research activities: quantification of ordering in self-assembled structures by image analysis techniques and optical sensors using hydrogel materials. Tim is the VISUM Palm Product Manager, as well as the Technical Lead at VISUM Ltd. and he brings vast optical integration and device instrumentation expertise.

DR. SEVERINE PHILIPPE (F) graduated with a 1st class Hons. B.Sc. in Applied Sciences (Physics and Physics Technology) from Dublin Institute of Technology, Ireland, in 2002 and received her PhD in Physics from Trinity College Dublin in 2007, for the study of polarisation effects in semiconductor optical amplifiers (SOA). She went on to work on the development and characterization of low noise SOAs, as well as THz generation by multimode lasers and their detection using four wave mixing in SOA, both in the Semiconductor Photonics Group, Trinity College Dublin, and the School of Electronic Engineering, Dublin City University. She previously worked as a R&D technician for the plastics industry and in quality control for electronic appliances. She joined VISUM in 2014 as an optical engineer and is currently involved in projects including various microscopy techniques, planar waveguide technologies and MIR spectroscopy.

NIKOLINA MIKULCIC (F) User Interface Designer. Nikolina comes from web design and development background with more than 15 years of experience in this field and she is an expert on UI/UX design strategy and front-end development (CSS/HTML). She studied graphic design and she started to learn web design due to her deeply interested for digital world. Since joining VISUM in September 2016, she has been creating a new User Interface for the VISUM product lines. Prior to VISUM, she has an extensive experience in the online world, having worked in companies such as EuroArt93 – digital agency, BlueBOX IT, Biroval. She also was a founder of design studio In Love With Blue and she worked as Project Manager, UI/UX designer and front-end developer. She was leader numerous digital projects for different clients as Erste Bank, Unicredit Bank, Telenor Bank, Komb Bank, ZAMP etc.

CHRIS DERMODY (M) Software Developer. Bachelor's degree in Product Design from Dublin Institute of Technology. Chris excels in creative problem-solving, using the latest technologies to transform ideas into real opportunities that provide exponential value to business operations. With a keen eye for user experience, Chris builds robust, responsive applications that stand the test of time. Since joining VISUM, Chris has been hard at work developing both front and backend features that push the VISUM product lines forward. Prior to VISUM, Chris enjoyed working across the full spectrum of business, founding startups and overseeing product development throughout the lifecycle of many successful businesses.

Experience

1. **PREVIEW**- PREdictiVe system to recommend Injection mold sEtup in Wireless sensor networks. (Grant Agreement number: 636892). VISUM Ltd. is responsible for the development of the Data Acquisition System and in in developing the Signal Conditioning, its integration with the ARM main board, as well as implementing a prototype.
2. **Phase 1 SME Instrument:** An IR-spectroscopy-based platform for the rapid integration and customisation of affordable, real-time monitoring solutions for the food and pharmaceutical industry (Grant Agreement number: 684767).
3. **Phase 2 SME Instrument:** Cloud-enabled IR spectroscopy platform for the rapid integration and customisation of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries (Grant Agreement number: 731043)
4. Development of a portable LIDAR device for the detection of pesticides for the University of Lerida in Spain (B2B contract)

Associated with document Ref. Ares(2018)1767830 - 03/04/2018

Development of a NIR device for at-line measurement of quality of oil in industrial frying equipment for bakery products (B2B contract).

Infrastructures and facilities

Spectroscopic Laboratory: equipped with several instruments, namely, an FTIR-ATR (Interspectrum), two UVVIS (PerkinElmer and Ocean Optics), one Extended-SWIR (StellarNet) and one NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc noncommercial chemometric tools (Matlab and Python toolboxes). Electronics, mechanical and optical workshops: equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others. ZEEMAX software for the design of optical systems and design setup by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

Participant No.	**7**
Short Name	**AIGUASOL**
Country	SPAIN
Organization Type	SME
Website	www.aiguasol.coop/en

Partner Profile

Founded in 1999, the SME AIGUASOL (www.aiguasol.coop/en) team is made up of more than 20 professionals with a high level of technical and scientific experience. They include engineers, PhDs, architects and physicists who have notable experience both in private companies and research centres, and distributed among offices in Barcelona, Basque Country, and Santiago de Chile.

AIGUASOL provides engineering and research services, promoting innovative solutions to reduce the impact associated with energy consumption (experience in urban planning, construction, industrial processes and power generation, with a focus on energy planning, savings measures, energy efficiency, process integration and renewable energy implementation). AIGUASOL is the official TRNSYS distributor for Spain and Portugal, and develop energy-advanced tools for third parties. AIGUASOL has participated in several Tasks of the International Energy Agency and is member of the Steering Committee of the ESTTP.

AIGUASOL have a wide experience on energetic plans (i.e. the Barcelona energy improvement plan), building thermal and lighting simulations, and renovation proposals definition and validation through cost optimal analysis (i.e. the TOBEEM project introducing a semi automatic cost optimal analysis for tertiary buildings, for more than 20,000 scenarios). We also have wide experience on monitoring HVAC systems (both, generation but also distribution and emission elements) in order to understand their operation, and to detect improvements (linked to emulation and simulation techniques) both to substitute physical elements (retrofit) and/or to operate the existing or the new ones in the correct way. Those methodologies are linked to BEMS-BEPS integration. In 2014 AIGUASOL founded the spinoff exsergy (www.exsergy.com) which aim is to promote the building retrofit operations under ESCO models, by offering a technological and consulting intermediation to the involved parts (properties and ESCO companies).

Role in the project

AIGUASOL employs a high quality scientific group of personnel that will lead the development of the Circular Economy methodology to easily evaluate circularity in building retrofit scenarios (WP2 leader), considering the transversal tasks related to the rest of the key objectives. AIGUASOL will participate in the technical coordination of activities in the Spanish demos site (WP4) and will support the development and application of the new modular facades (S10) by analysing the materials and methodologies to be implemented and sizing the energy savings expected to be reached. AIGUASOL will also offer oriented feedbacks related to national regulations and RES implementation, which will be part of several objectives and the core of the S11 service related to pay per use energy generation.

Description of detailed AIGUASOL's role in WP4. ***(Total Person-months: 21)***
Task 4.2: Engagement of the previously defined KPIs (in WP2) to be implemented and applied in the development of the technical specifications (TS) for any of the HOUSEFUL solutions *(2 Person-months)*.
Task 4.3: Developments in 4.3.3 subtask (Optimisation of Energy vector). With the collaboration of other partners, AIGUASOL will be in charge of the feasibility analysis of innovative both active and passive systems potentially installed as improved elements in building structure in all Frontrunner buildings. AIGUASOL with other partner's collaboration will promote the design and potential installation of optimal systems to be selected according to circular criteria as well as their cost (considering the cost optimal methodology and the circular principles established in previous WPs) *(4 Person-months)*.
Task 4.4: Contribution to the definition of the main energetic monitoring system, considering the different active and passive final solutions implemented and in coherence with the main cloud-based system to be developed *(1 Person-month)*.
Task 4.5: Collaboration in the definition of the execution documents related to the technical specifications for the HOUSEFUL solutions *(1 Person-month)*.
Task 4.6: Analysis on national, regional and local available financial resources. Analysis on existing funding schemes and validation of new one's related to circular economy business models *(3 Person-months)*.
Task 4.7: Collaboration with the local demos site (partner AHC) in the development of the execution technical documents required for the public tender and construction development procedure, specifically in those solutions related to reduce the energy consumption *(5 Person-months)*.
Task 4.8: Jointly with other partners, technical assistance and supervision of works for solutions potentially in-

stalled in S10, evaluating the resulted energy performance in S11. Continuous commissioning of the implemented solutions and report to the rest of the partners *(5 Person-months)*.

Associated with document Ref. Ares(2018)1767833 - 03/04/2018

Key personnel

Jordi Pascual (M), Physicist and PhD Engineering, senior expert

Since 1999, Project coordinator in the company at AIGUASOL in subjects related with building physics, and district networks. TRNSYS simulation of energetic systems and building behaviour. Since 2010, General Team Manager in AIGUASOL.

Oriol Gavaldà (M), Mechanical Engineer and Economist, senior expert

Since 2006, Project co-ordinator in the company AIGUASOL in subjects related with solar thermal energy, solar cooling, district networks and cogeneration. TRNSYS simulation of energy systems and buildings. Participation of TASK 25 and TASK 38 IEA Solar Heating & Cooling Program. Since 2010, research manager in AIGUASOL.

Àngel Carrera (M), Physicist, senior expert

Since 2005, Project co-ordinator at AIGUASOL in subjects related with energy systems, simulation of stochastic processes, statistical applied to the engineering field and surface-response methods (machine learning techniques). TRNSYS simulation of energy systems and buildings. Participation of TASK 25 and TASK 38 IEA Solar Heating & Cooling Program. Since 2012, General Manager in AIGUASOL.

Eva Majoral (F), Economist

Since 2007 Controller at AIGUASOL in subjects related with the management of several European and R+D projects.

Experience

Related publications:

In recent years, AIGUASOL has published over one hundred articles in indexed journals and conferences. Noteworthy because of the issues related to the proposal, the following:

- Vergleich verschiedener Konzepte für den Einsatz erneuerbarer Energien in Rechenzentren. Nirendra Lal Shrestha, Noah Pflugradt, Thorsten Urbaneck, Oscar Camera, Angel Carrera, Jaume Salom, Eduard Oro, Albert Garcia, Mieke Timmermann, Hans Trapman, Gilbert de Nijis, Joris van Dorp, Bastian Beerens. DKV-Tagung 2015, Dresden, Energieeffiziente Klimatisierung in Rechenzentren.
- Methodology for TRNSYS Large-Scale Simulation Studies using Experimental Design. Nirendra Lal Shrestha, Noah Pflugradt, Thorsten Urbaneck, Oscar Camera, Angel Carrera, Jaume Salom, Eduard Oro, Albert Garcia, Mieke Timmermann, Hans Trapman, Gilbert de Nijis, Joris van Dorp, Bastian Beerens. DKV-Tagung. 2015, Dresden, Energieeffiziente Klimatisierung in Rechenzentren.
- Evaluation of air leakage and its influence on thermal demands of office buildings in Madrid. Jordi Pascual, Oscar Cámara, Aleksandar Ivancic, Damien Tavan and Maria Casanova. REHVA HVAC European Journal. 01/2013.
- New calculation methodology for solar thermal systems. Angel Carrera, Oscar Camara, Maria Casanova, Merce Farré and Isabel Serra. Kassel (Germany) 2011. ISES Solar World Congress 2011.
- Pluridisciplinar analysis of energetic behaviour on glazed façades office buildings. How they work?. J. Pascual, M. Capdevila, B. Tena, D. Perez, E.M. Mitre. Palenc 2010, Rhodes (Greece). Second prize of the best paper of the congress.
- Energetic Approach of Zorrotzaurre Master Plan Energetic viability of variations to the actual master plan according to use, orientation, height and building intensity. Pascual J., Vallejo C. CISBAT 2007 International Scientific Conference, Lausanne, Switzerland.
- Barcelona Forum 2004. TRNSYS Simulations of a solar assisted District Heating and Cooling Network. L. Sisó, H. Schweiger, E. Wiemken, H.-M. Henning, H. Zinko, A.Ivancic, G. Torres), Proc. EuroSun2002-Congress. ISES-Europe. Bologna (Italia), June 2002.

At the other hand, AIGUASOL is the official distributor, for Spain and Portugal, of different energetic calculation tool (TRNSYS, EnergyPro, Meteonorm, Einstein, etc.), and produces and distributes their own tools (as TRANSOL).

Associated with document Ref. Ares(2018)1767830 - 03/04/2018

Related projects:

- **4RinEU** (H2020) [Started: 01-10-2016 (48 months)] is about Robust and Reliable technology concepts and business models for triggering deep Renovation of Residential buildings in EU. AIGUASOL will contribute by developing a tool that enables to define, in the first design stages, the best RES options. Also an expected contribution to the definition and adaptation of the prefabricated facades to the local demo is expected, including the energetic impact analysis, and the business model definition. Finally, AIGUASOL is in charge of the technical management of the solutions implementation for the local demo case.
- **InteGRIDy** (H2020) [Started: 15-01-2017 (48 months)] integrated Smart GRID Cross-Functional Solutions for Optimized Synergetic Energy Distribution, Utilization Storage Technologies. AIGUASOL will develop thermal-electrical models to evaluate the different measures to be undertaken and will design the monitoring scheme and its integration with the existing SCADA system and engineering, commissioning, follow-up (IPMVP).
- **GrowSmarter** (H2020) [Started: 15-01-2016 (60 months)] one of the Smart city projects which aims to transforming cities for a smart, sustainable Europe. AIGUASOL, as a subcontracted company of the DSO Gas Natural, has a specific role related to the technical definition and implementation of building the retrofitting solutions (including energy savings dimensioning and business impact definition). AIGUASOL also is responsible for the development of the Home Energy Management System (HEMS) to be implemented in the local demo cases.
- **Plug&Harvest** (H2020) [Started: 01-09-2017 (48 months)] Plug-n-play passive and active multi-modal energy Harvesting systems, circular economy by design, with high replicability for Self-sufficient Districts & Near-Zero Buildings. AIGUASOL will lead the development of the circular economy business models, considering the transversal tasks related to the rest of the key objectives. AIGUASOL will lead the technical coordination of activities in one of the demo sites and will support the development and application of the intelligent modular facades by analyzing the materials and methodologies to be implemented and sizing the energy savings expected to be reached. With its wide experience in DHC designs and operation, AIGUASOL will also contribute to the Plug-n-Play Optimal Energy Management System at District/Grid Level by offering oriented feedbacks related to national regulations and RES implementation.
- See other AIGUASOL's R+D projects at https://aiguasol.coop/solutions/r-d/

Infrastructures and facilities

N/A

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Participant No.	**8**
Short Name	**LGI**
Country	France
Organization Type	SME
Website	www.lgi-consulting.com

LGi sustainable innovation

Partner Profile

LGI is a French sustainable innovation SME, founded in 2005. It offers the following services: project development and management, in particular for EU projects, communication and dissemination, IT developments (webbased mainly) and strategy and innovation studies. LGI has a positioning in high tech sectors: energy, transport, security, ICT, health and environment. LGIs approach is usually multidisciplinary and European (LGIs consultants have competences in law, finance, engineering, chemistry, IT and are fluent in French, English, German, Spanish and Czech). LGI has more than ten years of experience with EU projects. LGI offers a wide range of services to support our clients' decision-making with a strategy base. The firm's strategic marketing approach is based on insight into both advanced technological developments and novel business models, with innovation at the heart of the process. LGI's approach combines the analysis of technological development with market and social drivers to support our customers with a global innovation approach. Our services include market research and forecasts, global industry analysis, stakeholder mapping, which enable our clients to address their key challenges and exploit their strategic opportunities. LGI has developed a recognised expertise on business modelling, technologies and market assessments. LGI is used to lead the work packages in the EU projects on innovation management, market surveys, IPR and result exploitation, financial analysis and technology assessments. It has its headquarters in Paris and branches in Valencia (Spain) and Hamburg.
Website: www.lgi-consulting.com

Role in the project

LGI will be in charge of the exploitation under the WP6 and will actively contribute to the communication and dissemination activities of the project. It is also envisaged to contribute to the LCC analysis in the WP5.

Key personnel

Martin Pecanka (M) contributes since 2010 to the development of LGI through his experience in building and managing collaborative projects. He drafts exploitation & IPR protection strategies and contributes to particular market and opportunity studies in a number of sectors (energy, climate, transport, security, health, etc.) for private clients and public organisations. Since 2014 he has been representing LGI in Spain, office in Valencia. Before joining LGI he worked for an engineering firm in Spain managing urban civil infrastructure projects cofunded from structural funds and in France supporting cooperation and infrastructure development projects on international level. He also spent 14 months on a post of junior transport attaché within the diplomatic services of the Czech Republic at the Czech Permanent Representation in Brussels. His economic and legal background allows him to contribute to a number of projects in the portfolio of LGI to enhance the dissemination of the results and increase the market uptake potential project outcomes. He was actively involved in the demEAUmed project finished in June 2017.

Camille Auriault (F) performs strategic marketing studies, conducts analyses of technological systems and value chains, performs competitive analyses and assists LGI's clients and partners with technology transfer and roadmapping activities in high-tech and scientific sectors. Prior to joining LGI, Camille worked for the international chemical group Solvay, performing market research and value chain analyses. She also worked in R&D at CEA, the French Alternative Energies and Atomic Energy Commission, and at Trinity College Dublin performing laboratory research. She holds a Masters degree in Intelligence Marketing from the French Business School HEC and a Chemical Engineering degree from the Dublin City University and ENSIACET.

Mathieu Salel (M) is an experienced professional with more than 4 years' experience in various industrial domains. He carries out strategic studies and analyses of innovative technologies particularly in the energy, transport and water sector. He has a background in aeronautics and energy sectors across France, Germany and China. Before joining LGI in 2016 he worked for the Smart Grids department of a French DSO (ENEDIS) where he analysed technical results of innovative smart grids solutions. In a previous research project on renewable marine energies in China he developed a model to estimate the global potential of electricity production from OTEC. Before joining the energy sector he spent three years working on behalf of Airbu, co-managing continuous improvement projects within the system engineer`s department. He holds a general engineering degree

(ICAM, Toulouse) and an power engineering degree (Tsinghua University, Beijing). He completed his education
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with a post-masters in future energy alternatives (Mines ParisTech, Paris). Mathieu was involved in the second part of the demEAUmed project, in particular on drafting the exploitation strategy.

Chloe Chavardès (F) has 10 years of experience in public communication. She is in charge of communication and dissemination activities throughout the entire life cycle of projects, develops targeted strategies and implements action plans. Her work covers online, print and multimedia communication as well as event management and planning. She actively contributed to the communication activities for many FP7 and H2020 projects, as well as international associations (ex: PROMETIA) and technological platforms (ex: SNETP). Her international background in technical and scientific sectors has allowed her to acquire the necessary knowhow to communicate at European scale to the general public, to decision-makers and to a more technical audience. She began her career in a communications research firm in Washington D.C. specialised in the energy sector. She then spent five years working in corporate communications at the European Space Agency in the Netherlands and France. Chloe is the official couch of LEGO Serious Play.

Experience

- Boo et al., 2014: "How Innovative Business Models Can Boost The Energy Efficient Buildings Market", 40th IAHS World Congress on Housing - Sustainable Housing Construction, December 2014
- Dunphy, N., Boo, E., Dallamaggiore, E., Morrissey, J., 2014: "Developing A Sustainable Housing Marketplace: New Business Models To Optimise Value Generation From Retrofit", 40th IAHS World Congress on Housing - Sustainable Housing Construction, December 2014
- Morrissey, J., Boo, E., Dallamaggiore, E., Dunphy, N. (2015). The Role of Innovative Business Models to Kindle Sustainability in the Built Environment. Round Table on Sustainable Cities: Research, Business and Local Government. Invited Paper, The Russell Group-China Workshop on Sustainable Cities. Xi'an Jiaotong-Liverpool University (XJTLU), Suzhou, China, 3-5 June.
- Joly, P., Boo, E., 2015: "Roadmap Actinides separation processes 2015" ISBN 9782919313099

Infrastructures and facilities

- (Since 2005) Involvement in approx. 30 FP7/H2020 projects – energy, transport and environment sectors (demEAUmed)
- (Since 2014) Support to PROMETIA association creation; PROMETIA secretariat coordination and communication since the association creation (www.prometia.org)
- (2012-2013) Value chain analyses in energy & transport, for UK & Spanish research houses
- (2012) Mapping, analyses and prospection on EU innovation schemes (current & foreseen)
- (Since 2011) Contribution to NUGENIA Association creation, development of online Open Innovation Platform (www.nugenia.org)

Participant No.	**9**	Associated with document Ref. Ares(2018)1767830 - 03/04/2
Short Name	**Reinberg**	
Country	**Austria**	
Organization Type	**SME**	ARCHITEKTURBÜRO REINBERG ZT GMBH A-1070 Wien, Lindeng. 39/ 8 Tel: (+43) 01-524 82 80 - 0 Fax: (+43) 01-524 82 80 - 15 Email: architekt@reinberg.net www.reinberg.net
Website	**www.reinberg.net**	

Partner Profile

Architect Reinberg is CEO and owner of “Architekturbüro Reinberg ZTGmbH” in Vienna.
He is “Ziviltechniker” (ZT), a special status for an engineering offices in Austria, a free business with specific obligations concerning high quality work, which is guaranteed by the chamber of architects.

The office is specialisesd on ecological buildings, passive house construction, closed cycles in constructions within ecological cycles.

The office is working as an independent planning office since 1980, the company has served as a civil-technical office since 1985 and as “Architekturbüro Reinberg” - limited company since 2006.

The company currently employs a staff of 7 people. Five of them have finished University studies.

The office has realized more than 100 projects with special high ecological qualities. The office is engaged in numerous research projects concerning alternative energy and sustainability. Architect Reinberg is teaching in numerous Universities in Europe and outside Europe. He is visiting Professor at the University of Krems (future building solutions) and at Politecnico di Milano for the academic year 2017-18

Role in the project

Assistance of the project partners in the field of architectural design, technical solutions in the field of architecture and ecological building (the office has experience more than 30 years in sustainable building). Architect Reinberg will be responsible for architectural and technical questions during the realization and the planning of the demo-projects (WP4). GWR will contribute with financial information, technical solutions and optimisations with details about architectural and aesthetic solutions. Architect Reinberg will work on optimised solutions to integrate the circular strategies within a high architectural standard.

Detailed description of GWR’s role in WP4. ***(Total Person-months: 24)***
Task 4.1: Support in circular economy audit for the Austrian Frontrunner buildings (Demo 3 and Demo 4), specifically addressing the material passports and possible deconstruction plans *(2 Person-months)*.
Task 4.2: Definition of the technical specifications (TS) for the solutions not including technological implementation (S4 and S9) in the Austrian Frontrunning buildings (Demo 3 and Demo 4) *(6 Person-months)*.
Task 4.7: Support in the selection process of companies for renovation works according to prize and trust for Demo 3 and Demo 4 *(1 Person-month)*.
Task 4.8: Implementation of the selected solutions for the Austrian demo buildings (Demo 3 and Demo 4) *(15 Person-months)*.

Key personnel

Primarily responsible: **Architect Georg W. Reinberg**, born 1950, has his office in Vienna. Study at Vienna Technical University and Syracuse University NY, USA. Received grants from Schütte and Fulbright. Arch. Reinberg leads a civil engineering office specialized in ecological green building. He has planned more than 330 projects; about 100 have been built. His office specialises in research, design, planning and supervision of ecological buildings since 1984. Reinberg is visiting professor at the Danube University Krems "Centre for Construction and Environment"; lectures at the Technical University of Vienna (MSc Program "Renewable Energy in Central and Eastern Europe", "Green Building Solution", inter alia.); and at the University of Applied Sciences in Vienna. Is member of the Board of Advisory Committee at the Danube University inter alia. He is also teaching and giving lectures internationally e.g. Latin-American, USA, Europe, China and Iran. He has several times received the Lower Austria building in wood Prize and the Prize for Green Building; the international prices Fassa Bortolo Prize for Sustainable Architecture 2003; Fassa Bortolo honourable mention 2008; Golden Medal from the City of Vienna 2010; Culture Prize from Upper Austria 1993, the Price of Architecture Burgenland 2002 and the Culture Prize Lower Austria 2012. He is an honorary Member of the IG Passive House and member of “Innovative Gebäude”. Since 2017 he is certificated Passive House Planner.
Others:
DI Wolfgang Csenar (M), specialist in building regulations and technical norms. He has many years of experi-

ence in design and execution plans and detailing ecological projects. He also is experienced in supervising duri construction.

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DI Marina Glaser (F), specialist in design and ecological architecture. Many years of experience in master plans, execution plans and detailing ecological buildings.
Marta Enriquez Reinberg (F), lic. Arq. Works within the office since 27 years and she is organising the office, specialist in communication and ecological design. She studied in Mexico and has special experience in passive cooling solutions.
DI. Martin Presich (M), was born 1961 in Eisenstadt, Austria. In 1991, he finished his studies of Architecture at the Technical University of Vienna. From 1988 on he was working as independent architect on numerous projects such as single-family houses, renovation projects in Passive-House Standard, hotel renovation in Moskau and a holiday Inn in Cuba. Since 1992 DI Presich is working in the Architecture Office Reinberg being assigned with many different projects like Passive-House Childrens day care center, total renovation of an apartment- and office building, Planning and Construction of a nursing home for the elderly and a home for the handicapped. He is employed for 25 years in the office, specialist in financial calculation, tenders and building inspection for sustainable buildings.

Experience

- List of up to 5 publications and/or products, services (including widely-used datasets or software), or other achievements (e.g. patents) relevant to the proposal:

1. "Ökologische Architektur – Entwurf, Planung, Ausführung / Ecological Architecture – Design, Planning, Realization",07. 2008 , Georg W. Reinberg, Matthias Boeckl 348 p., ISBN: 978-3-211-32770-8

2. "Georg W. Reinberg: Solar Architecture", 2003
   Hrsg. Adriana Labella, Librìa Verlag, Melfi (Italia), 112 p., ISBN 88-87202-48-6

3. Monitoring als Erfolgskontrolle (Monitoring as a performance control), 03 /2016, Greenbuilding #73, S. 24 - 29

4. Detail Green Book - Gebäudeintegrierte Solartechnik 11 2016
   ISBN 978-3-95553-325-0 - www.detail.de
   p. 80-91 (Design and Construction) Solarthermie
   p.90-91 - Gneis-Moos_56
   p. 91 - Schellenseegasse_172
   p. 110-111 - Kindergarten Dt. Wagram_215

5. Verwaltungsgebäude Windkraft Simonsfeld, Gebäudenutzung: GOLD 02/2017, Info-Heft Projekte 2015 und 2016, Auszeichnungsveranstaltung 2017, ÖGNB - Österr. Gesellschaft für nachhaltiges Bauen und klimaaktiv - ÖGUT Gmbh, S.16 Projekt Ernstbrunn_270

- List of up to 5 projects and activities which are connected to the subject of this proposal (show previous experiences in relation to the project´s tasks; participation in other (EU) research & innovation projects; activities related to the project´s subject etc.:

1. Straw-Clay Office Building: Tattendorf: http://www.reinberg.net/architektur/119
2. Solar Active House: Schellenseegasse: http://www.reinberg.net/architektur/172
3. Solar Active House Kraig: http://www.reinberg.net/architektur/207
4. Plus Energy Office Building: Ernstbrunn: http://www.reinberg.net/architektur/270
5. Active House Renovation: http://www.reinberg.net/architektur/288

Please have a look at our webpage www.reinberg.net, for some more information to our research work please have a look to the last page of CV Arch. Reinberg.

Infrastructures and facilities

Our Office is an architectural office with 8 Working Places, each with 2 monitors, we use AutoCAD 2016 and different other programs like Adobe CS (different versions), SketchUP, Archiphysik, Microsoft Office (different versions), Adobe Acrobat Professional

Participant No.	**10**	ssociated with (2018)1767830 -	document Ref. 03/04/2018
Short Name	**Turntoo**		
Country	The Netherlands		
Organization Type	SME		
Website	www.turntoo.com		

Partner Profile

A circular economy frontrunner offering solutions and consultancy services. Designing and implementing new business models that alleviate resource dependency by preventing the degradation of materials into waste. In Turntoo's vision, a product is a materials depot: its content is known and all materials remain the property of the manufacturer. Products are designed in such a manner that upon termination of use, materials can be recovered and reused in new products.

Turntoo's mission is to facilitate the continuity of life by designing a different relationship between mankind and earth. This results in, among others, innovative concepts, products, and services. In striving to create impact Turntoo works cross-sector and has coined many circular economy concepts and business models such as Light as a Service, washing on performance basis, raw materials passport or products as a materials depot.

Turntoo works for and with a vast array of organisations, both profit and non-profit. Its activities consist of:

1. Sharing vision and mission in a call to action and change of perspective; in lectures, workshops and publications.
2. Assisting organisations in the transition to a future-proof organisation, by consulting on the (re)design of business models, organisational processes, financing structures, contract types, services or products.
3. Designing and implementing circular economy showcases to inspire and elaborate on in the future.

Role in the project

Turntoo will:

- Definition of KPIs related to materials aspects (WP2)
- Lead the creation of Material Passports for every demo building, documenting and valorising materials in a globally accessible technology (WP4);
- Advice on architecture and sourcing of materials for the demo sites (WP4);
- Contribute to workshops for stakeholders in order to align propositions (WP7);
- Participate in brainstorm and reviewing the business model design and implementation (WP6).

Detailed description of TURNTOO's role in WP4. ***(Total Person-months: 5)***

Task 4.1: Providing the Madaster Platform to the 4 frontrunner buildings (the 4 Demos). Madaster will be used to produce the Material Passport. Turntoo will subcontract the Madaster service *(1 Person-month)*.

Task 4.2: Advise on design principles and choice in technical solutions. 2 countries*2 sites with: hours of preparation, meeting travel time, debriefing *(4 Person-months)*.

Key personnel

Thomas Rau (M) is the founder of RAU architects and Turntoo, both based in Amsterdam. He studied fine art and dance in Bonn and architecture at RWTH Aachen University. For years Rau has contributed to national and international discussions about sustainability. RAU Architects have been on the forefront of designing environmentally conscious buildings and healthy indoor climates since 1992. He founded Oneplanetarchitecture Institute
(OPAi) in 2008, focusing on social and economic questions of our time with regard to architecture. RAU and OPAi have contributed several innovations and set new standards first in the field of CO2-neutral and energy neutral building, later for energy positive building and more recently circular architecture.
Thomas Rau has performed as keynote speaker for large conferences as well as for smaller audiences of boards of directors, policy makers and politicians. He is ranked among the Top 5 Dutch key players in sustainability, a ranking published by national daily Trouw. He was elected as Dutch Architect of The Year 2013 and awarded with the ARC Oeuvre Award for his widespread contribution in both promoting and realising sustainable architecture in the last 20 years.

Marijn Emanuel (M) holds a Master's Degree in Architecture and Building sciences and built up 23 years of experience at RAU Architects. In his position as senior architect he has contributed to the design and development of various iconic buildings, including examples of energy positive buildings or material neutral buildings such as the Townhall in Brummen or Liander. In his role as project architect he both facilitates in design and materialisation assignments and in stakeholder and team management. Marijn Emanuel is involved in activities for both RAU Architects, Turntoo and Madaster Services.

Debbie Appleton (F), Master's Degree in Anthropology and since 2013 part of the Turntoo team in the role of new business developer. In this role she has helped develop the Philips Light as a Service proposition, several

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office interior as a service propositions and guided in the development of circular procurement for governmental bodies and institutions. Responsible process management and consultancy services; ranging from drafting strategies to implementing processes and designing roadmaps.

Experience

Related publications:

Book: Material Matters
Thomas Rau, Sabine Oberhuber. ISBN 9789461562258

Documentary: The End of Ownership
VPRO Backlight, https://youtu.be/oOO-pYUl9-w

Report: The path towards a circular category office furniture
https://www.pianoo.nl/sites/default/files/documents/documents/reportcircularcategoryenoktober2016.pdf

Related projects:

2010: First circular Light as a Service ("pay per lux") contract developed with Philips.
2012: Model for renting out washing machines on performance basis implemented with BSH.
2013: RAU Architects designed the first circular building as a raw materials depot in the municipality of Brummen and Turntoo developed the first version of a Materials Passport for it.
2014: Facilitation of the collaboration between Schiphol Airport, Philips and Cofely to realize circular Light as a Service at Schiphol Airport, including the contract design.
2015: The head office of Liander has opened its doors, a 20.000m2 building where more than 80% of all materials are used in a second cycle and which produces 1.5million kw/h of electricity per year. In this project Thomas Rau formulated the ambition to create the first energy positive, remountable circular building with a material passport thereby inspiring the whole team of builder, advisers and suppliers to deliver the most circular building possible - Turntoo was in lead of the development of the Materials Passport, an iteration of the first version for Brummen.

In addition to this Turntoo is one of the first members of the CE100 of the Ellen Mac Arthur Foundation, which is designed to advance the circular economy through collaboration and the sharing of knowledge

Infrastructures and facilities

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Participant No.	11	Associated with document Ref. Ares(2018)1767830 - 03/04/2018
Short Name	**NEUES LEBEN**	
Country	Austria	
Organization Type	**NPO building cooperative**	
Website	http://www.wohnen.at	

Partner Profile
The not-for-profit association "Neues Leben" was funded in 1950. Its purpose is to provide affordable and highquality housing for a broad share of the population. At the moment it manages about 11.500 dwellings. In the past three years about 200 new units were finished each year. The total assets of 2016 amounted to 545 mio. Euro. Neues Leben has a focus on high-quality architecture that accomodates to the need of the end-users. In the years 2003, 2010 and 2014 it was awarded with the Austrian builder-owner award.
Role in the project
NEUES LEBEN holds and manages the Austrian demo "Donaufelderstrasse" and will provide relevant data and will give access to scientists involved in HOUSEFUL for calculating circularity of buildings; organize stakeholder meetings and dialogs for co-creation of circular economy business options; will communicate project results and findings to partners; They want to include dismantling competences in future projects. **Detailed description of NEUES LEBEN's role in WP4.** ***(Total Person-months: 2)*** **Task 4.1:** Provide necessary information and access for BIM Modelling and Material Passport creation for Demo building 4 *(0.5 Person-month)*. **Task 4.2:** Definition of the type, number/quantity, origin and functionalities, among other main characteristics for those building materials available for renovation/refurbishment at local scale will be defined (S4). Results from Material Passport and BIM will be used as basis information for the development of Design for Disassembly (DfD) evaluation of Frontrunner buildings (S9) *(1 Person-month)*. **Task 4.3:** Subtask "Optimisation of energy vector": Co-creation between AIGUASOL and ITEC with stakeholders (including AHC, ALCN and NEUES LEBEN) will promote the design and potential installation of optimal systems to be selected according to circular criteria as well as their cost (installation, maintenance and renovation cost) *(0.5 Person-month)*.
Key personnel
Mag. Wolfgang Kiselka MSc. (Male, 32, project manager), Degree in Urban and Regional Planning and Economics. Experience in participatory projects on the local level. Employed at the Viennese Urban Renewal Office from 2011 to 2016. Working at NL since 2016, managing new building projects.
Experience

Related projects:

- **Global Park**
  Subsidized residential units. Passive house with integrated controlled ventilation. Intercultural inclusive project with intensive social support service. Built 2012-2014.
- **Sonnwendviertel**
  Passive house with integrated controlled ventilation. Broad range of shared common rooms like indoor cinema, wellness-center with pool and sauna, shared kitchen for events, boulder-area, rehearsal room, bicycle repair workshop, indoor playground etc. Built 2012 – 2014.
- **Paulasgasse 22**
  Wooden building with 60 subsidized residential units. Fully removable and recyclable elements. Built 2014 -2016.
- **Mühlgrundgasse**
  120 partly subsidized residential units. The whole building will be heated and cooled by a heat pump via thermo active building systems. This is the first residential building on such a scale to use this system. Planned start of construction: End of 2017.

Infrastructures and facilities

Full office building with all needed equipment. All services at the building through sub-contractors

Participant No.	**12**	ssociated with (2018)1767830 - document Ref. 03/04/2018
Short Name	**CECODHAS**	
Country	Belgium	
Organization Type	Non-profit	
Website	www.housingeurope.eu	

Partner Profile

Established in 1988, CECODHAS is the European Federation of Public, Cooperative & Social Housing - a network of 43 national and regional federations which together gather about 43.000 public, social and cooperative housing providers in 23 countries. Altogether, they manage over 26 million homes, about 11% of existing dwellings in the EU.

Social, public and co-operative housing providers have a vision of a Europe which provides access to decent and affordable housing for all in communities which are socially, economically and environmentally sustainable and where everyone is enabled to reach their full potential.

For CECODHAS, the priority of the European Union and its member states in the field of sustainability, energy and climate in the next decade should be to develop a positive business environment for the reduction of energy consumption in the housing sector, with a specific attention paid to the affordable housing sector.

Role in the project

CECODHAS participates in the WP 3 for the identification of stakeholders and the co-creation of CEBOs and their engagement,
Also, in the WP6 leading the task of policy recommendations to foster the circularity of housing sector.
CECODHAS will contribute to disseminate project results (WP7) among its wide network of contacts.

Detailed description of CECODHAS's role in WP4. ***(Total Person-months: 5)***
Task 4.6: Task Leader. Reviewing of the financial frameworks and existing funding policies for building renovation in the countries where the pilot projects will take place. This review will be in particular based on inputs by CECODHAS members.
Support for the development of financing plans (including alternative funding mechanisms) for the HOUSEFUL solutions, taking into account existing programs as well as potential future EU funding opportunities (ERDF and EIB).
Monitoring of the implementation of the plan for each Frontrunner building, including risk/uncertainty assessment, mitigation plans and the evaluation of different potential scenarios and impacts.
Organisation of an event where potential projects fulfilling HOUSEFUL criteria could meet potential financing institutions *(5 Person-month)*.

Key personnel

Sorcha Edwards (F), joined CECODHAS team in 2003. She is the Secretary General and Energy Co-ordinator and is in charge of the so-called CECODHAS Energy Expert Network, a group within CECODHAS gathering experts from 20 national housing federations from all across Europe.
She led the POWER HOUSE EUROPE Project, an initiative aimed at creating National and European Platforms of key stakeholders in order to promote energy efficiency and sustainable energy in housing. Sorcha is currently coordinating the POWER HOUSE Nearly Zero Energy Challenge project, aimed at engaging local housing organisations to share experiences and identify avoidable mistakes to meet the nearly-Zero obligations outlined in the Energy Performance of Building Directive.

After her Communication & Language degree during which she studied a year in Germany, she completed a Master in European Policies at the University of Leuven, Belgium. She has also worked for the Irish Institute for European affairs developing training seminars on EU policies and to the TACIS EU Technical Assistance Office to monitor the implementation of the TACIS programme.

Sorcha speaks English (mother tongue), French, German and Spanish.

Dijol Julien (M), Since 2007 is Policy Coordinator at CECODHAS, the European Social Housing Federation. He has been mostly working on the interlink between EU policies in the field of territorial cohesion and social inclusion and national housing policies. He has been working on various EU projects, in particular in the field of Cohesion Policy (URBACT, IEE).
He is an URBACT Thematic expert and was until recently external expert for the European Commission for a "study on sustainable regeneration in suburbs – promoting social integration in deprived neighbourhoods through housing interventions by ERDF".

Julien was also member of the Board of the Social Platform, the European network of social NGOs (2011-2013)

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Beyond this, he liaises with other EU organisations within the framework of EU projects in the field of urban development and social inclusion (EU programmes: URBACT, FP7, PROGRESS). He also coordinates CECODHAS publications on EU policies and housing as well as Structural Funds and housing, and is from times to times called to deliver training courses on EU policies related to housing to various audiences including local civil servants.

Julien is a sociologist and political scientist by education and has been trained at the Institute of political studies (IEP) in Bordeaux (France), the University of Stuttgart (Germany) and the Université Catholique de Louvain (Belgium).

Julien speaks French (mother tongue), English, German and Spanish.

Edit Lakato (F) is a Policy Officer at CECODHAS. She has been mostly working on the impact analysis and monitoring of the implementation of European public policies in relation to the social housing sector, in particular cohesion policy and its tools. Besides, she is also monitoring the activities of the European Parliament and proposals for action impacting the area.
Before joining CECODHAS in 2015, she gained experience on regional and urban policies at the European Commission and at different Regional Agencies in France and Hungary. Edit holds several degrees in social sciences (Law, political sciences and Economics) from the University of Reims (France) and University of Debrecen (Hungary) and she also made a specialisation in European and International law at the University Paul Cézanne (France). She speaks Hungarian (mother tongue), French, English and currently learning German.

Experience

Publications

- Publication: "POWER HOUSE NEARLY ZERO ENERGY CHALLENGE", published in June 2015 □ Publication: "ENERGY REFURBISHMENT FOR SUSTAINABLE SOCIAL, PUBLIC AND COOPERATIVE HOUSING: INSIGHTS ON THE CURRENT MARKET & TRENDS TOWARDS 2020", published
  in April 2014
- Publication: "EXEMPLARY FINANCING MODELS FROM ACROSS THE EUROPEAN UNION", published in January 2014
- Publication "STUDY ON FINANCING OF SOCIAL HOUSING IN 6 EUROPEAN COUNTRIES", published July 2013
- Publication "FAIR ENERGY TRANSITION TOWARDS NEARLY-ZERO ENERGY BUILDINGS", published in February 2013
- Publication: "THE HOUSING EUROPE REVIEW 2012 - THE NUTS AND BOLTS OF EUROPEAN SOCIAL HOUSING SYSTEM", published in 2012
- Chapter Social Housing in STUDY ON SOCIAL SERVICES OF GENERAL INTEREST FINAL REPORT - European Commission DG for Employment, Social Affairs and Inclusion", published in October 2011
- Publication "THE PLACE OF INHABITANTS RESIDENTS' PARTICIPATION IN CREATING SUSTAINABLE HOUSING & NEIGHBOURHOODS", published in May 2011
- Publication "URBAN REGENERATION IN EUROPE: THE PLACE OF SOCIAL HOUSING IN INTEGRATED URBAN POLICIES - CURRENT PERSPECTIVES", published in May 2009
- Publication "FINANCING SOCIAL HOUSING AFTER THE ECONOMIC CRISIS", published in September 2009
- Publication "WELFARE TRANSFORMATION AND DEMOGRAPHIC CHANGE IN EUROPE: CHALLENGES FOR THE SOCIAL HOUSING SECTOR", published in September 2007

Previous projects or activities relevant to the subject of this proposal.

POWER HOUSE Nearly Zero Energy Challenge (IEE) 2015 www.powerhouseeurope.eu

POWER HOUSE EUROPE (IEE) 2011 www.powerhouseeurope.eu

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E3SoHo (ICT-PSP) 2013 www.e3soho.eu

SHELTER (IEE) 2013 www.shelterproject-iee.eu

SF-Energy Invest (IEE) 2013 www.sf-energyinvest.eu

SUITE (URBACT) 2013 http://urbact.eu/en/projects/quality-sustainable-living/suite/homepage/

ERHIN Responsible Housing 2014 www.responsiblehousing.eu

HELPS 2015 hwww.helps-project.eu

HOST (AAL) 2014 www.host-aal.eu/cms/welcome-host/about-host-project-description

Infrastructures and facilities
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Participant No.	13
Short Name	WE&B
Country	SPAIN
Organization Type	SME
Website	www.weandb.org

Partner Profile

Water, Environment and Business for Development (WE&B) is a consulting company focused on the Water and Environment sectors, offering services in Social and Business Innovation. Our highly educated and globally experienced team participates in research projects to remain at the cutting edge of innovation while processing the skill and expertise to adapt to and enhance these innovations to bring them to our clients in the development sector. Under social innovation, WE&B brings its expertise in: (i) Socio-environmental site characterisation and analysis; (ii) Communication and stakeholder engagement strategies and implementation; (iii) Science – practice interface. Under Business innovation WE&B excels at (i) Sustainable Business and Economic Models; (ii) Market Studies / Analysis (iii) Cost – Benefit Analysis and (iv) Circular Economy and Economic Analysis. **WE&B** are members of the European Water Supply and Sanitation Technology Platform (WssTP), founding members and vice chair (together with UNESCO-IHE) of the WssTP working group on "Water Beyond Europe" and Administrators of the public authority network: *Network for Water in European Cities and Regions* (NetwercH20). WE&B are also members of the International Water Association and more specifically as active members of the specialist group Ecosystem Services. In addition, WE&B is part of the business networks The Practitioner Hub for Inclusive Business, Global Network on Sustainable Innovation and Entrepreneurship and Global Network on Sustainable Lifestyles. WE&B are implementing a number of Water, Climate Change and Environmental projects in Europe, Africa and Latin America on behalf of the World Bank, GGGI, European Commission, H2020, EuropeAid amongst others.

Role in the project

WE&B is the WP3 leader that will aim at developing and implementing the Social Engagement Strategies, where WE&B will work together with the other HOUSEFUL participants and the selected key stakeholders outside of the project consortium in the co-creation of the services for a circular economy at building scale. Within this WP, WE&B will lead the tasks Task 3.1. where stakeholders will be identified and analysed according to their interest and influence. Furthermore WE&B will lead task 3.2 on the engagement and co-creation strategies where WE&B will design and draw up the framework for 3 workshops using the innovative backcasting methodology for each of the case studies. WE&B will also lead task 5.4 that will assess the social impacts of the project through devising a number of social indicators and undertaking a baseline assessment to which the social impact can be assessed. WE&B will also participate in WP6 on business modeling and developing the HOUSEFUL SaaS functionalities. Under task WP7, WE&B will give valuable input into the communication strategy and coordinating the tasks related in WP3. WE&B will take an active role in the dissemination and communication of the project activities under WP7.

Key personnel

David Smith (M) is the director at WE&B. He has a MSc from Loughborough University (WEDC), United Kingdom, in Water and Environmental Management (2015), a Bachelors (2000) and Honours (2001) degree from the University of Cape Town, South Africa. His expertise in economic and social development is focused on environmental management, integrated water resource management, water technology, business models, economic viability for water and sanitation treatment, ecosystem services and capacity building. David was elected as a board member of the WssTP from 2012 - 2014. He is also the vice President and co-founder of NetwercH2O. He is currently a core member of the IWA Ecosystem Services Specialist group. David has also worked for four and a half years as a project coordinator at the research centre CTM in Spain, where he sourced and managed funding for consortiums of several water and environmental projects. David has worked for 5 years (2003 – 2008) for the water and wastewater treatment company Thames Water Utilities Ltd, in London England.
Beatriz Medina (F) is MSc in Environmental Sciences from the University of Alcalá de Henares and Master on Science, Environment and Health Communication degree. Beatriz is a partner and consultant at WE&B as an expert in the field of socio-environmental studies. Working experience in the field of science communication and knowledge brokering, public participation processes and analysis of risk perceptions in environmental issues. Related projects where she gained experience have been funded at Latin American level (WaterClima LAC by Europeaid, EcoQuito by AECID and Green Growth Potential Assesment in Peru). She developed broad experience in developing R&D&I projects by the European Commission (OBRA, PACHELBEL, MUSTANG, WaterDiss2.0, DEMOWARE, DESSIN, Water RtoM, INSITRATE, AfriAlliance, Run4LIFE). Additionally, she has published topic related articles and chapter books at scientific recipients.

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Moreover, Beatriz is research associate in the Sociology Department in the Autonomus University of Barcelona (UAB), working in the context of Honest project, which focuses on gaining understanding on how societies have engaged with nuclear energy, and how the nuclear energy sector has engaged with societies over the course of the past 60 years, focusing in the Spanish case study. Specially, collaborating in understanding perceptions and mechanisms of societal engagement.

Experience

- **Related publications**

- Development of a tool (Systematic Tool for Behavioural Assumption Validation and Exploration - STAVE) designed to support policy-making for sustainability in real-world settings in 6 countries, as part of the PACHEBEL project – Beatriz Medina
- **Medina, B**. (2014). *Research to Market, can we speed-up the knowledge transfer process? In. Herve-Bazin,* C. (ed.) Water Communication *Analysis of Strategies and Campaigns from the Water Sector. International Water Association Publishing, IWA,* ISBN: 9781780405216
- Amorsi, N.; **Medina B**.; Midgley, S.; Nickel, D., Stein, U.; von Toggenvrug, J. (2016). Analysis and uptake of FP water research results in Martinuzzi, R-A., Sedlacko, M. (eds.) *Knowledge Brokerage for a Sustainable Europe - Innovative Tools for Enhancing the Connectivity of Research and Policy Making on Sustainable Development Issues* (Greenleaf).

Related projects

- Innovation & demostration for a competative and innovative european water reuse sector (CAD, DEMOWARE, FP7 2016 - 2021)
- Green Growth Potential Assessment (GGPA) in Peru and Nepal (GGGI, 2016)
- Watershed and coastal management in latin america and the caribbean (WATERCLIMA LAC, EUROPEAID)
- Africa-EU innovation alliance for water and climate (AFRIALLIANCE, H2020, 2016 – 2021) - Stakeholder Mapping and Engagement Strategies for Water and Climate actions between Africa and Europe
- Low carbon business action in mexico (IDOM, EuropeAid 2015-2016)
- Recovery and utilisation of nutrients 4 low impact fertiliser (RUN4LIFE, H2020; 2017 - 2021)
- Demonstration of an environmentally-friendly desalination system concept: transforming seawater into valuable resources (LIFE+DREAMER; 2017 - 2020)
- to reduce the energy consumption in wastewater treatment plants and minimize its environmental impact. (LIFE+ BRAINYMEM; ACCIONA AGUA, 2017)

Infrastructures and facilities

Central offices in Barcelona (Sant Cugat) that can be used for project meetings.

Participant No.	**14**	
Short Name	ICONS	
Country	Italy	
Organization Type	Non-Profit Private Organisation	
Website	http://www.icube.global	

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Partner Profile

Fondazione iCons (ICONS) is a private non-profit organisation, whose mission is to improve human capital through public communication of European research and innovation achievements, cultural initiatives and citizens' participation for the benefit of society. ICONS brings together the know-how in market exploitation of the founder company iCons srl (www.icons.it) and the expertise in public research communication of its media spinoff youris.com (www.youris.com), established and managed since 2004 by ICONS founders and acknowledged today as one of the most important on-line information centres on research and innovation in Europe.

ICONS communication, dissemination and exploitation activities are managed and developed under the umbrella of the iCube Programme (www.icube.global), an innovative communication and exploitation programme for researchers and innovators aiming at maximizing public and professional impacts of communication and exploitation activities within public funded European and international projects. ICONS adopts an innovative integrated impact-based communication and exploitation approach. Its capability to effectively distribute contents and to assess, measure and demonstrate the effectiveness of communication and dissemination activities enables a scalable and optimized implementation of the project strategies to effectively reach the expected project impacts. To this end, ICONS studies and develops indicators to measure the research impacts on society in terms of awareness, acceptance and uptake.

ICONS partners with consortia in EU and public-funded projects and non-profit organizations (research centres, universities, associations, etc.) to enhance dialogue with stakeholders and the public at large. It supports research activities and outcomes through targeted communication services and strategies to increase outreach and impacts on society as a whole. ICONS masters a wide range of communication formats and channels including, among others, web-based knowledge transfer tools, public innovative communications formats (including, among others, journalistic articles, multimedia story-telling and various video formats) and social media strategies, global distribution via online, social and TV media and monitoring of outreach and engagement, design of dissemination materials, networking and dialogue with stakeholders' communities, exploitation modelling, coordination and management of the overall communication, dissemination and exploitation activities within EU projects' consortia.

The top management and key personnel of ICONS has successfully managed dissemination & communication activities in over 30 EU-funded projects (FP7 and H2020) in different research domains: Smart Cities, Bioeconomy, Environment, Security, Energy, Health, Transport, Nanotechnologies, Society, ICT. Out of them, fourteen projects cover energy efficiency domains.

Role in the project

The iCube department of the Fondazione iCons will carry out the overall management of communication and dissemination activities to maximise public and professional impacts of dissemination and exploitation activities within the HOUSEFUL project. ICONS will lead WP7 Communication and Dissemination for social and market acceptance, designing and implementing an impact-based Dissemination, Communication and Engagement strategy to support awareness, acceptance and replication at local and European level.

Key personnel

Elisabeth Schmid (F), iCube Progamme Director, has a four-year B.A. Hons in Linguistic Communication and Multilingual Mediation Sciences with a specialization in Translation. Elisabeth has been working as consultant for EC projects in various research domains and as a coordinator of EU project office activities since 2005. In 2009 she joined youris.com in charge of the coordination of youris.com communication activities. Since then she has worked as project manager in a number of EU-funded projects on energy efficiency, smart cities, environment, transport and society. In the past 6 years she has successfully initiated more than fifteen EU-funded projects and has coordinated the dissemination of 6 major EU-funded projects. She's currently managing the communication, dissemination and stakeholders' engagement activities of the SCC-01 Smart City Lighthouse project REMOURBAN and is coordinator of the FET-Open project EFFECT. In the Smart City area she also launched and coordinated the dissemination and communication activities of two FP7 Smart City projects, R2CITIES and CITyFiED and she has been one of the initiators of the MySmartCityDistrict clustering platform. She has been working as project manager in different communication and management activities in a number of Coordination and Support Actions (CSA) and Specific Support Actions (SSA) coordinated by ICONS founders on bioecon-

omy, environment, health, mobility and nanotechnologies.

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Marcello Bardellini (M) - Project Manager at iCube Programme has a Master Degree in Economics and Political Science at the University of Milan and a Master Diploma in Marketing and Communication Management. He has a professional background in market analysis, more than four years experience in European Projects with a focus on e-procurement and mistery shopping on e-government covering different European countries. He is in charge of project management & communication activities of EU-funded projects, stakeholders' relations management, development of dissemination and communication strategies and measurement of impacts. In the Smart City area he is dissemination and communication leader of the R2CITIES and CITyFiED (FP7) projects and he is directly involved in the citizen engagement activities and interaction with local correspondents in the H2020-SCC-01 REMOURBAN project. In the EeB and energy area he is responsible for exploitation activities in the BUILDHEAT project and for communication in the ISOBIO project. He is also the communication leader of the H2020 bioeconomy projects iMETland and SKIN.

Charlotte Michi (F) - iCube social media and community engagement strategist, holds a degree in physics and in classical piano. Combining her two passions, she worked as a researcher and as digital marketing specialist for brands of the classical music world. Over the past three years she has been working as social media and communication strategist in a number of EU-funded projects: EFFECT (H2020) in the FET area and in the Smart Cities projects CITyFiED (FP7), R2CITIES (FP7) and REMOURBAN (H2020). In the latter she has been involved in building citizens engagement strategies. Charlotte is now in charge of the communication activities of the UrbanGreenUP project (H2020 SCC-02). She joined Fondazione iCons from its foundation year (2016).

Experience

Related projects:

URBAN GreenUP "New Strategy for Re-Naturing Cities through Nature-Based Solutions" Funding programme: H2020-SCC-02-2016 Contract No. 730426.
Description: Leader of URBAN GreenUP Communication and Dissemination activities, also involved in the definition of strategies for citizen engagement and co-creation activities, replication and exploitation activities. Urban GreenUP aims at obtaining a tailored methodology (1) to support the co-development of Renaturing Urban Plans focused on climate change mitigation and adaptation and efficient water management, and (2) to assist in the implementation of NBS in an effective way.
Duration: 2017-2022
Website: www.urbangreenup.eu

REMOURBAN "Regeneration Model for accelerating the smart urban transformation" Funding programme: H2020-SCC-01-2014 Lighthouse project – GA 646511.
Description: leader of Dissemination and Communication activities with a key role in other specific tasks related to citizen engagement, replication and exploitation strategies. The project aims at the development and validation in three lighthouse cities of a sustainable urban regeneration model that leverages the convergence area of the energy, mobility and ICT sectors in order to accelerate the development of innovative technologies, organisational and economic solutions to increase resource and energy efficiency, improve the sustainability of urban transport and reduce greenhouse gas emissions.
Duration: 2015-2019
Website: www.remourban.eu

iMETland "A new generation of Microbial Electrochemical Wetland for effective decentralized wastewater treatment"
Funding programme: H2020-Water-1a-2014 – GA 642190.
Description: iMETland aims to construct and validate a full-scale application of a eco-friendly device to treat urban wastewater from small communities at zero-energy operation cost through the integration of Microbial Electrochemical Technologies (MET) with the biofilters used in constructed wetlands. Duration: 2015-2018
Website: www.imetland.eu

BuildHEAT "Standardized approaches and products for the systemic retrofit of residential Buildings, focusing on HEATing and cooling consumptions attenuatio

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Funding programme: H2020-EeB-08-2015 - GA 680658.
Description: BuildHeat develops standardised approaches, products and solutions for the systemic retrofit of residential Buildings, focusing on heating and cooling consumptions attenuation, improving residential wellbeing and bringing high-energy performance, comfort and healthiness.
Duration: 2015-2019
Website: www.buildheat.eu

CITyFiED "Replicable and Innovative Future Efficient Districts and cities" Funding programme: FP7-ENERGY-2013-8.8.1 – GA 609129.
Description: A replicable, systemic and integrated strategy to adapt European cities and urban ecosystems into the smart city of the future, by designing, constructing and managing large scale district renovation projects for achieving nearly zero energy cities. Leader of Dissemination and Communication activities, cooperation and collaboration framework through a dedicated communication and engagement approach towards the CITyFiED network of cities (cluster cities and a community of interest of 40 cities) to guarantee replication, development of exploitation-supporting tools. Duration: 2014-2019
Website: www.cityfied.eu

Infrastructures and facilities

youris.com public communication and distribution platform (www.youris.com).
The youris.com platform, established in 2004, is managed by ICONS. The platform will be made available for the distribution of the HOUSEFUL public communication contents. youris.com is one of the most important European primary information sources on European innovation for global media: its contents regularly taken up by online newspapers and magazines, journalists, information multipliers and social media networks, TV broadcasters. youris.com holds a record of more than 500 stories (videos, articles and interviews) produced on EU research and innovation, a global network of 120 national public TV broadcasters and an audience of over 150 million TV viewers worldwide on TV media. More than 10 million unique users worldwide have come across youris.com articles and videos online.

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Participant No.	15
Short Name	IDP
Country	Spain
Organization Type	SME
Website	www.idp.es

Partner Profile

IDP is an innovative and multidisciplinary SME (Small and Medium Enterprise) with services ranging from civil & infrastructure engineering, environmental sciences, ICT to project management & consultancy. With six offices between Spain and Latin America, **IDP** currently consists of a team of approximately 250 professionals working nationally and internationally in multidisciplinary projects in Electrical and Control Automation Engineering, Mechanical Engineering, Waste Treatment and Environmental Engineering, Information and Communication Technology, Chemical, Civil and Structural engineers, Architects and Project Managers.

INDUSTRIAL & ARCHITECTURE. More than 1,000 projects in all productive sectors (including Cement, Mining and Waste Industry) endorse IDP as a leading engineering company specialized in developing engineering, construction and consulting services for projects of industrial plants and logistic infrastructures. The multidisciplinary technical teams of IDP have the capability to design any type of facilities and is always looking for the most advanced solutions and efficient, technologically innovative and environmentally friendly. Therefore, the most innovative tools are early adopted and used in all our projects, like the know-how acquired through of **BIM technology (Building Information Technology),** making it able to address crosscutting and comprehensively any project with the aim of achieving the most efficient, cost effective and secure facilities.

ENVIRONMENT. IDP is a world leader in management and treatment of urban and industrial waste. Nowadays is working for multilateral organizations such as the Inter-American Development Bank (IDB) in areas related to the management, handling and treatment of Waste, as well as strategic planning, definition of collection systems, projection and recovery of treatment facilities, or research and development in treatment processes. In the water sector, IDP operates throughout the complete cycle (collection, purification, water supply, sanitation and treatment) with a shared objective to optimize water use and applications.

CIVIL WORKS. IDP has become lately in a world-leading company regarding the design and construction of civil works, including both railroads and roads as well as tunnels, bridges and so on using BIM methodology. IDP has invested money as well as human resources in order to internally develop this methodology in order to integrate it with GIS technology in order to develop GeoBIM models: BIM geo-located models that include all the necessary BIM information in addition to geo-location data. This has allowed IDP to produce highly accurate models of roads and railroads with all the physical info, which has enabled showing clients a very realistic view of the final outcome. In addition, these GeoBIM models allow simulating the performance of every sort of vehicles on these infrastructures, making clients sure that the civil works at issue will fulfil both the current regulations as well as user needs regarding visibility, driving comfort, etc.

SMART CITY. Thanks to the integration of BIM methodology and GIS technology, with GeoBIM models as an outcome, IDP has developed what we call CIM models (City Information Modelling). This new concept addresses the management of city infrastructures, roads/streets, public/private buildings, energy as well as water installations and production units and so on in a single city model. This aims at enabling city administrations to carry out an easier and unified management of city assets. The enormous amount of information that these models can take into account allows these administrations to carry out studies, simulations (new/different mobility routes, PV plants…), etc. of any action that is to be undertaken before actually carrying them out, thus having a better decision-making, saving future costs due to wrong decisions, etc.

Role in the project

The role of IDP in HOUSEFUL is twofold. Firstly, we will be responsible for the engineering of the implementation of the some of the proposed solutions inside all the Spanish demo-sites (WP4), which will afterwards be used for the execution of the works until their commissioning. In this way IDP will provide its expertise in both water and waste sector as well as in architecture and building services to act as the technology integrator who will develop the best solutions for Greywater Treatment, Biowaste disposal & valorization and Supply of high quality fertilizer/compost. Secondly, pairing with LEITAT, we will guide the advance from Lab to the pilot sites of Biogas production from blackwater and grinded biowaste until its effective implementation in the selected buildings.

Detailed description of IDP's role in WP4. ***(Total Person-months: 35)***

Task 4.2: Definition of the technical specifications (TS) for the solutions including technological implementation: Greywater and blackwater separation (S5.1), commercial O_3/UV hygienisation (S5.2), integration of water meters at household and building level for the control of water reuse (S5.3), integration in Building of NBS on facade or roof (S5.4), Treatment of liquid fraction with NBS indoor, at facade and outdoor (S6.1), AnMBR system (S7.1), and valorisation of biogas (S7.4) *(9 Person-months)*.

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Task 4.3: IDP will collaborate in the design of the compact and transportable AnMBR unit able to treat the organic load rate of 8-10 home equivalents. Then, IDP will carry out the integration of it and the other water and waste technological solutions such as commercial O_3/UV into the different buildings (in Task 4.8) *(5 Personmonths)*.

Task 4.4: Installation of water meters for quantification of water reuse in toilets at home and Building level after O_2/UV treatment in Demo buildings 1 and 2 *(5 Person-months)*.

Task 4.5: Preparing technical content of water and waste solutions for the definition of the technical execution projects in Demos 1, 2 and 3 *(1 Person-month)*.

Task 4.7: Provision of technical support to AHC for the preparation of tendering documents *(1 Person-month)*.

Task 4.8: Task Leader, in charge of the technical assistance and supervisión of the Works related to the technological innovations shown in S5, S6, S7, S8 and S9, as well as the implementation of the different solutions and monitoring elements in the building *(14 Person-months)*.

Key personnel

Eduard Loscos (M) BSc in Physics (UAB) specialized in Optics and Master in Environmental Engineering (UPC). After 10 years of experience as Sales Manager and CEO at ATIPIC TEN afterwards, he started as a Business Developer of R&D projects inside CETEMMSA Technology Centre. Now he is Deputy Business Development and Innovation Manager at IDP.

Vicenç Jimenez (M) Manager of Energy, Environment and Industry Department at IDP. He has an extensive experience as a project manager in Environmental: eg. Projects like a Municipal Solid Waste Facilities, Municipal Solid Waste Landfill at Catalonia (Refuse fraction from MBT's plants) Municipal Solid Waste Masterplan Santa Catarina (Brazil) or Industry: Inter- American Investment Corporation - Mexico Paper Manufacturing; Operations mechanical design for mining process and product processing industry or CEO Mechanical Engineer HVAC Laboratories (Medichem Combino Pharm) – Malta, etc.

Eduard Millet de Balanzó (M) MBA and Industrial Engineer with more than 15 years working on the field of waste management. He had worked in the URBASER company before he became a Project Manager in the Energy, Environment and Industry Department at IDP.

Jordi Hernández Sapé (M) MSc in Industrial Engineering (UPC) with more than 10 years in installations design and planning, with special focus on energy efficiency.

Andreu Cortada (M) Architect, Revit specialist and BIM Manager at IDP. He received his degree from UPC in Barcelona and afterwards finished Master Degree in Digital Architecture and Advanced Design from Universitat Internacional de Catalunya.

Mikel Borràs (M) Junior architect (UPC) after five years of experience carrying out projects in architectural rehabilitation, developing both draft and executive projects as well as construction management duties, he is now R&D projects deputy manager at IDP. He is responsible for the writing of R&D projects and their financial management within IDP.

Experience

Related publications:

- Master plan for the control and treatment of MSW in the Santa Catarina State, Brazil.
- Technical assistance for the supervision of the Project and Works for the Biomethanization and Compost Plant of Can Barba (Sabadell) Spain.
- Closure and Sanitation Project for a Municipal Solid Waste Landfill of 400 Ha, in Mexico
- Development of 13 public investment projects for the redevelopment and the recovery of the areas degraded by the solid waste (landfills) in the cities of Andahuaylas, Bagua, Huamanga, Oxapampa, Chancay, Pozuzo, San Juan Bautista, Yauyos, Abancay, Aymaraes, Huaura, Tarma and Chincha.
- Project, Works management and setting into operation of Disposal Center and Pneumatic Collection Center for MSW, Sabadell city centre.

Related projects:

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- **METGROW + -** Building upon its EIP RMC status and its core role in EIT Raw Materials, METGROW+ develops an industrially viable, flexible "New Metallurgical Systems" Toolbox, which consists of a broad range of already existing and newly developed "metallurgical unit operations". The unit operations follow a value-chain approach, from pre-treatment, to metal extraction, metal recovery and (residual) matrix valorisation. The industrially-driven, interdisciplinary METGROW+ consortium involves all relevant stakeholders along the value chain, from the (upstream) pretreatment to the final (downstream) residual matrix valorisation in building materials.
- **INTERMODEL EU** – H2020 Project under the call *MG-8.4a-2015: Smart governance, network resilience and streamlined infrastructure innovation*, which aims to develop a BIM-based methodology and its integration with physical simulation tools to improve the decision making regarding the design, construction and management of intermodal terminals. This shall aim at the their optimization from a holistic point of view, taking into account energy efficiency and environmental sustainability, Costs, facility management and maintenance and logistics & mobility performance, both in and outside the terminal.
- **DOF** - District of the Future, under the 7th Framework Program of Research and Development (ICT2013.6.4 Optimising Energy Systems in Smart Cities) of the European Union, which aims to develop a solution based on information technology and communications to facilitate the monitoring, management and optimization of energy production and consumption in three districts in three cities in Europe (Corby, Orleans and Sabadell).
- **IPAN** - Innovate Popular Low Density Structural Panel, under the 7th Framework Program of Research and Development (ENV-2012.6.3 Innovative resource efficient technologies, processes and services) of the European Union, which aims to design the process of large-scale production of industrial structural panels of wood, using both wood waste as the treetops (specifically fast growing trees), optimizing surface finish with innovative resins, minimizing emissions and resource consumption.
- **HERMES** - Development of Smart and Flexible Freight Wagons and Facilities for improved transport of Granular Multimaterials, under H2020 Program of Research (RIA, MG-2.2-2014), aims to improve Bulk Freight Transport. IDP leads the development of Bulk Cargo Terminals using BIM Methodology

Infrastructures and facilities

- **Autodesk – Revit.** A specific program for BIM (Building Information Modelling), consisting on the computer-aided design for 3D drawings.
- **Autodesk - Infraworks 360**. BIM preliminary design software helps to settle BIM 3D models using real environment data like Google Maps files.
- **Presto**. Software tool for budgets, measures, temporary planning, security and health, quality control, environment and costs control.
- **Autodesk – Plant 3D.** BIM Process & Instrumentation Design software.
- **CivilCad2000.** Software tool for the design, the calculation and the sizing of structures (usual structures, bridge abutments, reinforced bridges, prefabricated beams, mixed structures).

Participant No.	**16**	
Short Name	**Homebiogas**	
Country	Israel	
Organization Type	Small Medium Enterprise (SME)	
Website	https://homebiogas.com/	

Partner Profile

HOMEBIOGAS LTD (HBG) is an Israeli small enterprise founded in 2012. We develop and market technologically advanced, affordable and easily assembled biogas systems.
HOMEBIOGAS LTD (HBG) develops and manufactures market-leading, best of class, small scale biogas systems; i.e. integrated solutions for waste management and the generation of clean renewable energy. We aim to realise the huge market potential in this field and become a large, successful and highly-valued company.
Since 2006, the founders have been involved in international projects for creating renewable energy from waste, for "off-the-grid" communities in Africa, Asia and Latin America. The company's founders have been active in Israel since 2010 and formally incorporated HBG in February 2012. Based upon the knowhow and ideas that the founders accumulated during the previous years of field trials and scientific research, the team dedicated the last three years to develop and market family sized biogas systems.
Based on extensive field trials and user feedback, during 2014 we started a complete redesign of the product and a new approach to sales, with the strategic decision to tackle the business opportunity for HBG in the market of organic waste management and onsite renewable energy generation in Europe. We successfully adapted our products to high income markets through the development and commercialisation of HOMEBIOGAS TG1: a household oriented, convenient and technologically advanced biogas system. To date, we have sold more than 1000 units of HOMEBIOGAS TG1 to end users and distributors in over 73 countries (e.g. Australia, Israel, Italy, Spain, Portugal, Sweden, France, Chile, Kenya, Mexico, Nigeria, South Africa, USA). We are in the process of establishing our distribution channels and marketing presence. Revenues for 2016, the first year that we began offering our products for sale, exceeded €500K.
HBG employs 16 people and operates out of our 1,000 m^2 site on the campus of Hadassah Neurim in Beit Yanai, Israel. Up to now, our funding has come from investors who provided us with approx. €2.5 million, in addition to a support of approximately €700 thousand through programs originating from the Office of the Chief Scientist in Israel. Our investors include a Florida-based cleantech venture fund and Israeli, American and British individuals.
From its inception HBG decided to outsource all manufacturing work and has established long term relationships with large manufacturers who provide all of the product's components to the company for final testing, packaging and shipping. This concept reduces HBG's capital expenses and manpower outlays, and since these large manufacturers can manufacture in large volumes; lack of production resources will not be a bottleneck.

Role in the project

Homebiogas will demonstrate the feasibility of a conventional HOMEBIOGAS system for the valorisation of grinded biowaste together with solid fraction of domestic wastewater (WP4). Homebiogas will install this system an open area (a large garden) of a centenary house located in Austria. The system will be also feed with other kitchen wastes from neighbors. The biogas produced is planned to be stored for winter or be used for cooking in the same house that generates bio-waste and blackwater. The dried digestate can be used as fertilizer in the own garden (with a wintergarden installed). Homebiogas will demonstrate the feasibility of circular economic business models that underpin new services based on performance/functionality rather than ownership.
To achieve this objective, the innovation needs to progress from the current TRL6 (Technology demonstrated in a relevant environment) and CRL1 (Hypothetical commercial proposition) (Commercial Readiness Level, see section 1.3A for further information) to **TRL9** (Final technology system proven through successful implementation) and **CRL3** (Commercial scale up). The commercial and technical objectives will be achieved on time and on budget through the constant and effective management of all the technical, commercial, financial, legal and administrative aspects of the project.

Detailed description of HOMEBIOGAS's role in WP4. ***(Total Person-months: 15)***
Task 4.2: Definition of the technical specification (TS) for the solution S7 to be demonstrated in Demo 3 (TS 7.2), aimed at offering a compact, transportable and modular dry AD aimed at efficiently valorise the solid fraction of wastewater (WW) and bio-waste (BW) at continuous mode. The TS 7.2 will include a new sludge and liquid discharge system, based on a new suction unit. The suction unit will feed digestate to the compost system developed by ALCN (S8) *(2 Person-months)*.
Task 4.3: Modification of current dry AD HOMEBIOGAS according to TS 7.2. Preliminary tests of the new dry AD system using real wastes from Demo 3. Collaboration with ALCN and CARTIF in the valorisation of digestat as compost (S8) and biogas (S7) *(5 Person-months)*.

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Task 4.4: Collaboration in the definition of the monitoring program related to the biogas quality evaluation made by ALCN and digestate valorisation *(1 Person-month)*.
Task 4.5: Collaboration in the preparation of the execution project related with the innovation made in the dAD *(1 Person-month)*.
Task 4.8: Collaboration in the management of permits for implementation of the dAD system at building scale. Development of optimisation works of the dAD system by using WW and BW at continuous mode in Demo 3. Direct collaboration with ALCN in digestate and biogas valorisation *(6 Person-months)*.

Key personnel

Mr. Erez Lanzer (M): Project Manager; HBG Chief Operating Officer. He is a business savvy operator and organizer who combine management skills with business, sales and organizational skills. Previously, he was CEO at Lenopark, a $20 million private international B2C company manufacturing and selling consumer products. Education: B. A. Business Administration; Ruppin Academic Center, Israel.Mr Lanzer will be responsible for the overall running of the project, ensuring the different tasks in the Work Plan (Section 3.1) are performed according to deadlines and budget and that the project objectives are achieved. He will be the contact person for the project, being responsible for the communications and reports to the EC Project Officer.

Oshik Efrati (M): Technical Manager, HBG CEO and founder, is an entrepreneur who is a very focused and task oriented leader of people. He combines multi-disciplinary technology know-how and product management skills with people skills. Prior to HBG he was the Product Line Manager at SourceOutdoors (www.souceoutdoor.com), where he introduced and developed a product line which grew to be 50% of that company's income. Education: B. Sc. in Marine sciences, Ruppin Academic Center School of Marine Science, Israel.Mr Efrati will supervise all technical tasks of the HOMEBIOGAS project, ensuring the technical objectives are achieved and technical deliverables are produced and submitted on time.

Yair Teller (M): Chief Scientist Yair Teller is HBG Chief Scientist and founder, a visionary entrepreneur, committed to the cause of sustainability and fully motivated to help the poorer, underserved communities of the developing world. Yair became actively involved in sustainability, biogas and social change 9 years ago, during his stay in Mexico. Since then, he has become an internationally recognized evangelist, leader and authority on small biogas systems and in-situ recycling. Education: B. Sc. in Marine Sciences, Ruppin Academic Centre School of Marine Sciences, Israel. Mr Teller will play the leading role in the development and performance evaluation of HOMEBIOGAS TG6 alpha pilots and beta versions**.**

Experience

1. Patent 1: WO 2013190361 A1 *System and method for biogas and fertilizer production from household organic waste*

2. Patent 2: WO 2014203047 A1 *Lightweight assemblable appliance and respective method for production of biogas and liquid fertilizer*,

3. HBG has filed a **design patent,** which covers the physical look and feel of our product, in China, USA and Germany,

4. Our systems are the only small biogas systems that were tested and certified to meet the European **CE Standard** for safety,

HBG achieved world recognition by **meeting United Nations Secretary General Ban Ki Moon**, who wholeheartedly approved of the company's solutions for developing communities and blessed the collaboration with the

Infrastructures and facilities

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